AtkinsRéalis

South Waste Rock Pile at the Sunro Mine Site, Jordan River, BC

Teck Metals Ltd.

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PRELIMINARY TERRESTRIAL RISK ASSESSMENT

Signature Page

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Executive Summary

AtkinsRéalis Canada Inc.[1] (AtkinsRéalis) has completed this Preliminary Terrestrial Risk Assessment (PTRA) for Teck Metals Ltd. (Teck), for the South Waste Rock Pile at the former Sunro copper mine (the "Site") near Jordan River, BC. The purpose of the PTRA was to evaluate the potential for exposures and risks to people using this Site, as well as to terrestrial ecological receptors, from metals identified in soil. The results of the PTRA will be used to support risk management and remediation decision-making for this area of the Site. The scope of the PTRA is limited to the South Waste Rock Pile. The PTRA does not include an assessment of the North Waste Rock Pile, Concentrate Storage Area and tailings at the Site (including an area of piled tailings in the South Waste Rock Pile AEC) as, based on the available data and experience at similar sites, these areas are likely to pose unacceptable risks for human and/or ecological receptors and are also a source of metal loading to the adjacent aquatic environment of the Jordan River.

Human Health

The Site is accessible via an unpaved forest access road. While there is no formal day use area, signs of recreational use have been noted during Site visits. It was assumed that members of local Indigenous communities, as well as members of the general public, could use the Site for traditional or recreational purposes. No contaminants of potential concern (COPCs) for human health were identified in surface soil at the South Waste Rock Pile area of environmental concern (AEC), therefore no unacceptable risks to human health resulting from direct exposures to soil in this area are anticipated. Further, given the lack of COPCs in surface soil, the potential for significant exposures to soil COPCs via indirect exposure, including via consumption of country or traditional foods, is considered to be negligible.

Terrestrial Ecological Risks

The Site provides significant terrestrial habitat for a diverse array of species, potentially including species at risk (SAR). COPCs for terrestrial ecological receptors were identified in surface soil (copper and selenium) and subsurface soil (copper) at the South Waste Rock Pile, with copper identified as the COPC likely to drive risks.

An assessment of the terrestrial habitat at the South Waste Rock Pile was completed by Corvidae (2021). While elevated concentrations of copper have been identified in soil in this area, observations from SNC-Lavalin (2016) and Corvidae (2021) suggest that mature riparian forest ecosystem is present and was described as "intact" (Corvidae, 2021). While not conclusive, the biological observations for the South Waste Rock Pile AEC suggest that effects to terrestrial ecological receptors are not likely to be occurring at the population or community level. Further assessment of the area relative to reference areas could be conducted to confirm this finding.

The habitat assessment also identified potentially suitable habitat in the riparian corridor that bisects the North and South Waste Rock Piles for the Northern red-legged frog, an amphibian SAR documented within 1.4 km of the Site. Additionally, this habitat may also be suitable for the Wandering salamander, another amphibian SAR with documented occurrences in the area surrounding the Site. The presence of these amphibian SAR at the South Waste Rock Pile AEC is unknown and requires confirmation. The areas of suitable habitat for these amphibians in the AEC described by Corvidae (2021) relative to the areas being remediated within the AEC is also unclear. As a result, it is recommended that a survey of

[1] Formerly known as SNC-Lavalin Inc.

the riparian corridor be conducted to determine if the amphibian SAR are present, and that detailed mapping of habitat considered to be potentially suitable for SAR amphibians be performed. If these species are directly observed or are determined to have a high likelihood of being present based on the suitability of habitat within the contaminated area, additional assessment, followed by the completion of a Detailed Ecological Risk Assessment, is recommended.

Given the relatively small size of the areas of the Site evaluated herein and the vast productive habitat that surrounds these areas, consideration should be given to the disturbance to ecological receptors that would be required to complete physical remediation at the relatively inaccessible areas of the Site. Additionally, while the current PTRA does not include an evaluation of the potential for contributions of COPCs from these areas to nearby aquatic receiving environments, including the Jordan River, COPC loading from the South Waste Rock Pile should be considered in the decision-making process for contaminated soil management.

1. Introduction

AtkinsRéalis Canada Inc.[2] (AtkinsRéalis) has completed this Preliminary Terrestrial Risk Assessment (PTRA) for Teck Metals Ltd. (Teck), for the South Waste Rock Pile at former Sunro copper mine (the “Site”) near Jordan River, BC. The purpose of the PTRA was to evaluate the potential for exposures and risks to people using this Site, as well as to terrestrial ecological receptors, from metals identified in soil. The results of the PTRA will be used to support risk management and remediation decision-making for this area of the Site. This work was completed under purchase order 10560 and terms and conditions of the MSA TR00013 between Teck and AtkinsRéalis, as outlined in the work plan and cost estimate submitted to the Teck titled “2023 to 2024 Water Sampling and Risk Opinion Cost Estimate at the Sunro Mine, Jordan River, BC” (SNC-Lavalin, 2023).

The Site is located 3.5 km upstream of the community of Jordan River on Vancouver Island, BC and adjacent to the Jordan River (Drawing 631848-101). Where the Site borders the Jordan River, steep slopes and cliffs are present. Historical mining operations associated with the Site were accessed via a portal on the main haul tunnel (5100’ level) 1.2 km upstream of the community of Jordan River. Mine deposits are present on the east bank of the Jordan River, which are identified as areas of environmental concern (AECs) 1 and 2 and are referred to as the ‘North Waste Rock Pile’ and ‘South Waste Rock Pile’, respectively. Additional AECs include the Concentrate Storage Area (3), Tailings Pipeline (4), 5100 Portal (5), and 1963 Mine Blowout (6) (SNC-Lavalin, 2018). Drawings 631848-101 and 102, details the AECs and their location at the Site.

Mine wastes and soil have been characterized at these AECs during previous investigations (see **Section 1.2**), with elevated concentrations of select metals above applicable BC *Contaminated Sites Regulation*[3] (CSR) Standards (i.e., contamination). Previously, a Preliminary Risk Opinion (PRO; SNC-Lavalin, 2016) consisting of an exposure pathway analysis was provided for the North Waste Rock Pile, South Waste Rock Pile, and Concentrate Storage AECs; however, while potentially operable exposure pathways were identified, the PRO did not quantify exposures and associated risks. The PRO had also provided a qualitative assessment of potential risks to aquatic receptors, with a subsequent Preliminary Aquatic Ecological Risk Assessment completed by Azimuth Consulting Group Partnership (Azimuth, 2019).

The focus of the current PTRA is to further assess potential exposures and risks to human and ecological receptors associated with the soil contaminants identified at the South Waste Rock Pile. The scope of the PTRA is limited to the South Waste Rock AEC. The PTRA does not include an assessment of the North Waste Rock Pile, Concentrate Storage Area and tailings at the Site (including an area of piled tailings in the South Waste Rock Pile AEC) as, based on the available data and experience at similar sites, these areas are likely to pose unacceptable risks for human and/or ecological receptors and are also a source of metal loading to the adjacent aquatic environment of the Jordan River. As a result, these areas will have their own risk management plans/remediation plans.

Exposures to metals at the Site via the direct soil exposure pathways (i.e., incidental soil ingestion, skin contact and inhalation of particulates) are likely to drive risks to both human and ecological receptors and were the focus of this PTRA. A relatively small contribution from exposure to metals in other media (e.g., groundwater, sediment, surface water) would be anticipated for terrestrial receptors (including humans and terrestrial ecological receptors); therefore, additional routes of exposure were not evaluated herein. The potential for indirect exposure to mine-related contaminants though harvesting and ingestion

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[2] Formerly known as SNC-Lavalin Inc.
[3] *Contaminated Sites Regulation* (CSR), B.C. Reg. 375/96, includes amendments up to B.C. Reg. 133/2022, March 1, 2023.

of country foods (e.g., edible vegetation and/or wildlife) will be qualitatively discussed in the current evaluation.

The PTRA includes a problem formulation, comprised of the identification of contaminants of potential concern (COPCs) (i.e., based on receptor-pathway specific standards), receptors of concern, and the pathways by which receptors have the potential to be exposed to the COPCs. The PTRA also includes the quantitative evaluation of risks to humans (if applicable), using the Site for recreational or traditional purposes. Further, it evaluates the potential for unacceptable risks to terrestrial ecological receptors at the community or population level and considers risks to species at risk (SAR).

A limited biological survey/habitat assessment was conducted in July 2021 by Corvidae Environmental Consulting Inc. (Corvidae) to inventory and assess wildlife and vegetation in the waste rock area, including at the South Waste Rock Pile (Corvidae, 2021; Appendix A). The results of this assessment were reviewed and considered in the interpretation of the potential for soil quality at the Site to presents a potential risk to terrestrial ecological receptors at the Site. Additionally, Corvidae (2021) reviewed the potential for SAR to be present at the Site; while no SAR were identified during the Site visit, suitable habitat for listed amphibians was identified in the riparian corridor that bisects the South Waste Rock Pile. The implications of this will be considered in the determination of whether the evaluation of risks at the community to population level, versus the individual level, is appropriate for amphibians.

1.1 Regulatory Framework

The substances at the Site exceeding the applicable standards/guidelines (i.e., contaminants of concern [COCs][4]) were identified by SNC-Lavalin (2018). All COCs previously identified at the Site were carried forward in the PTRA as preliminary COPCs[5]. The COC in soil at the Site were assessed by analysis of soil and comparison of results to the reverted Wildlands (WL_R) standards outlined in the following:

- *Contaminated Sites Regulation* (CSR), B.C. Reg. 375/96, includes amendments up to B.C. Reg. 133/2022, March 1, 2023.

1.2 Previous Investigations and Risk Assessment

AtkinsRéalis completed investigations from 2015 to 2018 to enhance the understanding of conditions at the Site in support of risk management and remediation planning, completing three iterations of site investigation reports, as follows:

- Supplemental Stage 1 and Limited Stage 2 Preliminary Site Investigation (PSI) and Preliminary Risk Opinion, dated June 24, 2016 (SNC-Lavalin, 2016; the “2016 Stage 2 PSI”);
- Limited Detailed Site Investigation (DSI), Sunro Mine, Jordan River, BC dated July 10, 2017 (SNC-Lavalin, 2017; the “2017 DSI”);
- Supplemental Detailed Site Investigation, Sunro Mine, Jordan River, BC dated July 27, 2018 (SNC-Lavalin, 2018; the “2018 SDSI”); and

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[4] Contaminants of concern are defined as substance that is present in media at a site at levels that exceed generic numerical standards prescribed for that media.

[5] Contaminants of potential concern are defined as a substance that is retained for evaluation in a risk assessment.

- Preliminary Aquatic Ecological Risk Assessment – Jordan River adjacent to the 5100 Portal Area of the former Sunro Mine, BC dated August 2019 (Azimuth Consulting Group Partnership [Azimuth], 2019).

SNC-Lavalin identified on-Site and off-Site areas of potential environmental concern (APECs) with mine waste associated with the former operation for the Sunro mine as part the 2016 Stage 2 PSI. The Stage 2 PSI comprised soil, sediment, porewater and surface water sampling to identify AECs at the Site. Based on the results of the Stage 2 PSI, a PRO was conducted to qualitatively assess the potential for operable exposure pathways and thus potential risks to human, terrestrial and aquatic receptors. The PSI and PRO were intended to support regulator engagement for the Site.

During a desktop review, 13 suspected mine features were identified at the Site and seven were identified off-Site. Following the Site visit, six on-Site and six off-Site mine features were retained as APECs. Based on limited sampling results, six APECs were designated AECs, as described in **Table 1-1** below. The AECs at this Site are shown on Drawing 631848-102.

Table 1-1: Summary of Areas of Environmental Concern

AEC	Description
1	North Waste Rock Pile: located adjacent to the Jordan River adjacent to the 5100 Portal.
2	South Waste Rock Pile: located adjacent to the Jordan River downstream from the 5100 Portal.
3	Concentrate Storage: located adjacent to Unnamed Creek #2 and the mine access road to the 5100 Portal.
4	Tailings Pipeline: plastic pipe containing tailings assumed to extend from the 5100 Portal along the mine access road to the Site boundary where it becomes an off-Site APEC.
5	5100 Portal: Closed portal discharging water from the underground workings into the Jordan River.
6	1963 Mine blowout to surface extending to the Jordan River past AEC 3 and north of the 5100 Portal Area.

The investigation reports summarize the elevated concentrations of metals in mine wastes and soils, acid generation and leaching of metals from waste rock and/or tailings, as well as the discharge of impacted groundwater and surface water into aquatic receiving environments, including the Jordan River. Azimuth (2019) included a preliminary evaluation of risks to the Jordan River.

The investigations have identified mine wastes containing concentrations of metals exceeding the BC CSR Standards and Protocol 11 Upper Cap Concentrations (UCC) (BC Ministry of Environment and Climate Change Strategy (ENV), 2023a) for the protection of human health and/or environmental health in soil, groundwater, porewater and surface water at the Site. This included potential exposures to UCCs in soil mine wastes, porewater and surface water at the North Waste Rock Pile (AEC 1); soil at the South Waste Rock Pile (AEC 2); and soil only the Concentrate Storage Area (AEC 3) of the Site. Protocol 12 “high risk” conditions (i.e., where UCCs exceedances and complete exposure pathways per BC ENV [2023b] Protocol 12 are present) were identified at the Site in soil/mine wastes for the protection of ecological receptors only. On this basis, the Site is classified as a “high-risk” Site based on the current evaluation of the exposure pathways. At the North Waste Rock Pile, for example, the 95% upper confidence limit of the mean copper concentration was more than five times greater than the UCC (SNC-Lavalin, 2018). As assessment of the loading of dissolved copper from the Site to Jordan River was also completed by SNC-Lavalin (2018) to estimate the mass of copper discharging into the river. Based on the “high risk” conditions at the Site, proximity to the Jordan River and subsequent copper discharge, as well as presence of high sulphide waste rock, low sulphide waste rock and tailings at the North Waste Rock Pile and Concentrate Storage Area, these areas of the Site are to be remediated in the future. The tailings present at South Waste Rock Pile are also slated for remediation.

A biological survey/habitat assessment was also conducted in July 2021 by Corvidae Environmental Consulting Inc. (Corvidae) to inventory and assess wildlife and vegetation in the waste rock area, including in the South Waste Rock Pile (Corvidae, 2021). The assessment reviewed the potential for SAR to be present at the Site. While no SAR were identified, Corvidae (2021) concluded that there is suitable habitat for listed amphibians in the riparian corridor that bisects the North and South Waste Rock Piles.

1.3 Approach and Guidance

The PTRA has been completed in accordance with guidance and principles for risk assessment prescribed by the BC ENV CSR and associated guidance documents and protocols, as well as by the Canadian Council of Ministers of the Environment (CCME) and Health Canada. Risk assessment methods described in the guidance used are consistent with methods commonly used by regulatory agencies across Canada and the United States, including the United States Environmental Protection Agency (US EPA). The PTRA includes consideration of potential health risks to persons spending time at the Site, either directly or indirectly exposed to impacts reported in soil (i.e., the only media under evaluation in the current PTRA) as per BC ENV (2023c) and Health Canada (2010 and 2021). Ecological receptor exposure via soil pathways was evaluated accordance with BC ENV (2023c) and CCME (2020).

2. Problem Formulation

The Problem Formulation identifies the Site setting, COPCs, receptors of concern (ROCs), and the potentially operable exposure pathways between the COPCs and the receptors.

2.1 Site Setting

The historical mine comprising the Site, located on the southwest coast of Vancouver Island adjacent to the Jordan River and 3.5 km north of the community of Jordan River, encompasses an area of 250 hectares.

As previously described, historical mine operations occurred throughout the Site. As the Site is expected to naturalize overtime, the current and anticipated future land use of all AECs within the Site, including the South Waste Rock Pile, is reverted wildlands, associated with the application of BC CSR WL_R standards. The Site is accessed via an unpaved forest access road, which merges with Highway 14. There is no day use area, however, indications of recreational use, as evident by the presence of makeshift seating, beer bottles and ammunition casings, have been noted during Site visits. Fishing may also occur in the Jordan River and recreational foraging for vegetation/fungi may also take place in the region.

2.2 Ecological Setting

2.2.1 Surface Water Bodies

The nearest surface water body is the Jordan River, which is adjacent to the Site and flows into the Pacific Ocean (Strait of Juan de Fuca). Additionally, two unnamed creeks have been observed flowing over waster rock and tailings on either side of the South Waste Rock Pile, which subsequently flow into the Jordan River. Theses creeks flow year-round and are referred to as unnamed Creek #1 and Creek #2. The current PTRA includes the evaluation of the potential for risks to human and ecological receptors associated with direct or indirect terrestrial exposures to COPCs at the South Waste Rock Pile only. While the current PTRA does not include an evaluation of the potential for contributions of COPCs from these areas to nearby aquatic receiving environments, including the Jordan River, COPC loading from these areas should be considered in the decision-making process for how to manage or remediate soils at the South Waste Rock Pile.

2.2.2 Regional Ecology

The Site is located within the Coastal Western Hemlock very wet maritime (CWHvm1) subzone of BC. As stated in Meidinger and Pojar (1991), the CWHvm1 has a wet, humid climate with cool summers and mild winters with relatively little snow. Precipitation is high. Forests are typically dominated by Western hemlock and Western red cedar, as well as Amabilis fir and Yellow-cedar in wetter parts. The CWHvm1 understory generally features a well-developed shrub layer dominated by red huckleberry and Alaskan blueberry, and a well-developed moss layer dominated by *Hylocomium splendens* and *Rhytidiadelphus loreus*. Herbs are sparse and include minor amounts of Deer fern, Five-leaved bramble, Bunchberry, and Queen's cup. Subdued terrain on the west coast and northern end of Vancouver Island features very old successional stages of forest.

Wildlife that may be present include black-tailed deer, black bear, Douglas squirrel, Deer mouse and Brown bat. An incredible diversity of bird species is present, such as Blue grouse, Lewis woodpecker, American robin, Purple finch, Spotted owl, Northern flicker, Common raven, Chestnut-backed chickadee and Winter wren. Amphibian species may also be present in the damp litter of mature forests and near surface water, including Northwestern salamander, Western toad and Northern red-legged frog.

2.2.2.1 Habitat Assessments

The habitat assessments described below were conducted for the larger Site. Although the focus of the PTRA is the South Waste Rock Pile, observations provided for areas outside of the South Waste Rock Area are provided for context and information on surrounding available habitat that may influence the presence of wildlife species at the South Waste Rock Pile.

A Site visit was completed by SNC-Lavalin in November 2015 by a Registered Professional Biologist[6]. Generally, the natural vegetation at the Site was noted to have been significantly modified by historical development and excavation activities. The dominant climax species in areas outside of the waste rock piles, including the South Waste Rock Pile, consisted primarily of Western hemlock followed by lesser amounts of Western red cedar. The South Waste Rock Pile had an abundance of tree cover, shrub layer, and undergrowth. Established stands included Western hemlock, Western red cedar, and the occasional Big-leaf maple, the shrub layer consisted of Salal, Oval-leafed blueberry and Swordfern thickets.

The ecological setting, as well as the identification of the potential use of the Site by species at risk, was described and evaluated by Corvidae (2021)[7]. Their findings are summarized in a letter report with photos, provided as Appendix A. A Site visit was completed in July 2021 to document the ecological features of the waste rock areas, such as vegetation and habitat types and wildlife presence. The areas of waste rock described in the report, which appears to include areas within both the North and South Waste Rock Pile AECs (as it was not specified in the report), was found to be mostly barren, consisting of cemented rock with no vegetation growth and erosion of neighbouring soils into the Jordan River. A riparian corridor is present between the waste rock and an unnamed tributary to the Jordan River, which was occupied by Western hemlock and Western redcedar, Salmonberry, Sword fern and Huckleberry, while wetland vegetation was present in the stream and on the stream boundary. The extent and the size of this corridor and the overlap with identification contamination at the South Waste Rock Pile AEC have not yet been documented and should be determined through future habitat assessment of this area. Numerous wildlife species are noted to inhabit the area, however, the piled and compacted waste rock is not suitable for habitat due to the steep slope and barren nature. The forest riparian areas however are suitable for wildlife species, such as birds, mammals, reptiles and amphibians. Bear and deer scat were observed on Site, as well as American robin, Chipping sparrow, Dark-eyed junco and Red-tailed hawk. Critical habitat for Little Brown myotis (*Myotis lucifugus)* and Northern myotis *(Myotis septentrionalis*) were noted to overlap with the Site, and roosting habitat in the forest (on cedar snags) was also observed.

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6 Andrew Wan, MET, RPBio (#1936).

7 Jessica Harvey, MSc, RPBio (#2556).

2.2.3 Species at Risk

No species at risk were observed during the 2021 wildlife and vegetation survey (Corvidae, 2021). A query of the BC CDC iMap tool by Corvidae (2021) yielded occurrences of Northern red-legged frog (*Rana aurora*) approximately 1.4 km to the east of the waste rock area in the Desolation Creek watershed and in the ditch near the forest service road. Corvidae (2021) also concluded that the riparian corridor that bisects the waste rock area in the South Waste Rock Pile is considered suitable habitat for the Northern red-legged frog. A BC CDC iMAP (2023) query was also completed herein, which identified the same species (occurrence 32313) as well as potential presence of ermine (*Anguinae* sub species; occurrence 126698) and Wandering salamander (*Neides vagrans*; occurrence 103421) in the Jordan River area. See Appendix A for the 2023 species search output.

2.3 Contaminants of Potential Concern

SNC-Lavalin (2016, 2018) provides a summary of previous investigations at the Site, which includes the collection and analysis of soil samples from the Site in 2016 to 2018. Additionally, groundwater, sediment, porewater and surface water collected from the Site were also characterized. As previously described and further documented in **Section 2.4**, the scope of the current PTRA is limited to the evaluation of potential risks to human and terrestrial ecological receptors associated with exposure to contamination in soils at the South Waste Rock Pile. The area of piled tailings within the South Waste Rock Pile AEC (see Drawings 631848-102 and 103) has been excluded from evaluation, as this area will be physically remediated. While the current PTRA does not include an evaluation of the potential for contributions of COPCs from these areas to nearby aquatic receiving environments, including the Jordan River, COPC loading from these areas should be considered in the decision-making process for how to manage or remediate soils at the South Waste Rock Pile at the Site.

2.3.1 Soil COPC Screening

All metals identified at concentrations that exceeded the BC CSR WL_R numerical standards in soil at the Site were retained as preliminary COPCs for pathway specific screening in the PTRA to identify final COPCs. Soil concentrations of preliminary COPCs were compared to BC ENV CSR Schedule 3.1 Standards protective of human health for human intake of contaminated soil and for the environment for toxicity to soil invertebrates and plants. As noted, the most relevant land use to be evaluated at the Site, as defined in the CSR, and considered for this PTRA included WL_R. Soil parameters in exceedance of the CSR standards for the protection of human health and terrestrial ecological receptors, as well as the BC ENV Protocol 4 (BC ENV, 2023d) background concentrations, were retained as final COPCs requiring further assessment and were carried forward into the PTRA (**Section 3**), when operable exposure pathways to receptors were identified. CSR Industrial Land Use (IL) standards apply to depths of 3 m and greater and thus were used to screen for COPCs in soil samples collected beyond this depth.

All available soil sample locations located within the South Waste Rock Pile (AEC 2) were included the soil dataset for the current assessment which included 20 soil samples, 13 of which were for surface soil (≤ 1 m bgs).

Final soil COPCs are summarized below in **Table 2-1** for the South Waste Rock Pile (AEC2).As presented in **Table 2-1**, no final COPCs were identified for the evaluation of risks to human health. Final COPCs identified for the evaluation of risks to terrestrial ecological ROCs include copper and selenium.

Table 2-1: Summary of Soil COPCs

Preliminary COPC	Maximum Soil Concentration (µg/g) (and Location)	Protocol 4 Background Concentration (µg/g)	CSR UCC		CSR Screening Standards		Retained as a Final COPC for:	
			Human Health (µg/g)	Terrestrial Ecological (µg/g)	Human Health (µg/g)[a]	Terrestrial Ecological (µg/g)[a]	Human Health?	Terrestrial Ecological?
AEC 2 South Waste Rock Pile, Surface Soil (≤ 1 m bgs)								
Chromium	43 (BH18-10)	65	2,500	2,000	250	200	No	No
Cobalt	21.7 (WR15-05)	30	250	450	25	45	No	No
<u>Copper</u>	**<u>4,020 (SS17-07)</u>**	100	75,000	1,500	7,500	**150**	No	**<u>Yes</u>**
Iron	46,300 (SS17-08)	70,000	350,000	NS	35,000[b]	NS	No*	No*
Selenium	**5.73 (SS17-07)**	4	4,000	15	400	**1.5**	No	**Yes**
Vanadium	74.9 (BH18-03)	200	4,000	1,500	400	150	No	No
AEC 2 South Waste Rock Pile, Subsurface Soil (> 1 m bgs)								
Chromium	103 (BH18-03 DUP)	65	2,500	2,000	250	200	No	No
Cobalt	28.6 (BH18-02)	30	250	450	25	45	No*	No
<u>Copper</u>	**<u>1,510 (BH18-02)</u>**	100	75,000	1,500	7,500	**150 (300[c])**	No	**<u>Yes</u>**
Iron	62,500 (BH18-03)	70,000	350,000	NS	35,000[b]	NS	No*	No*
Selenium	0.75 (BH18-10)	4	4,000	15	400	1.5	No	No
Vanadium	237 (BH18-03 DUP)	200	4,000	1,500	400	150 (300[c])	No	No

Notes:

Bold	Indicates parameter retained as a COPC for further evaluation in the HHERA.
<u>Underlined</u>	Exceeds CSR UCC
NS	No standard
UCC	Upper Cap Concentration, BC CSR Protocol 11
WL_R	Wildlands reverted land use
( )	Brackets, comparison to IL standard as sample is > 3 m bgs and exceeds WL_R standard; if concentrations less than IL, not retained as a COPC
a	CSR Schedule 3.1 Part 1 Matrix Standard, Soil Standard for human intake of contaminated soil or toxicity to soil invertebrates and plants, for WL_R
b	CSR Schedule 3.2 Part 2 Generic Numerical Soil Standards to Protect Human Health
c	CSR Schedule 3.1 Part 1 Matrix Standard, Soil Standard for human intake of contaminated soil or toxicity to soil invertebrates and plants, for IL > 3 m bgs
*	Maximum concentration below Protocol 4 background concentration in soil for Vancouver Island

2.4 Receptors of Concern and Exposure Pathways

2.4.1 Human Receptors of Concern and Exposure Pathway Analysis

Potential receptors of concern (ROCs) considered for evaluation in the current PTRA were based on assumed current/future uses at the Site and are consistent with ENV and Health Canada (2021) guidance. As previously described, the Site is accessed via an unpaved forest access road, which merges with Highway 14. Access is not restricted, and while there is no formal day use area, signs of recreational use including makeshift seating, beer bottles and ammunition casings, have been noted during Site visits. Potential receptor groups at the Site associated with wildlands use were considered for further evaluation in the PTRA, including:

- Members of local Indigenous communities; and
- Recreational land users.

From the list of human ROCs above, the groups with the highest potential exposure to COPCs at the Site were selected for evaluation of potential risk where operable exposure pathways were identified. Those selected receptors serve as surrogates for the groups that are not directly assessed. Justification for retaining and excluding receptor groups is presented for each group, below:

- Members of Local Indigenous Communities: Retained.
  - The Site is located within the ancestral land of the Pacheedaht First Nation, therefore members of local Indigenous communities were retained as ROCs. People may be exposed to COPCs present at depths ≤ 1 m bgs through direct contact with soil including incidental ingestion, dermal contact and soil particulate (i.e., dust) inhalation. Ultimately, no soil COPCs were identified for the protection of human health therefore no unacceptable risks to this receptor group are predicted.
- Recreational Land User: Exposures assumed to be similar or less than those for members of local Indigenous Communities.
  - At this time, it was assumed that Indigenous community member use of the Site would be similar to or greater than that of a recreational land user. As such, as with the local Indigenous communities, no unacceptable risks are predicted for this receptor group.

It is acknowledged that human receptors at the Site could be exposed to COPCs in other media or via other pathways that are not directly or evaluated in the current PTRA; however significant exposures are not anticipated. This includes consumption of surface water, as well as consumption of traditional or country foods harvested from the Site, such as during hunting, fishing and foraging. As no COPCs were identified in soil for the protection of human health, significant plant uptake of contaminants from soil is considered unlikely. As such, bioaccumulation of contaminants up the food chain with subsequent human exposure is considered to be an insignificant exposure pathway. Based on the lack of identification of soil COPCs for the evaluation of risks to human health at the South Waste Rock Pile, the potential for human health risks is considered to be negligible based on the currently available data. This conclusion is considered appropriate to support risk management and remediation decision-making for the South Waste Rock Pile AEC.

2.4.2 Ecological Receptors of Concern and Exposure Pathway Analysis

As per ENV Protocol 1 (2023c) and CCME (2020), ERAs are conducted to assess the potential for contamination to pose an unacceptable level of risk to ecological receptors, with the goal for the continued presence of a biologically diverse, functional, self-sustaining and interdependent community or ecosystem. Where the potential for unacceptable risks is identified, remediation or risk management may be required to mitigate current or future risks.

Protection of ecological receptors at the community or population level is important for the continued existence of individual species and the health of ecosystems as a whole. Though the primary focus of the ERA is to assess potential effects at the community or population level, in some circumstances, such as for species of special concern, protection at the individual level may be required.

For the ecological portion of the PTRA, listed and sensitive species, populations, communities, habitats or ecosystems that are potentially exposed to COPCs are defined as ROCs. Ecological ROCs have been selected based on the information compiled from field observations and desktop data reviews, as outlined in Corvidae (2021) and summarized in **Section 2.2**.

The following section identifies potential ecological ROCs and evaluates potentially operable exposure pathways at the Site, where soil COPCs were identified for the protection of terrestrial ecological receptors. Exposure pathways are the route(s) by which ecological receptors are exposed to COPCs identified at the AECs.

2.4.2.1 Terrestrial Ecological Receptors and Exposure Pathways

As described in **Section 2.2**, the biogeoclimatic zone which spans the Site support a tremendous diversity and density of plants and wildlife. As per BC ENV (2023c) and CCME (2020), the following ecological receptor groups were considered for inclusion as ROCs in the terrestrial environment and thus, have been identified as preliminary terrestrial ROCs for evaluation in the PTRA:

- Terrestrial vegetation communities (including trees, shrubs and herbs);
- Soil invertebrate communities;
- Terrestrial bird populations (including herbivores, insectivores and carnivores);
- Terrestrial mammal populations (including herbivores, insectivores and carnivores);
- Reptiles;
- Amphibians; and
- Federal/provincially-listed species at-risk as individuals.

Three species of special concern were identified to either have been present in the vicinity of the Site, or to have the potential to be present at the Site. These species include the following:

- Northern Red-legged Frog (*Rana aurora*, BC blue listed);
- Ermine (*Mustela richardsonii anguinae*, BC blue listed); and
- Wandering salmander (*Aneides vagrans*, BC blue listed).

Exposure pathways are the route(s) by which ecological receptors are exposed to contaminants in the environment (e.g., soil, water, food). The terrestrial ecological ROCs listed above have the potential to be directly exposed to COPCs identified in Site soils; the direct contact with soil COPC exposure pathway is anticipated to be the pathway that would contribute most significantly to a terrestrial organism's overall exposure. Soil COPCs for terrestrial ecological receptors were identified at the South Waste Rock Pile AEC. Most ecological receptors are only anticipated to come into contact with COPCs at-grade (i.e., surface soil $\leq$ 1 m bgs), with the exception of deep-rooting vegetation which may contact subsurface soil COPC up to a depth of $\leq$ 3 m bgs (CSAP, 2013).

CCME (2020) provides potential ecological exposure pathways for the above-listed ROCs. Exposure pathways can be “direct” or “indirect”. Direct contact occurs when a receptor is exposed directly to the media containing the COPCs. An example would be a soil invertebrate (e.g., earthworm) exposed to COPCs in soil. An example of an indirect exposure would be an animal consuming prey that has accumulated COPCs from contaminated media. In the current PTRA, potential exposures and risks to terrestrial ecological receptors are qualitatively evaluated in **Section 3**.

2.5 Conceptual Site Model

The conceptual models for the human health risk and ecological risk components of the PTRA are summarized in **Table 2-2** and **Table 2-3**, respectively. Based on the information presented in the Problem Formulation, no COPCs were identified for the evaluation of risks to human receptors and thus, no unacceptable risks are predicted.

Based on the identification of final COPCs and potentially operable exposure pathways for terrestrial ecological receptors at the South Waste Rock Pile AEC, the PTRA includes a qualitative evaluation of risks to these receptors.

Table 2-2: Conceptual Site Model for Human Health

Land Use	Receptor of Concern	Age Group	Preliminary Risk Conclusion
Wildlands	Indigenous Community Members	All ages	No final COPCs identified and thus, no unacceptable risks are predicted
	Recreational Land User	All ages	Exposure pathways and assumptions assumed to be equivalent to or less than those associated with Indigenous community members and thus, no unacceptable risks are predicted

Note:
BOLD Exposure pathway retained for further evaluation in the Preliminary Terrestrial Risk Assessment (PTRA).

Table 2-3: Conceptual Site Model for Ecological Receptors

Land Use	Receptor of Concern	Receptor Type	COPCs and Operable Pathways
Wildlands	Ecological Receptors	Terrestrial Populations/ Communities	COPCs for terrestrial ecological receptors identified in soils at the South Waste Rock Pile AEC. **Potentially operable exposure pathways include those associated with direct contact with soil COPCs (including dermal contact and ingestion) and indirect exposure (e.g., through ingestion of contaminated food items).**
		Terrestrial SAR: Northern red-legged frog, Ermine and Wandering salamander identified as SAR with the potential to be present at the Site	COPCs for terrestrial ecological receptors identified in soils at the South Waste Rock Pile AEC. **Potentially operable exposure pathways include those associated with direct contact with soil COPCs (including dermal contact and ingestion) and indirect exposure (e.g., through ingestion of contaminated food items).**

Note:
BOLD Exposure pathway retained for further evaluation in the Preliminary Terrestrial Risk Assessment.

3. Preliminary Terrestrial Risk Assessment

Protection of ecological receptors at the community or population level is important for the continued existence of individual species and the health of ecosystems as a whole. Though the primary focus of the ERA is to assess potential effects at the community or population level, in some circumstances, such as for SAR, protection at the individual level may be required. The scope of the PTRA is limited to the evaluation of potential risks to terrestrial ecological receptors at the South Waste Rock Pile AEC. While the current PTRA does not include an evaluation of the potential for contributions of COPCs from these areas to nearby aquatic receiving environments, including to the Jordan River, COPC loading from these areas should be considered in the decision-making process for how to manage or remediate soils at the South Waste Rock Pile.

Soil COPCs for terrestrial ecological receptors were identified at the South Waste Rock Pile and terrestrial ecological receptors, including vegetation (trees, shrubs, herbs), soil invertebrates, birds, mammals, reptiles and amphibians, as well as SAR, have the potential to be directly and indirectly exposed to the COPCs identified. Potential exposures of the COPCs to terrestrial ecological receptors at the South Waste Rock Pile will be briefly discussed and qualitatively evaluated in **Section 3.1**, below.

3.1 South Waste Rock Pile

Copper and selenium were identified as COPCs for terrestrial ecological receptors in soils at the South Waste Rock Pile (AEC 2), with copper exceeding the BC CSR UCC in both surface and subsurface soil. As previously described in **Section 2.4.2**, at-grade receptors are most likely to come into contact with surficial soils, with the exception of deep rooting vegetation which may be exposed to soils up to 3 m bgs. The subsurface soil sample exceedances at the South Waste Rock Pile AEC 2 are very deep, which were identified at a depth > 11 m bgs. No operable exposure pathways exist between terrestrial ecological receptors and contamination at depths > 3 m bgs, and thus, there are no direct contact risks associated with this deep contamination.

As the goal of the PTRA is to identify COPCs that will drive risk assessment outcomes to inform remedial decision-making, 95% upper confidence limits of the mean (UCLMs) COPC concentrations were calculated using US EPA ProUCL Version 5.1, with the selection of 95% UCLM values determined by sample size, the distribution of data and number of non-detect samples in each data set as per recommendations by Helsel and Gilroy (2012). Appendix B provides the UCLM outputs.

The surface soil 95% UCLM concentration selenium was estimated to be 2.68 µg/g. While the 95% UCLM concentrations exceed the BC CSR WL_R soil standards for toxicity to plants and soil invertebrates, the selenium concentration is less than the BC ENV (2023d) Protocol 4 regional background concentration of 4 µg/g. On this basis, no unacceptable risks are predicted for selenium for terrestrial ecological receptors at the community to population level.

The surface soil 95% UCLM concentration for copper was estimated to be 1,695 µg/g. The 95% UCLM concentration for copper exceeds the both the BC CSR WL_R standard for toxicity to soil invertebrates and plants, as well as the UCC. Though there are elevated concentrations of COPCs identified in piled waste rock at the South Waste Rock Pile AEC, the bioavailability of copper in the waste rock pile is anticipated to be low as the copper toxicity characteristic leaching procedure (TCLP) results have been below standards (SNC-Lavalin, 2018) and due to the physical characteristics (hardened nature) of waste rock. Copper concentrations in excess of the UCC were also identified outside of the South Waste Rock Pile, such as at SS17-07 which is adjacent to the Jordan River (see Drawing 631848-103 and attached analytical tables). Based on the magnitude of exceedances noted in this AEC, copper is likely to drive risks to terrestrial ecological receptors and has been discussed in further detail, below.

The total area of the South Waste Rock Pile AEC is 4,300 m^2, with the area of tailings within this AEC that is slated to be remediated encompassing approximately 500 m^2. Corvidae (2021) noted that outside of the area of piled waste rock material, mature riparian forest ecosystem is present and was described as “intact” (Corvidae, 2021). While not conclusive, the biological observations from SNC-Lavalin (2016) and Corvidae (2021) for the South Waste Rock Pile AEC suggest that effects to terrestrial ecological receptors, particularly to the habitat provided by terrestrial vegetation, are not likely to be occurring at the population or community level. Additional observations comparing the vegetation at the South Waste Rock Pile to reference areas, with consideration of species distribution and other vegetation assessment metrics and would provide further support to this conclusion.

Three SAR were identified with the potential to be present in the vicinity (< 1.5 km) of the Site, including the Northern red-legged frog, Wandering salamander and Ermine (see **Section 2.2.3**). Corvidae (2021) identified suitable habitat for the Northern red-legged frog in the riparian corridor adjacent the South Waste Rock Pile . It is likely that this habitat would also be suitable for the Wandering salamander (BC CDC, 2023), while for the Ermine significant use is unclear as this species inhabits a variety of habitat subtypes, including wooded areas adjacent to wetlands and river systems, as well as meadows and agricultural fields. The home range for the Ermine is > 10,000 m^2 (FCSAP, 2012) therefore it is likely this species would not be present for long-periods of time at the South Waste Rock Pile AEC. Home ranges for the Northern red-legged frog have been recorded as 100 m^2 to 200 m^2 (Haggard, 2000; Schuett-Hames, 2004) and based on Maxcy and Richardson (1999) and COSEWIC (2012, 2014) the home range for the Wandering salamander is also likely small (<100 m^2), with many staying within 2 m to 10 m of capture sites.

The Northern red-legged frog is most commonly found in shallow ponds, along streams, in wetlands and moist forest (BC CDC, 2008). They require standing water for breeding, in which eggs are attached to emergent vegetation below surface water. This species is negatively associated with sloped hills and elevation and tend to be more abundant in riparian areas near breeding locations (BC MOE, 2004). The most critical component for terrestrial habitat is likely sufficient cover and, on a larger scale, connectivity and distance between wetlands to maintain metapopulation dynamics (BC MOE, 2004). As mentioned previously, most frogs don’t move any significant distance beyond capture points (< 10 m), however significant movement (> 300 m) has been recorded over short periods, likely due to displacement. Potential exposures to COPC in the South Waste Rock Pile to the Northern red-legged frog are therefore likely only to be through soil contact and invertebrates (diet) and in flatter areas; the unnamed creeks are unlikely to be breeding habitat based on the absence of stagnant pooled water and emergent vegetation, but this should be confirmed during a future Site visit.

Conversely, the Wandering salamander is completely terrestrial throughout its life cycle, lacking an aquatic larval stage (COSEWIC, 2014). From egg to adult, the species requires moist microhabitats, and is almost always found inside cavities or cracks within decaying logs or under loose bark covering logs or stumps, and rarely in contact with the ground (COSEWIC, 2014). As noted in the BC CDC iMap search, extensive tracts of suitable habitat exist in nearby forested lands and along Juan de Fuca Provincial Park, therefore the species is likely to be present in the region. The Wandering salamander is thought to be tolerant of some habitat disturbance because it has been found along the edges of forests and within clearings, burned-over areas, and transmission corridors (COSEWIC, 2014). As noted, this species has a high site fidelity and low rates of active dispersal and movement. Potential exposures to the Wandering salamander are therefore likely only to be through invertebrates (diet) as contact with soil is expected to be minimal, and direct contact with surface water is also unlikely.

While concentrations of copper in surface soil measured in this South Waste Rock Pile AEC are likely to drive potential risks to amphibian SAR if present, the bioavailability of copper in AEC surface soil is not currently well understood and based on experience at other mine sites, has the potential to be relatively low. Additionally, the areas of suitable habitat for these amphibians in the AEC described by Corvidae (2021) relative to the areas being remediated within the AEC is also unclear. As a result, it is recommended that a survey of the riparian corridor be conducted to determine if the amphibian SAR are present, and that detailed mapping of habitat considered to be potentially suitable for SAR amphibians be performed. If these species are directly observed or are determined to have a high likelihood of being present based on the suitability of habitat within the contaminated area, additional surface soil sampling targeting areas of the AEC considered suitable habitat for amphibians is recommended. Co-located samples of invertebrates and plant tissue could also be collected from these locations, to enable an evaluation of the bioavailability of copper in AEC surface soil and to allow for direct inputs of invertebrate tissue concentrations into amphibian exposure estimates. This additional information could be used to support the completion of a DERA for amphibian SAR with the potential to be present in this AEC, if required.

4. Conclusions

The objective of the current PTRA was to evaluate potential exposures and risks to human and terrestrial ecological receptors exposed to soil in the South Waste Rock Pile, to support remedial and risk management decision-making for this AEC. The results of the risk evaluation for human health and terrestrial ecological receptors are summarized below.

4.1 Human Health Risks

The Site is accessible via an unpaved and gated forest access road. While there is no formal day use area, signs of recreational use have been noted during Site visits. It was assumed that the Site could be used for traditional or recreational purposes. No COPCs for human health were identified in surface soil at the South Waste Rock Pile AEC, therefore no unacceptable risks to human health resulting from direct exposures to soil in this area are anticipated. Given the lack of COPCs in surface soil, the potential for significant exposures to soil COPCs via indirect exposure, including via consumption of country or traditional foods, is considered to be negligible.

4.2 Terrestrial Ecological Risks

The Site provides significant terrestrial habitat for a diverse array of species, potentially including SAR. COPCs for terrestrial ecological receptors were identified in surface soil (copper and selenium) and subsurface soil (copper) at the South Waste Rock Pile, with copper identified as the COPC likely to drive risks.

An assessment of the terrestrial habitat at the South Waste Rock Pile was completed by Corvidae (2021). While elevated concentrations of copper have been identified in soil in this area, observations from SNC-Lavalin (2016) and Corvidae (2021) suggest that mature riparian forest ecosystem is present and was described as “intact” (Corvidae, 2021). While not conclusive, the biological observations for the South Waste Rock Pile AEC suggest that effects to terrestrial ecological receptors are not likely to be occurring at the population or community level. Further assessment of the area relative to reference areas could be conducted to confirm this finding.

The habitat assessment also identified potentially suitable habitat in the riparian corridor that bisects the North and South Waste Rock Piles for the Northern red-legged frog, an amphibian SAR documented within 1.4 km of the Site. Additionally, this habitat may also be suitable for the Wandering salamander, another amphibian SAR with documented occurrences in the area surrounding the Site. The presence of these amphibian SAR at the South Waste Rock Pile AEC is unknown and requires confirmation. The areas of suitable habitat for these amphibians in the AEC described by Corvidae (2021) relative to the areas being remediated within the AEC is also unclear. As a result, it is recommended that a survey of the riparian corridor be conducted to determine if the amphibian SAR are present, and that detailed mapping of habitat considered to be potentially suitable for SAR amphibians be performed. If these species are directly observed or are determined to have a high likelihood of being present based on the suitability of habitat within the contaminated area, additional assessment, followed by the completion of a DERA, is recommended.

Given the relatively small size of the areas of the Site evaluated herein and the vast productive habitat that surrounds these areas, consideration should be given to the disturbance to ecological receptors that would be required to complete physical remediation at the relatively inaccessible areas of the Site. Additionally, while the current PTRA does not include an evaluation of the potential for contributions of COPCs from these areas to nearby aquatic receiving environments, including the Jordan River, COPC loading from the South Waste Rock Pile should be considered in the decision-making process for contaminated soil management.

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TABLES

TABLE 1: Soil Log
631848-2015 Sunro Mine Stage 1 and Ltd. Stage 2

Area of Environmental Concern	Sample Location	Sample ID	Sample Date (yyyy mm dd)	Sample Type[a]	Description	North (m)	East (m)	Depth (m)
South Waste Rock Pile	WR15-05	WR15-05-151117	2015 11 17	WR	WASTE ROCK, 50% gravel, angular, some red staining (iron mineralization), no apparent copper mineralization, trace fine grained sulfide minerals (2% pyrite), highly weathered	5364751.64	422234.14	0.35
	SS17-01	SS17-01	2017 12 13	SS	SILT and ORGANICS, black, loose, damp, rootlets	-	-	0.0 - 0.1
	SS17-02	SS17-02	2017 12 13	SS	SILT and ORGANICS, trace sand, fine grained, black, loose, damp, rootlets	-	-	0.0 - 0.1
	SS17-03	SS17-03	2017 12 13	SS	SAND, fine grained, some silt, brown, loose, damp, trace organics, rootlets	-	-	0.0 - 0.1
	SS17-04	SS17-04	2017 12 13	SS	SAND, fine grained, some silt, poorly graded, brown, loose, moist, trace organics	-	-	0.0 - 0.1
	SS17-05	SS17-05	2017 12 13	SS	GRAVEL, fine, subrounded, sandy, poorly, graded, grey, loose, moist, waste rock	-	-	0.0 - 0.1
	SS17-06	SS17-06	2017 12 13	SS	SAND, fine grained, some silt, trace gravel, fine, subrounded, poorly graded, dark brown, loose, moist, rootlets, waste rock	-	-	0.0 - 0.1
	SS17-07	SS17-07	2017 12 13	SS	SAND and GRAVEL, fine to coarse grained sand, fine gravel, angular, poorly graded, orange brown, loose moist, waste rock	-	-	0.0 - 0.1
	SS17-08	SS17-08	2017 12 14	SS	SAND and GRAVEL, fine to coarse grained sand, fine to coarse gravel, angular, trace cobbles, angular, poorly graded, grey with orange staining on some rock, some cementation observed with iron and copper staining, loose, moist, waste rock	-	-	0.0 - 0.1

[a] SS - Surface sample
TL = Tailings
WR = Waste rock

TABLE 2: Summary of Analytical Results for Metals in Soil

Area of Environmental Concern	Sample Location	Sample ID	Sample Date (yyyy mm dd)	Depth Interval (m)	pH	Aluminum	Antimony	Arsenic	Barium	Beryllium	Boron	Cadmium	Chromium	Cobalt	Copper	Iron	Lead	Lithium	Manganese	Mercury	Molybdenum	Nickel	Selenium	Silver	Strontium	Thallium	Tin	Tungsten	Uranium	Vanadium	Zinc
						Total Metals																									
					pH	µg/g	µg/g	µg/g	µg/g	µg/g	µg/g	µg/g	µg/g	µg/g	µg/g	µg/g	µg/g	µg/g	µg/g	µg/g	µg/g	µg/g	µg/g	µg/g	µg/g	µg/g	µg/g	µg/g	µg/g	µg/g	µg/g
BC Standard																															
CSR Reverted Wildlands Land Use (WLR)[a]					n/a	40,000	20	10	350	1-150[b]	15,000	1-30[b]	60[c]	25	75-150[b]	35,000	120	65	2,000	25	15	70-150[b]	1	20	20,000	9	50	25	30	100	200-450[b]
CSR Reverted Wildlands Land Use Upper Cap Concentration (WLR UCC)					n/a	400,000	200	250	7,000	1,500	150,000	300	2,000	250	1,500	350,000	1,200	650	20,000	250	800	1,500	15	200	200,000	90	500	250	2,500	1,500	4,500
CSR Industrial Land Use (IL)[a]					n/a	250,000	40	10	350	1-350[b]	1,000,000	1-70[b]	60	25	75-300[b]	150,000	120-1,000[b]	450	2,000	75	15	70-250[b]	1	40	150,000	25	300	200	30	100	200-450[b]
CSR Industrial Land Use Upper Cap Concentration (IL UCC)					n/a	1,000,000	400	400	15,000	3,500	1,000,000	750	2,500	2,000	3,000	1,000,000	10,000	4,500	20,000	750	1,500	2,500	20	400	1,000,000	250	3,000	2,000	20,000	3,000	4,500
South Waste Rock Pile	SS17-01	SS17-01	2017 12 13	0.0 - 0.1	4.08	9,720	0.26	2.49	49.8	0.11	2.6	0.185	23.5	8.70	**1,660**	16,900	8.33	5.70	136	0.123	0.91	20.2	**1.09**	0.47	21.1	< 0.10	0.54	0.22	0.244	38.9	51.0
	SS17-02	SS17-02	2017 12 13	0.0 - 0.1	4.31	6,550	0.18	1.40	42.4	< 0.10	4.1	0.142	13.7	8.76	**433**	11,500	4.69	3.99	259	0.132	0.39	15.7	0.58	0.21	34.0	< 0.10	0.31	0.83	0.131	25.2	38.0
	SS17-03	SS17-03	2017 12 13	0.0 - 0.1	4.72	16,800	0.24	2.69	38.2	0.16	< 2.0	0.076	37.9	11.2	**1,010**	31,200	9.99	11.0	255	< 0.040	1.66	21.3	**1.90**	0.80	11.0	< 0.10	0.62	0.38	0.149	63.2	34.4
	SS17-04	SS17-04	2017 12 13	0.0 - 0.1	4.16	17,100	0.58	4.87	47.8	0.17	2.0	0.104	32.6	12.0	**1,890**	31,200	12.0	9.68	258	0.077	2.42	22.4	**2.48**	1.28	19.3	< 0.10	0.74	0.45	0.272	64.4	38.7
	SS17-05	SS17-05	2017 12 13	0.0 - 0.1	6.40	16,700	0.11	2.88	49.5	0.22	< 2.0	0.073	34.4	11.1	144	24,300	2.53	11.2	381	< 0.040	0.36	26.0	< 0.20	0.11	18.9	< 0.10	0.44	< 0.20	0.267	59.3	42.3
	SS17-06	SS17-06	2017 12 13	0.0 - 0.1	5.59	21,400	0.14	3.57	80.2	0.31	2.1	0.113	40.6	12.2	52.9	27,700	3.86	16.5	489	0.059	0.39	31.5	< 0.20	< 0.10	25.2	< 0.10	0.41	< 0.20	0.348	66.5	54.4
	SS17-07	SS17-07	2017 12 13	0.0 - 0.1	4.02	15,300	0.49	1.60	15.4	< 0.10	< 2.0	0.116	32.8	20.2	**4,020**	**44,300**	9.69	6.88	267	< 0.040	1.07	24.3	**5.73**	2.29	10.4	< 0.10	1.67	1.68	0.077	55.4	42.4
	SS17-08	SS17-08	2017 12 14	0.0 - 0.1	4.30	15,300	0.48	1.35	14.6	< 0.10	< 2.0	0.079	34.3	15.5	**2,140**	**46,300**	17.9	3.53	258	< 0.040	9.87	18.4	**4.60**	2.45	14.0	< 0.10	1.34	0.76	0.052	71.5	33.2
	WR15-05	WR15-05-151117	2015 11 17	0.4	4.2	-	0.8	1.9	22	0.3	< 2	0.07	35.4	21.7	**1,890**	-	14.8	5.5	361	0.09	1	25.1	**4.8**	1.7	12.4	< 0.1	2.6	-	< 0.1	66.3	45
	BH18-01	BH18-01-01	2018 02 28	0.2 - 0.3	5.26	18,400	0.30	3.72	60.3	0.25	< 2.0	0.076	33.0	10.9	**691**	25,700	3.53	11.4	324	< 0.040	0.55	24.8	0.50	0.24	17.6	< 0.10	0.37	0.26	0.308	59.5	43.5
		BH18-01-04	2018 02 28	3.4 - 3.7	6.78	14,000	0.15	1.81	28.1	0.14	< 2.0	0.047	34.2	10.6	52.8	23,100	1.24	7.87	306	< 0.040	0.19	23.7	< 0.20	< 0.10	11.2	< 0.10	0.27	< 0.20	0.113	62.0	40.1
		BH18-01-05	Duplicate	3.4 - 3.7	6.78	13,200	0.14	1.86	26.7	0.14	< 2.0	0.055	31.8	10.4	49.5	22,500	1.45	7.52	268	< 0.040	0.20	21.4	< 0.20	< 0.10	11.2	< 0.10	0.26	< 0.20	0.111	57.4	36.5
		QA/QC RPD%			0	6	*	3	5	*	*	*	7	2	6	3	16	5	13	*	*	10	*	*	0	*	*	*	*	8	9
		BH18-01-09	2018 02 28	13.1 - 13.7	5.98	24,800	0.19	3.09	47.4	0.21	< 2.0	0.084	42.7	12.7	74.8	23,200	2.58	12.0	243	< 0.040	0.93	27.4	< 0.20	< 0.10	24.2	< 0.10	0.33	1.26	0.255	67.3	41.8
		BH18-01-10	2018 02 28	15.2 - 15.5	5.24	20,500	0.20	3.07	48.9	0.23	< 2.0	0.086	39.5	9.77	41.0	22,800	2.73	13.4	212	0.042	0.61	24.0	0.43	< 0.10	12.6	< 0.10	0.39	0.37	0.284	63.4	41.1
		BH18-01-12	2018 02 28	18.8 - 18.9	7.73	27,100	0.24	1.99	48.1	0.21	4.6	0.047	**72.2**	24.9	146	34,100	1.25	10.1	510	< 0.040	0.55	26.7	< 0.20	< 0.10	33.2	< 0.10	0.54	2.92	0.116	95.0	58.3
	BH18-02	BH18-02-03	2018 03 01	3.8 - 4.3	7.94	16,000	2.40	1.84	24.7	0.15	< 2.0	0.063	35.8	12.8	121	25,500	1.38	8.44	317	< 0.040	0.22	29.6	< 0.20	< 0.10	17.6	< 0.10	0.37	0.28	0.129	59.9	45.5
		BH18-02-04	Duplicate	3.8 - 4.3	8.01	13,800	0.35	1.71	24.5	0.14	< 2.0	0.064	30.8	12.2	**369**	22,700	1.92	7.33	274	< 0.040	0.16	24.8	< 0.20	0.10	14.0	< 0.10	0.30	0.27	0.135	55.5	41.0
		QA/QC RPD%			1	15	*	7	1	*	*	*	15	5	101	12	33	14	15	*	*	18	*	*	23	*	*	*	*	8	10
		BH18-02-06	2018 03 01	11.7 - 12.0	4.83	34,900	0.21	3.03	73.6	0.24	2.5	0.113	48.2	21.5	**1,510**	26,500	2.75	13.0	381	0.042	1.02	35.6	0.37	< 0.10	25.4	< 0.10	0.36	2.72	0.193	67.3	51.9
		BH18-02-08	2018 03 01	17.1 - 17.4	6.15	25,100	0.19	2.51	77.2	0.22	< 2.0	0.131	42.5	22.4	**788**	26,000	1.59	12.4	438	< 0.040	0.89	29.5	< 0.20	< 0.10	27.0	< 0.10	0.31	3.02	0.201	64.4	52.3
		BH18-02-10	2018 03 01	19.7 - 20.1	6.91	24,600	0.16	3.41	98.4	0.20	2.8	0.255	41.5	**28.6**	**1,660**	30,200	2.15	14.3	515	< 0.040	0.32	31.4	< 0.20	0.27	30.6	< 0.10	0.32	0.35	0.148	**110**	55.4
	BH18-03	BH18-03-01	2018 03 01	0.3 - 1.2	4.75	12,800	0.39	0.73	7.8	< 0.10	< 2.0	0.066	28.1	21.3	**1,190**	34,800	4.92	2.67	236	< 0.040	1.49	20.4	**1.66**	0.92	11.5	< 0.10	0.70	0.69	< 0.050	74.9	32.9
		BH18-03-04	2018 03 01	5.2 - 5.5	6.52	25,100	0.21	5.04	39.5	0.29	< 2.0	0.100	47.8	12.6	148	24,100	5.38	24.2	262	0.042	0.33	33.2	0.28	0.17	8.88	< 0.10	0.36	0.22	0.393	54.3	52.8
		BH18-03-06	2018 03 01	7.2 - 7.4	7.48	23,300	0.18	1.54	58.7	0.24	8.2	0.063	25.4	16.7	263	31,200	1.23	9.41	913	< 0.040	0.78	15.5	< 0.20	< 0.10	18.5	< 0.10	0.33	0.38	0.259	63.6	47.6
		BH18-03-08	2018 03 01	8.2 - 8.5	7.18	31,800	< 0.10	0.54	44.0	0.32	9.3	0.062	59.2	24.2	**494**	46,000	0.73	9.66	552	< 0.040	0.51	27.6	< 0.20	< 0.10	20.0	< 0.10	0.37	< 0.20	0.271	**124**	93.8
		BH18-03-12	2018 03 01	11.4 - 11.6	7.46	25,400	0.12	0.92	14.0	0.32	10.2	< 0.040	**86.1**	20.6	116	62,500	0.54	6.89	475	< 0.040	0.47	24.7	< 0.20	< 0.10	18.0	< 0.10	0.48	< 0.20	0.171	**204**	71.8
		BH18-03-16	Duplicate	11.4 - 11.6	7.56	29,600	< 0.10	0.93	16.4	0.41	12.4	0.041	**103**	23.2	126	72,000	0.51	8.05	588	< 0.040	0.55	29.5	< 0.20	< 0.10	21.5	< 0.10	0.58	< 0.20	0.185	**237**	77.4
		QA/QC RPD%			1	15	*	*	16	*	19	*	18	12	8	14	*	16	21	*	*	18	*	*	18	*	*	*	*	15	8
	BH18-10	BH18-10-01	2018 02 28	0.6 - 0.9	5.89	22,100	0.21	3.67	63.8	0.26	< 2.0	0.084	43.0	13.4	**304**	28,900	2.95	12.8	363	< 0.040	0.47	32.4	< 0.20	0.11	18.5	< 0.10	0.44	0.52	0.328	74.0	45.8
		BH18-10-03	2018 02 28	4.3 - 4.6	8.09	29,000	0.13	1.49	75.7	0.21	< 2.0	< 0.040	**60.9**	**25.5**	92.4	36,400	1.25	6.91	629	0.043	1.23	29.2	< 0.20	< 0.10	35.3	< 0.10	0.35	12.1	0.167	**120**	48.8
		BH18-10-05	2018 02 28	10.4 - 10.7	5.38	37,400	0.15	6.98	64.5	0.43	3.7	0.116	54.4	19.1	**96.0**	33,700	3.82	21.9	351	0.089	0.94	37.8	0.75	0.11	11.0	< 0.10	0.45	0.35	0.583	82.6	45.4
		BH18-10-06	2018 02 28	15.5 - 15.9	8.31	73,300	< 0.10	1.18	41.3	0.12	3.9	0.078	**147**	**25.2**	132	31,500	0.83	10.6	638	< 0.040	0.70	74.9	< 0.20	0.10	82.2	< 0.10	< 0.20	3.02	0.096	47.0	70.3
	BH18-11	BH18-11-01	2018 02 26	0.2 - 0.5	5.75	18,000	0.12	1.80	30.9	0.17	< 2.0	0.112	42.3	16.2	**526**	28,500	1.25	9.55	376	< 0.040	0.22	29.5	< 0.20	< 0.10	17.6	< 0.10	0.48	0.29	0.142	73.0	42.6
		BH18-11-03	2018 02 26	2.4 - 2.7	7.26	19,200	< 0.10	2.00	33.2	0.20	< 2.0	0.095	45.2	19.1	**702**	31,600	1.20	9.86	413	< 0.040	0.25	31.9	< 0.20	< 0.10	18.6	< 0.10	0.32	< 0.20	0.148	83.2	48.0
		BH18-11-05	2018 02 26	7.5 - 7.8	6.67	15,300	0.13	1.89	31.3	0.15	< 2.0	0.065	34.1	12.6	81.2	24,200	1.37	8.97	312	< 0.040	0.19	24.6	< 0.20	< 0.10	12.5	< 0.10	0.26	< 0.20	0.135	62.8	38.2
		BH18-11-09	2018 02 26	9.8 - 9.9	5.45	18,500	0.17	3.76	36.9	0.16	< 2.0	0.188	46.2	24.0	**1,030**	24,700	2.27	11.4	274	< 0.040	1.69	42.4	0.44	0.12	15.9	< 0.10	0.32	0.41	0.206	63.9	80.1
		BH18-11-11	2018 02 26	16.5 - 16.8	8.53	24,200	0.26	0.65	28.8	0.15	< 2.0	0.080	42.8	17.6	**441**	35,100	0.47	3.56	501	< 0.040	1.18	21.5	0.38	0.13	46.9	< 0.10	0.30	6.96	0.069	**117**	34.3
		BH18-11-13	2018 02 26	19.2 - 19.5	9.02	17,800	< 0.10	0.38	12.1	0.12	2.1	0.048	41.1	18.4	137	39,400	0.32	2.21	450	< 0.040	1.34	20.8	< 0.20	0.16	25.7	< 0.10	0.41	6.20	< 0.050	**163**	36.1

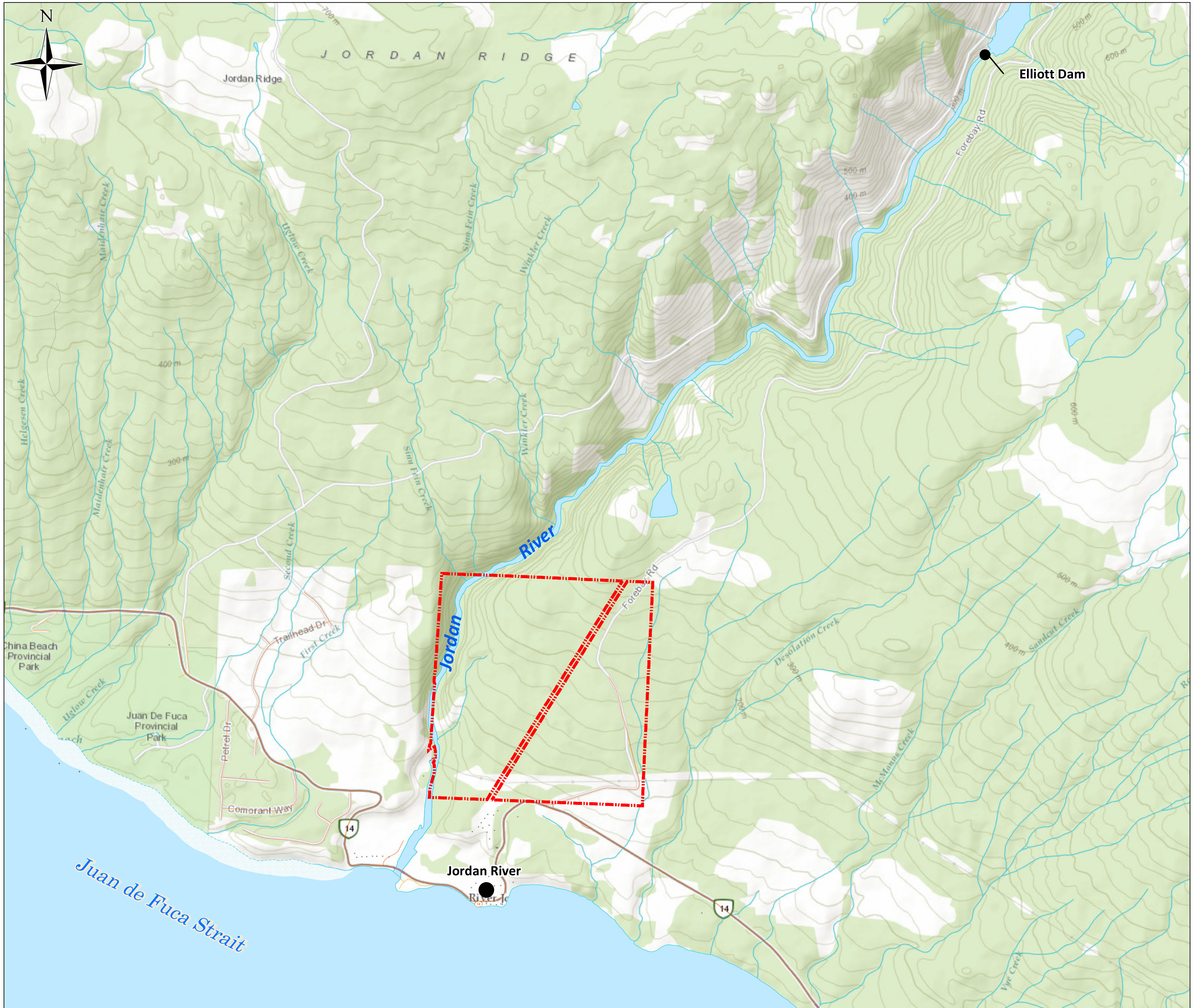
Associated Caro file(s): 5111639, 6072283, 8011451, 8031548.
All terms defined within the body of SNC-Lavalin's report.
< Denotes concentration less than indicated detection limit or RPD less than indicated value.
\- Denotes analysis not conducted.
n/a Denotes no applicable standard/guideline.
RPD Denotes relative percent difference.
* RPDs are not calculated where one or more concentrations are less than five times RDL.

BOLD Concentration greater than CSR Reverted Wildlands Land Use (WLR) standard or greater than CSR Industrial Land Use (IL) standard.
SHADED Concentration greater than CSR Reverted Wildlands Land Use Upper Cap Concentration (WLR UCC) standard or greater than CSR Industrial Land Use Upper Cap Concentration (IL UCC) standard

[a] The site-specific factors used for determining the matrix standards for this site include: intake of contaminated soil, groundwater used for drinking water, toxicity to soil invertebrates and plants, and groundwater flow to surface water used by freshwater aquatic life (whichever is most stringent). Samples with depths below 3.0 m are compared to Industrial Land Use (IL) standards.
[b] Standard varies with pH.
[c] Individual standards exist for Cr +3 and Cr +6. Reported value represents more stringent standard.

DRAWINGS

- 631848-101 – Site Location Plan
- 631848-102 – Areas of Environmental Concern
- 631848-103 – Soil Sample Location Plan



Legend

Site Boundary

NOTES:
1. Original in colour.
2. Numerical scale reflects full-size print. Print scaling will distort this scale, however scale bar will remain accurate.
3. Intended for illustration purposes, accuracy has not been verified for construction or navigation

REFERENCES:
1. Service Layer Credits: Content may not reflect National Geographic's current map policy. Sources: National Geographic, Esri, Garmin, HERE, UNEP-WCMC, USGS, NASA, ESA, METI, NRCAN, GEBCO, NOAA, increment P Corp.
Sources: Esri, HERE, Garmin, Intermap, increment P Corp., GEBCO, USGS, FAO, NPS, NRCAN,

REVISIONS:
0 - AO - 2018-06-26 - DRAFT - LH
2 - AO - 2018-07-25 - FINAL - JC

0 0.5 1 2 Kilometers

CLIENT:
Teck Resources Limited

PROJECT LOCATION:
Sunro Mine, Jordan River

AtkinsRéalis

Site Location Plan

BY: AO	SCALE: 1:30,000	DATE: 2018/07/25	REF No:	REV: 0
CHK'D: JC	Proj Coord Sys: NAD 1983 UTM Zone 10N		631848-101	

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Project Path: \\Sli2606\projects\Current\\Sli2606\projects\Current Projects\Teck Metals Ltd\631848 Sunro Mine\GIS_CAD\Exports

Legend

- Flow Direction
- Water Features
- Mine Access Road
- Former Mine Buildings
- Property Boundary
- Jordan River
- AEC 1: North Waste Rock Pile
- AEC 2: South Waste Rock Pile
- AEC 3: Concentrate Storage Area
- AEC 4: Tailings Pipeline
- AEC 5: 5100 Portal
- AEC 6: 1963 Mine Blowout
- Inferred Extent of Ferricrete
- Inferred Extent of Tailings
- 1 AEC ID

REFERENCES:
1. Service Layer Credits: © 2018 Microsoft Corporation © 2018 DigitalGlobe ©CNES (2018) Distribution Airbus DS
2. BCGOV ILMB Crown Registry and Geographic Base Branch (CRGB)
(data accessed through www.GeoBC.gov.bc.ca)
3. GPS Data Collected using an eTrex. Accuracy expected to be approximately +/- 3.5 to 10m depending on forest cover.

NOTES:
1. Original in colour.
2. Numerical scale reflects full-size print. Print scaling will distort this scale, however scale bar will remain accurate.
3. Intended for illustration purposes, accuracy has not been verified for construction or navigation purposes.

REVISIONS:
0 - AO - 2018-07-17 - DRAFT -JC
1 - AO - 2018-07-25 - FINAL - JC

0 25 50 100 150 200 Meters

CLIENT:
Teck Resources Limited

PROJECT LOCATION:
Sunro Mine, Jordan River

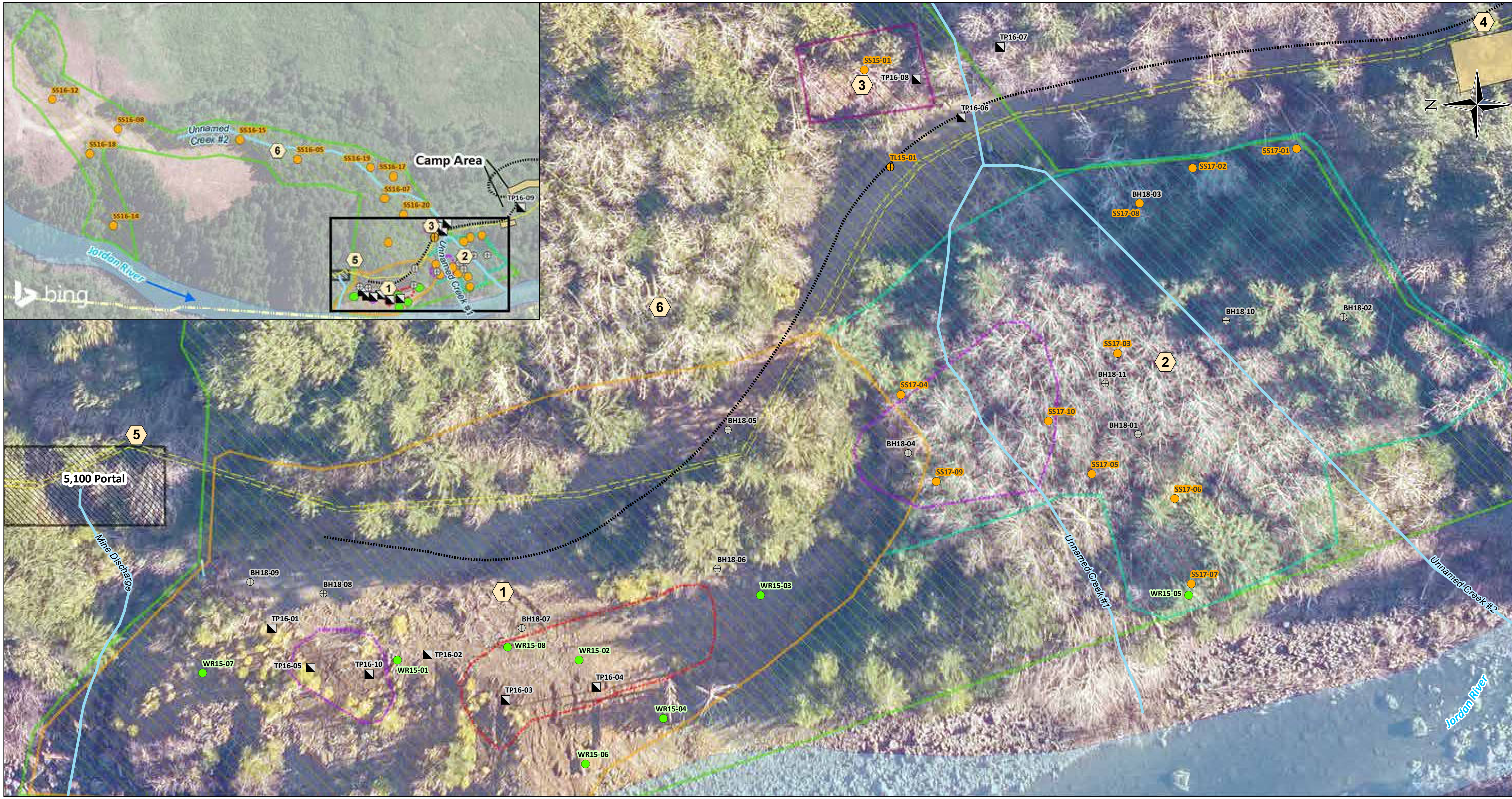
AtkinsRéalis

Areas of Environmental Concern

BY: AO	SCALE: 1:2,000	DATE: 2018/07/25	REF No: 631848-102	REV: 0
CHK'D: JC	Proj Coord Sys: NAD 1983 UTM Zone 10N			

MXD Path: \\Sli2606\projects\Current Projects\Teck Metals Ltd\631848 Sunro Mine\GIS_CAD\MapSeries\631848-102~AreasofEnvironmentalConcern.mxd

Project Path: \\Sli2606\projects\Current Projects\Teck Metals Ltd\631848 Sunro Mine\4.0 Execution\4.5 GIS and Drawings\Exports\PDF\DSI



Legend

- Borehole
- Tailings Sample
- Soil Sample
- Waste Rock Sample
- Testpit
- Flow Direction
- Water Features
- Mine Access Road
- Former Mine Buildings
- Property Boundary
- Jordan River
- AEC 1: North Waste Rock Pile
- AEC 2: South Waste Rock Pile
- AEC 3: Concentrate Storage Area
- AEC 4: Tailings Pipeline
- AEC 5: 5100 Portal
- AEC 6: 1963 Mine Blowout
- Inferred Extent of Ferricrete
- Inferred Extent of Tailings
- 1 AEC ID

REFERENCES:
1. Service Layer Credits: © 2018 Microsoft Corporation © 2018 DigitalGlobe ©CNES (2018) Distribution Airbus DS
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(data accessed through www.GeoBC.gov.bc.ca)
3. GPS Data Collected using an eTrex. Accuracy expected to be approximately +/- 3.5 to 10m depending on forest cover.

NOTES:
1. Original in colour.
2. Numerical scale reflects full-size print. Print scaling will distort this scale, however scale bar will remain accurate.
3. Intended for illustration purposes, accuracy has not been verified for construction or navigation purposes.

REVISIONS:
0 - AO - 2018-07-17 - DRAFT -JC
1 - AO - 2018-07-25 - FINAL - JC

0 5 10 20 30 40 Meters

CLIENT:
Teck Resources Limited

PROJECT LOCATION:
Sunro Mine, Jordan River

AtkinsRéalis

Soil Sample Location Plan

BY: AO	SCALE: 1:600	DATE: 2018/07/25	REF No:	REV: 0
CHK'D: JC	Proj Coord Sys: NAD 1983 UTM Zone 10N		631848-103	

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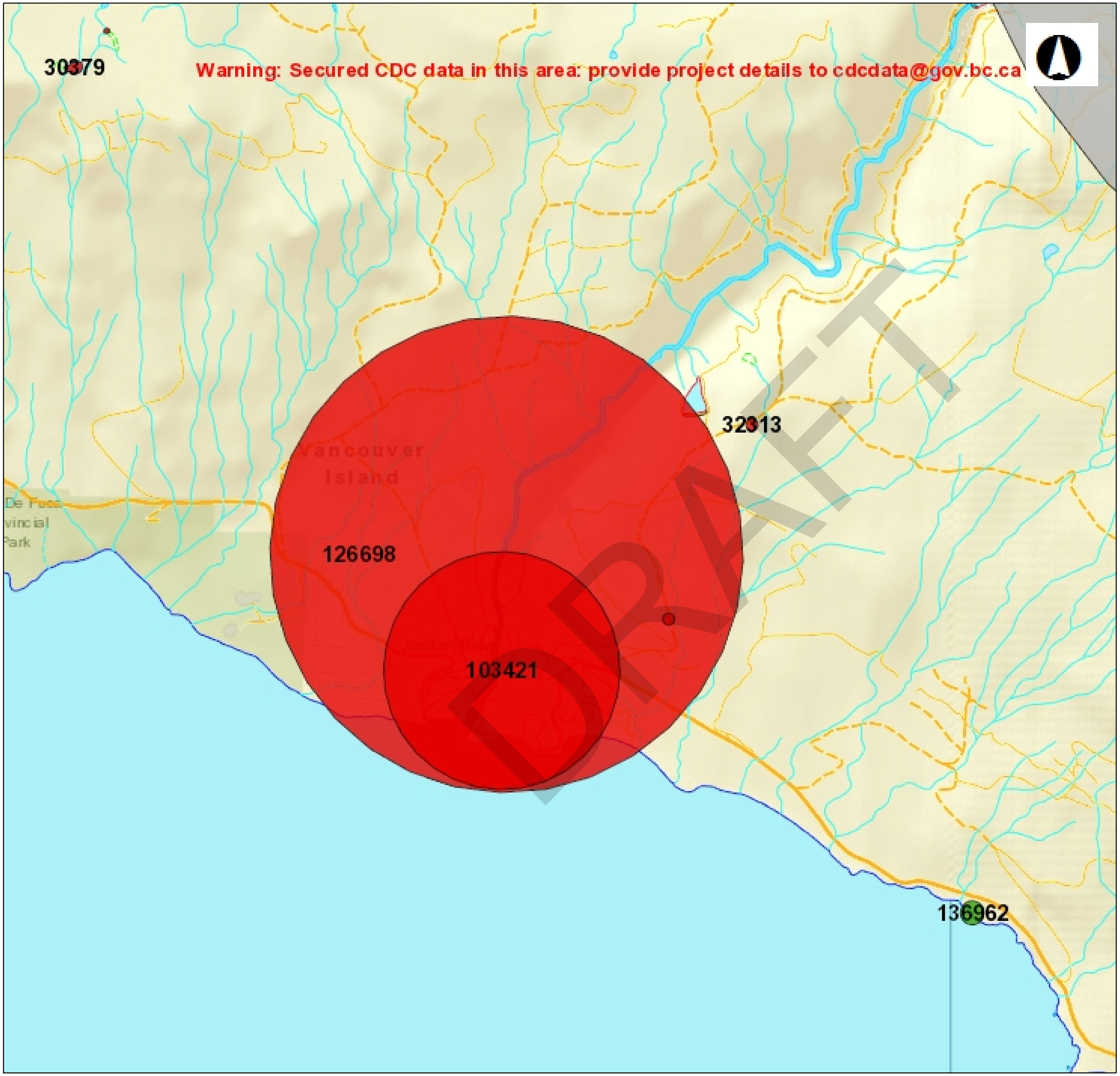
APPENDIX A

Habitat Assessment and Species Search Results



BRITISH COLUMBIA

CDC Occurrence Map

Legend

Species and Ecosystems at
Available Occurrences - CD

FEATURE_CODE

- Animal - Vertebrate
- Animal - Invertebrate
- Plant - Vascular
- Plant - Non-vascular
- Ecological Community

Species and Ecosystems at
Secured) Publicly Available

Species and Ecosystems at
and Historical) Publicly Avail
CDC

FEATURE_CODE

- Animal - Vertebrate

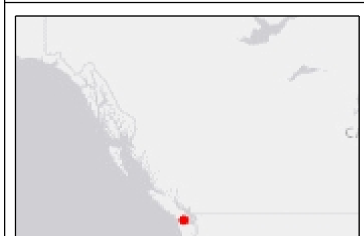
0 1.47 2.94 km

1: 72,224

Datum: NAD83

Projection: WGS_1984_Web_Mercator_Auxiliary_Sphere

Key Map of British Columbia



VEGETATION AND WILDLIFE SURVEYS AND IMPACT ASSESSMENT

FOR SNC LAVALIN SUNRO MINE RECLAMATION

PREPARED FOR:
J.Croston and C.Pinnell
SNC LAVALIN
#3-520 LAKE STREET
NELSON BC V1L 4C6

CORVIDAE PROJECT #2021-082
SEPTEMBER 2021

CORVIDAE
ENVIRONMENTAL CONSULTING INC
6526 Water Street, Sooke, BC

SOLUTION ORIENTED. PROTECTION OF THE ENVIRONMENT. ABSOLUTE INTEGRITY. OPEN COMMUNICATION. RESPECT.

TABLE OF CONTENTS

LIST OF TABLES

LIST OF FIGURES

LIST OF PHOTOS

CAVEAT

This report has been prepared with the best information available at the time of writing, including, communications with the client and regulators, site visits, review of site plans and design drawings and other documentation relevant to the project. This assessment has been developed to assist the project in remaining in compliance with relevant environmental regulations, acts and laws pertaining to the project and to identify and mitigate the expected impacts of the project and reclamation activities directly related to the project.

1 INTRODUCTION

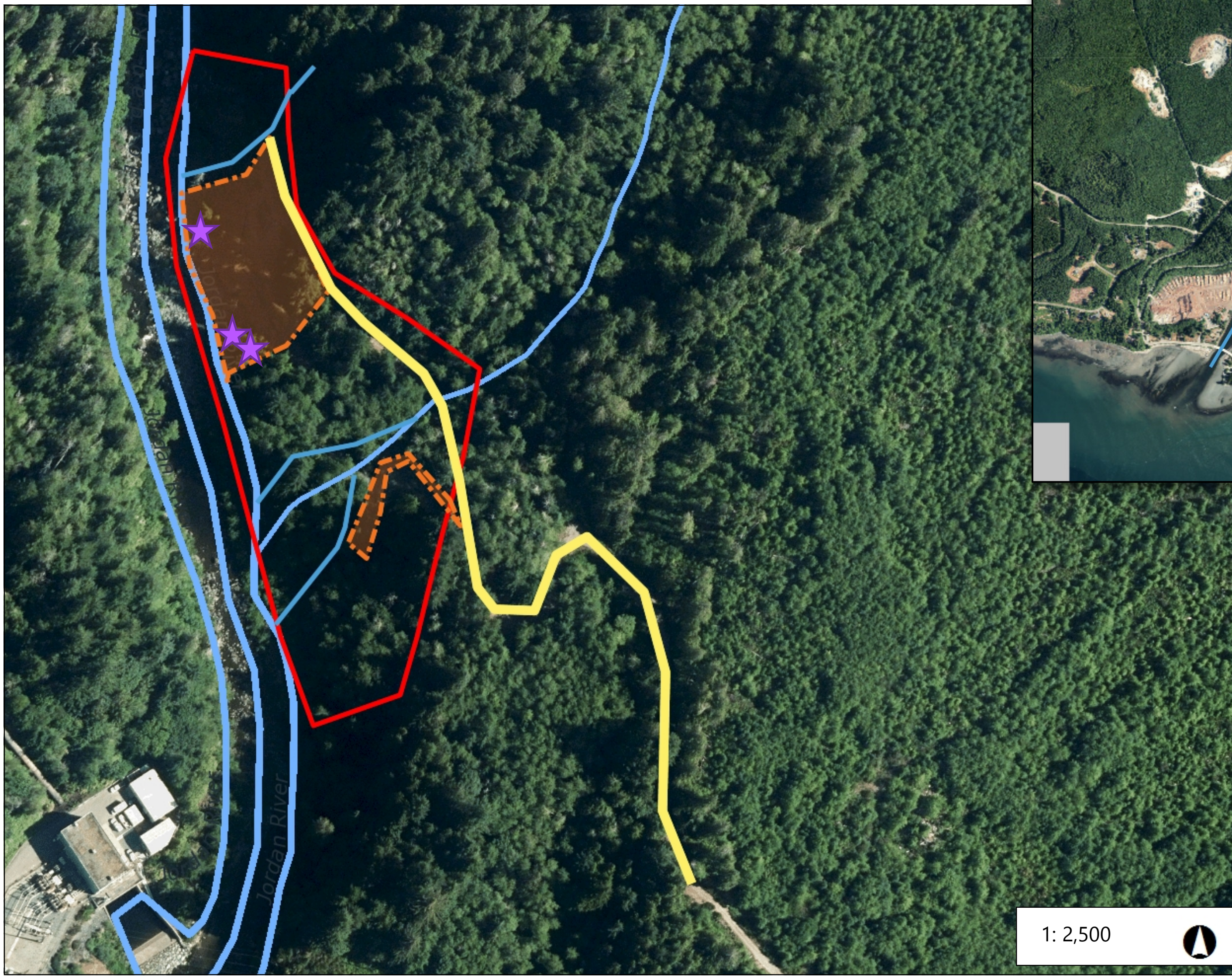
Corvidae Environmental Consulting Inc. (Corvidae) is pleased to provide this Wildlife and Vegetation Surveys and Impact Assessment for the proposed reclamation of the waste rock area at the Sunro Mine located along the east bank of the Jordan River (Latitude 48.427º N; Longitude 124.051º W). The mine is permanently closed. Currently planned activities tailings management and waste rock area reclamation.

The area to be reclaimed is approximately 2 hectares (ha) in size and is situated on along the bank of Jordan River (Figure 1). Steep slopes formed of waste rock tailings form a large area in the north half of the area, and a smaller portion of the southern half of the area. The area is accessed via a forest service road from Highway 14.

SNC Lavalin is planning to reclaim the mine by removing or capping the piled tailings, decommissioning the roads and planting native vegetation. This document provides an inventory and assessment of the wildlife and vegetation in the waste rock area, potential impacts of the proposed reclamation on vegetation and wildlife, and recommendations on the protection of environmentally sensitive features and methods to minimize impacts on the vegetation and wildlife.

Making a difference...together

Figure 1. Waste rock area at the Sunro Mine in Jordan River, BC



Legend

- Waste Rock Area
- Access road (FSR)
- Target areas for waste rock reclamation
- Watercourses
- Standing dead trees to retain

127.0 0 63.5 127.0 Meters

NAD_1983_UTM_Zone_10N

Important: This map is for general information purposes only. The Capital Regional District (CRD) makes no representations or warranties regarding the accuracy or completeness of this map or the suitability of the map for any purpose. This map is not for navigation The CRD will not be liable for any damage, loss or injury resulting from the use of the map or information on the map and the map may be changed by the CRD at any time.

2 SCOPE OF WORK

Corvidae completed a survey of the waste rock area, documenting the ecological features in the area including

- Areas of sensitivity, habitat and biodiversity values;
- Plant communities and plant species on site;
- Potential wildlife presence and wildlife habitat.

Background information was reviewed, including applicable databases. Mitigations to minimize the impacts of the reclamation activities on the environment have been provided in Section 6.

3 METHODS

3.1 DESKTOP REVIEW

Baseline biophysical conditions were compiled by reviewing the best available data and information including existing reports for the area and conducting searches of online provincial and federal databases:

- BC Conservation Data Centre (BC CDC 2021a and 2021b);
- BC HabitatWizard (Province of BC 2021);
- Aerial photographs of the property (Google Earth 2021) and
- Capital Region District (CRD) mapping and database (CRD 2021)

3.2 FIELD ASSESSMENT

A field assessment of the property was completed by Qualified Environmental Professionals (QEPs) from Corvidae. The assessment included characterization of vegetation and habitat types, wildlife sign and species observations, and wildlife habitat features.

4 ENVIRONMENTAL SITE ASSESSMENT

Corvidae QEP's completed a site visit on July 19, 2021. Appendix A shows photos of the waste rock area.

4.1 CLIMATE AND BIOGEOCLIMATIC ZONE

The project is located within the Coastal Western Hemlock (CWH) biogeoclimatic zone, and specifically in the Submontane variant of the Very Wet Maritime subzone (classified as CWHvm1; BC CDC 2021a). The CWHvm1 occurs from sea-level to 650m of elevation on Vancouver Island. The CWHvm1 has a wet, humid climate with cool summers and mild winters featuring relatively little snow. Growing seasons are long (Green and Klinka 1994).

4.2 TERRAIN AND SOILS

Podzolic soils dominate most of the coastal region, the interior wet belt, and the mountain systems of British Columbia. Podzolic soils generally form under coniferous forest in temperate, wet or cold, moist climates. Podzols are typically well drained, coarse textured, and undergo intense leaching of clay, organic matter, iron, and aluminum from upper to lower mineral horizons (Meidinger and Pojar 1991). This soil is present in the areas adjacent to the disturbed waste rock area.

The waste rock area is mostly barren with steep slopes composed of mine tailings. With no vegetation growth in these areas the slopes are prone to erosion and pose a risk to the Jordan river that flows along the base of the slopes.

4.3 VEGETATION

The CWHvm1 is typically dominated by western hemlock, amabilis fir, and lesser amounts of western red cedar. The understorey generally features a well-developed shrub layer dominated by red huckleberry and Alaskan blueberry, and a well-developed moss layer dominated by *Hylocomium splendens* and *Rhytidiadelphus loreus*. Herbs are sparse and include minor amounts of deer fern, five-leaved bramble, bunchberry, and queen's cup (Green and Klinka 1994). This is reflected in the intact forested ecosystems adjacent to the waste rock area.

The property has been disturbed from mining activities, tailings placement and road construction. The waste rock area consists of compacted rock and is therefore a poor growth medium for vegetation. Some coniferous trees and small shrubs are present but appear to be stunted and in poor condition.

There is a riparian corridor of mature second growth forest remaining between the rock waste areas where an unnamed tributary to the Jordan River crosses the waste rock area. The riparian area is dominated by western hemlock and western redcedar with an understory of salmonberry, sword fern and red huckleberry. The stream boundary and stream contain wetland vegetation including skunk cabbage, sedges, and rushes.

A second unnamed tributary of Jordan river at the north end of the waste rock area. The stream has been directed into a rocky spillway to the Jordan River. There is mature riparian vegetation located along the stream and spillway banks including salmonberry and sword fern, however, this stream is steep, and the channel/spillway is not vegetated.

Table 1 provides a list of the vegetation species that were found in the waste rock area during the site visit. Five invasive species were observed: bull thistle, Himalayan blackberry, common tansy, and scotch broom, and tansy ragwort. Measures to removed and prevent invasive species are discussed in Section 5 of this report.

Table 1. Plant species observed on site during field visit in July 2021

Common Name	Scientific Name	BC Provincial Status[1]	*SARA* Schedule 1 Status[2]
Alsike clover	*Trifolium hybridum*	Exotic	--
Bigleaf maple	*Acer macrophyllum*	Yellow	--
Bracken fern	*Pteridium aquilinum*	Yellow	--
Bull thistle	*Cirsium vulgare*	**Invasive**; Exotic	--
Cleavers	*Galium aparine*	Yellow	--

Common Name	Scientific Name	BC Provincial Status[1]	*SARA* Schedule 1 Status[2]
Common foxglove	*Digitalis purpurea*	Exotic	--
Common horsetail	*Equisetum arvense*	Yellow	--
Common rush	*Juncus hesperius*	Yellow	--
Common snowberry	*Symphoricarpos albus*	Yellow	--
Common tansy	*Tanacetum vulgare*	**Invasive**; Exotic	--
Common velvet grass	*Holcus lanatus*	Exotic	--
Creeping buttercup	*Ranunculus repens*	Exotic	--
Douglas-fir	*Pseudotsuga menziesii*	Yellow	--
Douglas' sagewort	*Artemisia douglasiana*	Unknown	--
Dwarf red raspberry	*Rubus pubescens*	Yellow	--
Fireweed	*Chamaenerion angustifolium*	Yellow	--
Grand fir	*Abies grandis*	Yellow	--
Himalayan blackberry	*Rubus armeniacus*	**Invasive**; Exotic	--
Lady fern	*Athyrium filix-femina var. cyclosorum*	Yellow	--
Oceanspray	*Holodiscus discolor var. discolor*	Yellow	--
Osoberry	*Oemleria cerasiformis*	Yellow	--
Purple cudweed	*Gamochaeta ustulata*	Unknown	--
Purple-leaved willowherb	*Epilobium ciliatum*	Yellow	--
Red alder	*Alnus rubra*	Yellow	--
Red elderberry	*Sambucus racemosa*	Yellow	--
Red-flowering currant	*Ribes sanguineum var. sanguineum*	Yellow	--
Red huckleberry	*Vaccinium parvifolium*	Yellow	--
Red raspberry	*Rubus idaeus ssp. strigosus*	Yellow	--
Salal	*Gaultheria shallon*	Yellow	--
Salmonberry	*Rubus spectabilis*	Yellow	--
Scotch broom	*Cytisus scoparius*	**Invasive**; Exotic	--
Sedge sp.	*Carex sp.*	--	--
Siberian miner's lettuce	*Claytonia sibirica*	Yellow	--
Skunk cabbage	*Lysichiton americanus*	Yellow	--
Slender cudweed	*Pseudognaphalium thermale*	Yellow	--
Stinging nettle	*Urtica dioica*	Yellow	--
Sword fern	*Polystichum munitum*	Yellow	--
Trailing blackberry	*Rubus ursinus*	Yellow	--
Wall lettuce	*Mycelis muralis*	Exotic	--
Western hemlock	*Tsuga heterophylla*	Yellow	--
Western redcedar	*Thuja plicata*	Yellow	--
Wild carrot	*Daucus carota*	Exotic	--

[1] BC CDC 2021b

[2] Government of Canada 2021

4.4 WILDLIFE

The Coastal Western Hemlock biogeoclimatic zone is home to many wildlife species. Black-tailed deer, black bear, marten and gray wolf are the most common large mammals in this zone on Vancouver Island. Great horned owl, barred owl, ruffed grouse, northern flicker, hairy woodpecker, common raven, Steller's jay, chestnut-backed chickadee, red-breasted nuthatch, varied thrush, red-tailed hawk, and Townsend's warbler are commonly occurring bird species. Amphibians that may occur in this biogeoclimatic zone include western toad, Pacific treefrog, western redbacked salamander (Pojar et al. 1991).

The barren rocky parts of the waste rock area have no notable wildlife habitat due to the steep slope and lack of structure and vegetation. There are several large western redcedar trees at the base of the slope that are standing dead. These wildlife trees may provide nesting habitat for birds and daytime roosting habitat for bats (Figure 1). The forested riparian areas have suitable habitat for birds, mammals, reptiles, and amphibians. Table 2 provides a list of the species and wildlife sign observed during the site visit at the waste rock area.

Table 2. Wildlife species observed on site during field visit in July 2021

Common Name	Scientific Name	BC Provincial Status[1]	*SARA* Schedule 1 Status[2]
MAMMALS			
Black bear (scat)	*Ursus americanus*	Yellow	--
Mule Deer (scat)	*Odocoileus hemionus*	Yellow	--
BIRDS			
American robin	*Turdus migratorius*	Yellow	--
Chipping sparrow	*Spizella passerina*	Yellow	--
Dark-eyed junco	*Junco hyemalis*	Yellow	--
Red-tailed hawk	*Buteo jamaicensis*	Yellow	--

[1] BC CDC 2021a

[2] Government of Canada 2021

4.5 SPECIES AT RISK

No species at risk were observed during the wildlife and vegetation survey. A query of the BC CDC iMap tool yielded occurrences of northern red-legged frog (Rana aurora) approximately 1.4 km to the east of the waste rock area (BC CDC 2021b). There is suitable habitat for this species in the riparian corridor that bisects the waste rock area.

CRITICAL HABITAT

A 50 km grid square of critical habitat for *Myotis lucifugus* and *Myotis septentrionalis* (Little Brown Myotis and Northern Myotis) overlaps the waste rock area (Figure 2). Note, critical habitat mapping is done at a high level to indicate areas in which the biophysical attributes of critical habitat are known to or may occur. For example, the 50 km^2 polygon for bats contains a known occurrence of the species and therefore, it is assumed that additional populations may occur in suitable habitat.

As noted above in Section 4.4, potential bat day-time roosting habitat was observed (cedar snags). No permanent roosting features were observed (e.g., rock crevasses).

5 POTENTIAL ENVIRONMENTAL IMPACTS AND RECOMMENDED MITIGATION MEASURES

The potential impacts of the proposed development of the property on the environment are:

- Impacts on sensitive ecosystem areas such as riparian habitat,
- Spread of invasive plant species,
- Sensory disturbance to wildlife, and
- Erosion and sediment transport within and around the project area.

The residual environmental impacts of the activities on the property will be reduced by the implementation of the mitigation and restoration measures recommended below. The mitigation measures provided in this report are designed to protect the environment and to minimize the potential impacts of the restoration activities on vegetation and wildlife species in the waste rock area. These recommendations were developed based on professional experience and in accordance with Procedures for Mitigating Impacts on Environmental Values (Environmental Mitigation Procedures) (BC Ministry of Environment [MOE] 2014).

RIPARIAN ENVIRONMENT

The removal of vegetation in the riparian area may result in the loss of features, functions and conditions that are vital for maintaining bank stability and fish habitat conditions. Vegetation in the riparian area controls surface water run-off from the upland areas, preventing excessive silt and surface run-off pollution from entering the aquatic environment.

No clearing of vegetation or trees should occur within the remaining riparian forest habitat in the waste rock area. No reclamation is needed in these areas as mature native vegetation is well established.

REVEGETATION

Open or disturbed areas should be reclaimed and revegetated in order to prevent establishment of invasive plants not only in the disturbed area but also in adjacent plant communities. Once the waste rock area has been remediated and restored, the area should be replaced with native trees and shrubs and/or seeded with native seed mix (Table 3). Overall plant density should be approximately one plant per 1 to 2 m^2.

Table 3. Recommended native vegetation to plant in disturbed area

Common Name	Species
Salal	*Gaultheria shallon*
Salmonberry	*Rubus spectabilis*
Nootka rose	*Rosa nutkana*
Oceanspray	*Holodiscus discolor*
Sword fern	*Polystichum munitum*
Western redcedar	*Thuja plicata*
Bigleaf maple	*Acer macrophyllum*
Coastal Revegetation Mix by Pacific Premier or equivalent	

*shrubs and ferns should be at least 1 gallon size; trees should be 3 gallon minimum size

INVASIVE SPECIES

Invasive plants establish readily in disturbed areas as they have a wide ecological tolerance and grow and propagate quickly. Invasive weed control is difficult for established populations. Immediate eradication of new and existing infestations should be a high priority during the reclamation of the waste rock area. Species should be removed using the most appropriate methods, at the correct time of year, and plant material must be disposed of correctly (burned or bagged and disposed of properly in a landfill) to avoid re-establishment or spread (Table 4). Following removal, re-seed bare soil with desirable, competing vegetation. Chemical control is not recommended due to the sensitive aquatic ecosystems of the adjacent Jordan River.

Table 4. Removal and disposal methods for invasive species

Species	Removal Method	Removal Timing	Plant Disposal
Bull thistle	Regular cutting or pulling can help wear down plant reserves, reduce plant growth, and reduce populations, but is not likely to eradicate the species.	Cutting and pulling are best done before flowering to eliminate seed production.	If plants are cut prior to flowering, the plant material can be left on the site to decompose. If plants are cut post flowering, all plant parts, including flower heads, should be bagged and disposed of properly in a landfill.
Common tansy	Can be removed by hand-pulling. If possible, dig out the roots, paying careful attention not to damage nearby vegetation.	Cutting and pulling are best done before flowering to eliminate seed production.	Burned or bagged and disposed of properly in a landfill. Do not compost.
Himalayan blackberry	Can be removed by pulling or cutting the canes from the ground. If possible, dig out the roots, paying careful attention not to damage nearby vegetation.	Removal should occur in the spring and early summer before they produce berries as canes that are cut as the plant is producing flowers are least likely to re-sprout.	Burned or bagged and disposed of properly in a landfill. Do not compost.
Scotch broom	Avoid disturbing the soil which can stimulate dormant broom seeds to sprout. Small broom plants can be pulled easily from the ground by hand without disturbing the soil. Larger plants should be cut below the root crown using loppers or a pruning saw.	Scotch broom removal should occur mid-April through early June, before the flowers go to seed.	Bagged and disposed of properly in a landfill or burn on site. Do not ‘recycle’ garden debris or compost.

WILDLIFE AND WILDLIFE HABITAT

Reclamation activities accomplished with heavy machinery can cause displacement of wildlife, use of less suitable habitat, reduced foraging ability, increased energy expenditure and lower reproductive success.

The following measures should be taken to minimize impacts on wildlife and wildlife habitat in and adjacent to the waste rock area:

- If work occurs within the sensitive time period for breeding and nesting birds (mid-March to end of August; Government of Canada 2021b), and nesting bird activity is detected in or adjacent to the reclamation areas, then work should stop and a QEP must recommend the appropriate mitigation, such as protective buffers.
- Where suitable, retain habitat that provides shelter for wildlife, such as standing dead trees. In standing dead trees must be removed, then a QEP must observe the trees at several times during the 48 hours before removal to ensure that the tree is not being used by birds or bats.
- In the event that an amphibian or reptile is encountered during clearing or construction, the QEP will recommend the appropriate mitigation, such as avoidance or relocation. All salvage must be done by the QEP and with the appropriate wildlife permit.

EROSION AND SEDIMENT CONTROL

The primary focus of erosion and sediment control planning is erosion control; if there is no erosion then there is no sediment. Erosion control is far more cost effective to implement and manage than sediment control.

Erosion controls, including the recommendations listed below, are recommended to be maintained for the duration of the project.

- Install a sediment barrier at the streamside edges of the reclamation area to prevent sediment laden runoff from entering the Jordan River or its tributaries.
- Store materials and soils in dry, flat areas at least 15m outside the edge of the SPEA.
- Heed weather advisories and scheduling work to avoid wet and rainy periods that may result in high surface water flow volumes and/ or increase erosion and sedimentation.
- Regularly monitor the aquatic environment for signs of sedimentation during all phases of the work, undertaking or activity and taking corrective action if required.

6 CONCLUSION

This report provides the results of the wildlife and vegetation surveys of the waste rock area and an assessment of the potential effects of the reclamation activities and recommended mitigation measures to minimize the impacts of the activities on wildlife and vegetation in the adjacent environments.

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APPENDIX A – SITE PHOTOGRAPHS

Photo 1. Access road into waste rock area. July 2021.

Photo 2. Steep waste rock slope with old equipment and old silt fencing. July 2021.

Photo 3. Steep waste rock slope with old equipment and old silt fencing. July 2021.

Photo 4. Steep waste rock slope with the Jordan River at the base. July 2021.

Photo 5. Steep waste rock slope with the Jordan River at the base. July 2021.

Photo 6. Cedar snags in the waste rock area. July 2021.

Photo 7. Stunted, yellowed trees growing in the waste rock area. July 2021.

Photo 8. Smaller south waste area with mature riparian forest adjacent. July 2021.

Photo 9. Tributary 1 at the north end of the waste rock area. July 2021.

Photo 10. Tributary 2 to Jordan River where culverted under the FSR. July 2021.

APPENDIX B

ProUCL Output

General Statistics on Uncensored Data

Date/Time of Computation	ProUCL 5.2 2023-10-13 2:43:48 PM
User Selected Options	
From File	WorkSheet.xls
Full Precision	OFF

From File: WorkSheet.xls

General Statistics for Censored Data Set (with NDs) using Kaplan Meier Method

Variable	NumObs	# Missing	Num Ds	NumNDs	% NDs	Min ND	Max ND	KM Mean	KM Var	KM SD	KM CV
Aluminum	13	1	13	0	0.00%	N/A	N/A	16105	19043877	4364	0.271
Antimony	14	0	13	1	7.14%	0.1	0.1	0.314	0.0403	0.201	0.639
Arsenic	14	0	14	0	0.00%	N/A	N/A	2.476	1.336	1.156	0.467
Barium	14	0	14	0	0.00%	N/A	N/A	39.71	430	20.74	0.522
Beryllium	14	0	10	4	28.57%	0.1	0.1	0.182	0.0054	0.0735	0.404
Boron	14	0	4	10	71.43%	2	2	2.2	0.301	0.549	0.25
Cadmium	14	0	14	0	0.00%	N/A	N/A	0.0994	0.00109	0.0331	0.333
Chromium	14	0	14	0	0.00%	N/A	N/A	34.06	69.06	8.31	0.244
Cobalt	14	0	14	0	0.00%	N/A	N/A	14.45	20.79	4.56	0.316
Copper	14	0	14	0	0.00%	N/A	N/A	1189	1139838	1068	0.898
Iron	13	1	13	0	0.00%	N/A	N/A	29454	89031026	9436	0.32
Lead	14	0	14	0	0.00%	N/A	N/A	6.974	27.48	5.242	0.752
Lithium	14	0	14	0	0.00%	N/A	N/A	8.59	15.94	3.992	0.465
Manganese	14	0	14	0	0.00%	N/A	N/A	312.6	8169	90.38	0.289
Mercury	14	0	5	9	64.29%	0.04	0.04	0.0601	9.9592E-4	0.0316	0.525
Molybdenum	14	0	14	0	0.00%	N/A	N/A	1.504	6.203	2.491	1.656
Nickel	14	0	14	0	0.00%	N/A	N/A	24.56	27.57	5.251	0.214
Selenium	14	0	9	5	35.71%	0.2	0.2	1.739	3.515	1.875	1.078
Silver	14	0	11	3	21.43%	0.1	0.1	0.777	0.656	0.81	1.042
Strontium	14	0	14	0	0.00%	N/A	N/A	17.86	39.77	6.306	0.353
Thallium	14	0	0	14	100.00%	0.1	0.1	N/A	N/A	N/A	N/A
Tin	14	0	14	0	0.00%	N/A	N/A	0.784	0.428	0.654	0.834
Tungsten	13	1	10	3	23.08%	0.2	0.2	0.514	0.159	0.399	0.777
Uranium	14	0	12	2	14.29%	0.05	0.1	0.184	0.0108	0.104	0.564
Vanadium	14	0	14	0	0.00%	N/A	N/A	62.52	227.6	15.08	0.241
Zinc	14	0	14	0	0.00%	N/A	N/A	42.3	41.59	6.449	0.152

General Statistics for Raw Data Sets using Detected Data Only

Variable	NumObs	# Missing	Minimum	Maximum	Mean	Median	Var	SD	MAD/0.675	Skewness	CV
Aluminum	13	1	6550	22100	16105	16800	19043877	4364	2372	-0.896	0.271
Antimony	13	0	0.11	0.8	0.331	0.26	0.0429	0.207	0.193	1.031	0.626
Arsenic	14	0	0.73	4.87	2.476	2.245	1.336	1.156	1.105	0.548	0.467
Barium	14	0	7.8	80.2	39.71	40.3	430	20.74	20.61	0.226	0.522
Beryllium	10	0	0.11	0.31	0.215	0.21	0.00421	0.0649	0.0667	0.0489	0.302
Boron	4	0	2	4.1	2.7	2.35	0.94	0.97	0.445	1.598	0.359
Cadmium	14	0	0.066	0.185	0.0994	0.0895	0.00109	0.0331	0.0267	1.479	0.333
Chromium	14	0	13.7	45.2	34.06	34.35	69.06	8.31	7.265	-1.068	0.244
Cobalt	14	0	8.7	21.7	14.45	12.8	20.79	4.56	4.522	0.458	0.316
Copper	14	0	52.9	4020	1189	856	1139838	1068	937	1.479	0.898
Iron	13	1	11500	46300	29454	28900	89031026	9436	4744	0.031	0.32
Lead	14	0	1.2	17.9	6.974	4.805	27.48	5.242	5.248	0.832	0.752

	A	B	C	D	E	F	G	H	I	J	K	L	M
54		Lithium	14	0	2.67	16.5	8.59	9.615	15.94	3.992	4.388	0.19	0.465
55		Manganese	14	0	136	489	312.6	295.5	8169	90.38	92.66	0.0736	0.289
56		Mercury	5	0	0.059	0.132	0.0962	0.09	9.4770E-4	0.0308	0.046	0.0905	0.32
57		Molybdenum	14	0	0.22	9.87	1.504	0.73	6.203	2.491	0.526	3.344	1.656
58		Nickel	14	0	15.7	32.4	24.56	24.55	27.57	5.251	6.301	0.125	0.214
59		Selenium	9	0	0.5	5.73	2.593	1.9	3.849	1.962	1.957	0.592	0.757
60		Silver	11	0	0.11	2.45	0.962	0.8	0.743	0.862	0.875	0.746	0.896
61		Strontium	14	0	10.4	34	17.86	18.05	39.77	6.306	5.263	1.223	0.353
62		Thallium	0	0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
63		Tin	14	0	0.31	2.6	0.784	0.51	0.428	0.654	0.245	2.047	0.834
64		Tungsten	10	1	0.22	1.68	0.608	0.485	0.187	0.433	0.319	1.881	0.712
65		Uranium	12	0	0.052	0.348	0.206	0.197	0.0102	0.101	0.108	-0.0412	0.49
66		Vanadium	14	0	25.2	83.2	62.52	65.35	227.6	15.08	10.23	-1.342	0.241
67		Zinc	14	0	32.9	54.4	42.3	42.5	41.59	6.449	6.153	0.159	0.152
68													
69						Percentiles using all Detects (Ds) and Non-Detects (NDs)							
70													
71		Variable	NumObs	# Missing	10%ile	20%ile	25%ile(Q1)	50%ile(Q2)	75%ile(Q3)	80%ile	90%ile	95%ile	99%ile
72		Aluminum	13	1	10336	13800	15300	16800	18400	18880	20960	21680	22016
73		Antimony	14	0	0.113	0.132	0.15	0.25	0.458	0.484	0.553	0.657	0.771
74		Arsenic	14	0	1.365	1.52	1.65	2.245	3.398	3.61	3.705	4.123	4.721
75		Barium	14	0	14.84	19.36	24.23	40.3	49.73	54	62.75	69.54	78.07
76		Beryllium	14	0	0.1	0.1	0.103	0.17	0.243	0.254	0.288	0.304	0.309
77		Boron	14	0	2	2	2	2	2	2.04	2.45	3.125	3.905
78		Cadmium	14	0	0.0709	0.0748	0.076	0.0895	0.113	0.114	0.134	0.157	0.179
79		Chromium	14	0	24.88	30.8	32.65	34.35	39.93	41.28	42.79	43.77	44.91
80		Cobalt	14	0	9.402	11.02	11.13	12.8	18.38	19.54	20.97	21.44	21.65
81		Copper	14	0	192	381.4	456.3	856	1833	1890	2065	2798	3776
82		Iron	13	1	18380	24860	25700	28900	31600	33520	42400	45100	46060
83		Lead	14	0	1.634	2.782	3.095	4.805	9.915	10.79	13.96	15.89	17.5
84		Lithium	14	0	3.668	4.896	5.55	9.615	11.15	11.28	12.38	14.1	16.02
85		Manganese	14	0	241.7	256.8	258	295.5	372.8	378	403.4	439.6	479.1
86		Mercury	14	0	0.04	0.04	0.04	0.04	0.0725	0.0822	0.113	0.126	0.131
87		Molybdenum	14	0	0.283	0.378	0.39	0.73	1.385	1.558	2.192	5.028	8.901
88		Nickel	14	0	18.94	20.32	20.63	24.55	28.63	30.3	31.78	32.08	32.34
89		Selenium	14	0	0.2	0.2	0.2	0.835	2.335	3.328	4.74	5.126	5.609
90		Silver	14	0	0.1	0.106	0.11	0.355	1.19	1.448	2.113	2.346	2.429
91		Strontium	14	0	11.15	12.04	12.8	18.05	19.2	20.02	23.97	28.28	32.86
92		Thallium	14	0	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
93		Tin	14	0	0.335	0.394	0.418	0.51	0.73	0.98	1.571	1.996	2.479
94		Tungsten	13	1	0.2	0.208	0.22	0.38	0.69	0.732	0.816	1.17	1.578
95		Uranium	14	0	0.0595	0.0908	0.108	0.149	0.271	0.286	0.322	0.335	0.345
96		Vanadium	14	0	43.85	57.74	59.35	65.35	72.63	73.4	74.63	77.81	82.12
97		Zinc	14	0	33.56	36.56	38.18	42.5	45.6	46.68	50.1	52.19	53.96

UCL Statistics for Data Sets with Non-Detects

User Selected Options	
Date/Time of Computation	ProUCL 5.2 2023-10-13 2:44:18 PM
From File	WorkSheet.xls
Full Precision	OFF
Confidence Coefficient	95%
Number of Bootstrap Operations	2000

Aluminum

General Statistics

Total Number of Observations	13	Number of Distinct Observations	12
		Number of Missing Observations	1
Minimum	6550	Mean	16105
Maximum	22100	Median	16800
SD	4364	Std. Error of Mean	1210
Coefficient of Variation	0.271	Skewness	-0.896

Normal GOF Test

Shapiro Wilk Test Statistic	0.934	**Shapiro Wilk GOF Test**
1% Shapiro Wilk Critical Value	0.814	Data appear Normal at 1% Significance Level
Lilliefors Test Statistic	0.196	**Lilliefors GOF Test**
1% Lilliefors Critical Value	0.271	Data appear Normal at 1% Significance Level

Data appear Normal at 1% Significance Level

Assuming Normal Distribution

95% Normal UCL		**95% UCLs (Adjusted for Skewness)**	
95% Student's-t UCL	18263	95% Adjusted-CLT UCL (Chen-1995)	17775
		95% Modified-t UCL (Johnson-1978)	18212

Gamma GOF Test

A-D Test Statistic	0.677	**Anderson-Darling Gamma GOF Test**
5% A-D Critical Value	0.734	Detected data appear Gamma Distributed at 5% Significance Level
K-S Test Statistic	0.24	**Kolmogorov-Smirnov Gamma GOF Test**
5% K-S Critical Value	0.237	Data Not Gamma Distributed at 5% Significance Level

Detected data follow Appr. Gamma Distribution at 5% Significance Level

Gamma Statistics

k hat (MLE)	11.42	k star (bias corrected MLE)	8.832
Theta hat (MLE)	1411	Theta star (bias corrected MLE)	1824
nu hat (MLE)	296.8	nu star (bias corrected)	229.6
MLE Mean (bias corrected)	16105	MLE Sd (bias corrected)	5419
		Approximate Chi Square Value (0.05)	195.6
Adjusted Level of Significance	0.0301	Adjusted Chi Square Value	191.1

Assuming Gamma Distribution

95% Approximate Gamma UCL	18912	95% Adjusted Gamma UCL	19355

Lognormal GOF Test

Shapiro Wilk Test Statistic	0.839	**Shapiro Wilk Lognormal GOF Test**	
10% Shapiro Wilk Critical Value	0.889	Data Not Lognormal at 10% Significance Level	
Lilliefors Test Statistic	0.261	**Lilliefors Lognormal GOF Test**	
10% Lilliefors Critical Value	0.215	Data Not Lognormal at 10% Significance Level	

Data Not Lognormal at 10% Significance Level

Lognormal Statistics

Minimum of Logged Data	8.787	Mean of logged Data	9.642
Maximum of Logged Data	10	SD of logged Data	0.335

Assuming Lognormal Distribution

95% H-UCL	19654	90% Chebyshev (MVUE) UCL	20795
95% Chebyshev (MVUE) UCL	22867	97.5% Chebyshev (MVUE) UCL	25744
99% Chebyshev (MVUE) UCL	31393		

Nonparametric Distribution Free UCL Statistics

Data appear to follow a Discernible Distribution

Nonparametric Distribution Free UCLs

95% CLT UCL	18096	95% BCA Bootstrap UCL	17692
95% Standard Bootstrap UCL	17996	95% Bootstrap-t UCL	17950
95% Hall's Bootstrap UCL	17824	95% Percentile Bootstrap UCL	17862
90% Chebyshev(Mean, Sd) UCL	19736	95% Chebyshev(Mean, Sd) UCL	21381
97.5% Chebyshev(Mean, Sd) UCL	23664	99% Chebyshev(Mean, Sd) UCL	28148

Suggested UCL to Use

95% Student's-t UCL	18263		

Note: Suggestions regarding the selection of a 95% UCL are provided to help the user to select the most appropriate 95% UCL.
Recommendations are based upon data size, data distribution, and skewness using results from simulation studies.
However, simulations results will not cover all Real World data sets; for additional insight the user may want to consult a statistician.

Note: For highly negatively-skewed data, confidence limits (e.g., Chen, Johnson, Lognormal, and Gamma) may not be reliable. Chen's and Johnson's methods provide adjustments for positvely skewed data sets.

Antimony

General Statistics

Total Number of Observations	14	Number of Distinct Observations	14
Number of Detects	13	Number of Non-Detects	1
Number of Distinct Detects	13	Number of Distinct Non-Detects	1
Minimum Detect	0.11	Minimum Non-Detect	0.1
Maximum Detect	0.8	Maximum Non-Detect	0.1
Variance Detects	0.0429	Percent Non-Detects	7.143%
Mean Detects	0.331	SD Detects	0.207
Median Detects	0.26	CV Detects	0.626
Skewness Detects	1.031	Kurtosis Detects	0.572
Mean of Logged Detects	-1.285	SD of Logged Detects	0.628

	A	B	C	D	E	F	G	H	I	J	K	L
101	**Normal GOF Test on Detects Only**											
102					Shapiro Wilk Test Statistic	0.902	**Shapiro Wilk GOF Test**					
103					1% Shapiro Wilk Critical Value	0.814	Detected Data appear Normal at 1% Significance Level					
104					Lilliefors Test Statistic	0.174	**Lilliefors GOF Test**					
105					1% Lilliefors Critical Value	0.271	Detected Data appear Normal at 1% Significance Level					
106	**Detected Data appear Normal at 1% Significance Level**											
107												
108	**Kaplan-Meier (KM) Statistics using Normal Critical Values and other Nonparametric UCLs**											
109					KM Mean	0.314					KM Standard Error of Mean	0.0558
110					90KM SD	0.201					95% KM (BCA) UCL	0.411
111					95% KM (t) UCL	0.413					95% KM (Percentile Bootstrap) UCL	0.408
112					95% KM (z) UCL	0.406					95% KM Bootstrap t UCL	0.441
113					90% KM Chebyshev UCL	0.482					95% KM Chebyshev UCL	0.558
114					97.5% KM Chebyshev UCL	0.663					99% KM Chebyshev UCL	0.87
115												
116	**Gamma GOF Tests on Detected Observations Only**											
117					A-D Test Statistic	0.238	**Anderson-Darling GOF Test**					
118					5% A-D Critical Value	0.739	Detected data appear Gamma Distributed at 5% Significance Level					
119					K-S Test Statistic	0.117	**Kolmogorov-Smirnov GOF**					
120					5% K-S Critical Value	0.238	Detected data appear Gamma Distributed at 5% Significance Level					
121	**Detected data appear Gamma Distributed at 5% Significance Level**											
122												
123	**Gamma Statistics on Detected Data Only**											
124					k hat (MLE)	2.956					k star (bias corrected MLE)	2.325
125					Theta hat (MLE)	0.112					Theta star (bias corrected MLE)	0.142
126					nu hat (MLE)	76.85					nu star (bias corrected)	60.45
127					Mean (detects)	0.331						
128												
129	**Gamma ROS Statistics using Imputed Non-Detects**											
130	GROS may not be used when data set has > 50% NDs with many tied observations at multiple DLs											
131	GROS may not be used when kstar of detects is small such as <1.0, especially when the sample size is small (e.g., <15-20)											
132	For such situations, GROS method may yield incorrect values of UCLs and BTVs											
133	This is especially true when the sample size is small.											
134	For gamma distributed detected data, BTVs and UCLs may be computed using gamma distribution on KM estimates											
135					Minimum	0.01					Mean	0.308
136					Maximum	0.8					Median	0.25
137					SD	0.217					CV	0.704
138					k hat (MLE)	1.6					k star (bias corrected MLE)	1.305
139					Theta hat (MLE)	0.192					Theta star (bias corrected MLE)	0.236
140					nu hat (MLE)	44.8					nu star (bias corrected)	36.53
141					Adjusted Level of Significance (β)	0.0312						
142					Approximate Chi Square Value (36.53, α)	23.7					Adjusted Chi Square Value (36.53, β)	22.33
143					95% Gamma Approximate UCL	0.475					95% Gamma Adjusted UCL	0.504
144												
145	**Estimates of Gamma Parameters using KM Estimates**											
146					Mean (KM)	0.314					SD (KM)	0.201
147					Variance (KM)	0.0403					SE of Mean (KM)	0.0558
148					k hat (KM)	2.452					k star (KM)	1.974
149					nu hat (KM)	68.66					nu star (KM)	55.28
150					theta hat (KM)	0.128					theta star (KM)	0.159

80% gamma percentile (KM)	0.471	90% gamma percentile (KM)	0.613
95% gamma percentile (KM)	0.749	99% gamma percentile (KM)	1.049

Gamma Kaplan-Meier (KM) Statistics

Approximate Chi Square Value (55.28, α)	39.19	Adjusted Chi Square Value (55.28, β)	37.4
95% KM Approximate Gamma UCL	0.443	95% KM Adjusted Gamma UCL	0.465

Lognormal GOF Test on Detected Observations Only

Shapiro Wilk Test Statistic	0.966	**Shapiro Wilk GOF Test**
10% Shapiro Wilk Critical Value	0.889	Detected Data appear Lognormal at 10% Significance Level
Lilliefors Test Statistic	0.118	**Lilliefors GOF Test**
10% Lilliefors Critical Value	0.215	Detected Data appear Lognormal at 10% Significance Level

Detected Data appear Lognormal at 10% Significance Level

Lognormal ROS Statistics Using Imputed Non-Detects

Mean in Original Scale	0.311	Mean in Log Scale	-1.396
SD in Original Scale	0.212	SD in Log Scale	0.732
95% t UCL (assumes normality of ROS data)	0.412	95% Percentile Bootstrap UCL	0.404
95% BCA Bootstrap UCL	0.411	95% Bootstrap t UCL	0.442
95% H-UCL (Log ROS)	0.525		

Statistics using KM estimates on Logged Data and Assuming Lognormal Distribution

KM Mean (logged)	-1.358	KM Geo Mean	0.257
KM SD (logged)	0.637	95% Critical H Value (KM-Log)	2.263
KM Standard Error of Mean (logged)	0.177	95% H-UCL (KM -Log)	0.47
KM SD (logged)	0.637	95% Critical H Value (KM-Log)	2.263
KM Standard Error of Mean (logged)	0.177		

DL/2 Statistics

DL/2 Normal		**DL/2 Log-Transformed**	
Mean in Original Scale	0.311	Mean in Log Scale	-1.407
SD in Original Scale	0.213	SD in Log Scale	0.757
95% t UCL (Assumes normality)	0.411	95% H-Stat UCL	0.541

DL/2 is not a recommended method, provided for comparisons and historical reasons

Nonparametric Distribution Free UCL Statistics

Detected Data appear Normal Distributed at 1% Significance Level

Suggested UCL to Use

95% KM (t) UCL	0.413		

Note: Suggestions regarding the selection of a 95% UCL are provided to help the user to select the most appropriate 95% UCL.

Recommendations are based upon data size, data distribution, and skewness using results from simulation studies.

However, simulations results will not cover all Real World data sets; for additional insight the user may want to consult a statistician.

Arsenic

General Statistics

Total Number of Observations	14	Number of Distinct Observations	14

	A–E	F	G–K	L
201			Number of Missing Observations	0
202	Minimum	0.73	Mean	2.476
203	Maximum	4.87	Median	2.245
204	SD	1.156	Std. Error of Mean	0.309
205	Coefficient of Variation	0.467	Skewness	0.548
206				
207	**Normal GOF Test**			
208	Shapiro Wilk Test Statistic	0.957	**Shapiro Wilk GOF Test**	
209	1% Shapiro Wilk Critical Value	0.825	Data appear Normal at 1% Significance Level	
210	Lilliefors Test Statistic	0.16	**Lilliefors GOF Test**	
211	1% Lilliefors Critical Value	0.263	Data appear Normal at 1% Significance Level	
212	**Data appear Normal at 1% Significance Level**			
213				
214	**Assuming Normal Distribution**			
215	**95% Normal UCL**		**95% UCLs (Adjusted for Skewness)**	
216	95% Student's-t UCL	3.023	95% Adjusted-CLT UCL (Chen-1995)	3.033
217			95% Modified-t UCL (Johnson-1978)	3.031
218				
219	**Gamma GOF Test**			
220	A-D Test Statistic	0.196	**Anderson-Darling Gamma GOF Test**	
221	5% A-D Critical Value	0.739	Detected data appear Gamma Distributed at 5% Significance Level	
222	K-S Test Statistic	0.125	**Kolmogorov-Smirnov Gamma GOF Test**	
223	5% K-S Critical Value	0.229	Detected data appear Gamma Distributed at 5% Significance Level	
224	**Detected data appear Gamma Distributed at 5% Significance Level**			
225				
226	**Gamma Statistics**			
227	k hat (MLE)	4.672	k star (bias corrected MLE)	3.718
228	Theta hat (MLE)	0.53	Theta star (bias corrected MLE)	0.666
229	nu hat (MLE)	130.8	nu star (bias corrected)	104.1
230	MLE Mean (bias corrected)	2.476	MLE Sd (bias corrected)	1.284
231			Approximate Chi Square Value (0.05)	81.57
232	Adjusted Level of Significance	0.0312	Adjusted Chi Square Value	78.92
233				
234	**Assuming Gamma Distribution**			
235	95% Approximate Gamma UCL	3.161	95% Adjusted Gamma UCL	3.267
236				
237	**Lognormal GOF Test**			
238	Shapiro Wilk Test Statistic	0.967	**Shapiro Wilk Lognormal GOF Test**	
239	10% Shapiro Wilk Critical Value	0.895	Data appear Lognormal at 10% Significance Level	
240	Lilliefors Test Statistic	0.111	**Lilliefors Lognormal GOF Test**	
241	10% Lilliefors Critical Value	0.208	Data appear Lognormal at 10% Significance Level	
242	**Data appear Lognormal at 10% Significance Level**			
243				
244	**Lognormal Statistics**			
245	Minimum of Logged Data	-0.315	Mean of logged Data	0.796
246	Maximum of Logged Data	1.583	SD of logged Data	0.509
247				
248	**Assuming Lognormal Distribution**			
249	95% H-UCL	3.354	90% Chebyshev (MVUE) UCL	3.545
250	95% Chebyshev (MVUE) UCL	4.02	97.5% Chebyshev (MVUE) UCL	4.679

	A	B	C	D	E	F	G	H	I	J	K	L
251					99% Chebyshev (MVUE) UCL	5.973						
252												
253	**Nonparametric Distribution Free UCL Statistics**											
254	**Data appear to follow a Discernible Distribution**											
255												
256	**Nonparametric Distribution Free UCLs**											
257					95% CLT UCL	2.985					95% BCA Bootstrap UCL	3.047
258					95% Standard Bootstrap UCL	2.973					95% Bootstrap-t UCL	3.065
259					95% Hall's Bootstrap UCL	3.052					95% Percentile Bootstrap UCL	3.007
260					90% Chebyshev(Mean, Sd) UCL	3.403					95% Chebyshev(Mean, Sd) UCL	3.823
261					97.5% Chebyshev(Mean, Sd) UCL	4.406					99% Chebyshev(Mean, Sd) UCL	5.55
262												
263	**Suggested UCL to Use**											
264					95% Student's-t UCL	3.023						
265												
266	Note: Suggestions regarding the selection of a 95% UCL are provided to help the user to select the most appropriate 95% UCL.											
267	Recommendations are based upon data size, data distribution, and skewness using results from simulation studies.											
268	However, simulations results will not cover all Real World data sets; for additional insight the user may want to consult a statistician.											
269												
270												
271	Barium											
272												
273	**General Statistics**											
274					Total Number of Observations	14					Number of Distinct Observations	14
275											Number of Missing Observations	0
276					Minimum	7.8					Mean	39.71
277					Maximum	80.2					Median	40.3
278					SD	20.74					Std. Error of Mean	5.542
279					Coefficient of Variation	0.522					Skewness	0.226
280												
281	**Normal GOF Test**											
282					Shapiro Wilk Test Statistic	0.976	**Shapiro Wilk GOF Test**					
283					1% Shapiro Wilk Critical Value	0.825	Data appear Normal at 1% Significance Level					
284					Lilliefors Test Statistic	0.0989	**Lilliefors GOF Test**					
285					1% Lilliefors Critical Value	0.263	Data appear Normal at 1% Significance Level					
286	**Data appear Normal at 1% Significance Level**											
287												
288	**Assuming Normal Distribution**											
289	**95% Normal UCL**						**95% UCLs (Adjusted for Skewness)**					
290					95% Student's-t UCL	49.52					95% Adjusted-CLT UCL (Chen-1995)	49.18
291											95% Modified-t UCL (Johnson-1978)	49.58
292												
293	**Gamma GOF Test**											
294					A-D Test Statistic	0.299	**Anderson-Darling Gamma GOF Test**					
295					5% A-D Critical Value	0.742	Detected data appear Gamma Distributed at 5% Significance Level					
296					K-S Test Statistic	0.129	**Kolmogorov-Smirnov Gamma GOF Test**					
297					5% K-S Critical Value	0.23	Detected data appear Gamma Distributed at 5% Significance Level					
298	**Detected data appear Gamma Distributed at 5% Significance Level**											
299												
300	**Gamma Statistics**											

	A–E	F	G–K	L
301	k hat (MLE)	3.147	k star (bias corrected MLE)	2.52
302	Theta hat (MLE)	12.62	Theta star (bias corrected MLE)	15.75
303	nu hat (MLE)	88.12	nu star (bias corrected)	70.57
304	MLE Mean (bias corrected)	39.71	MLE Sd (bias corrected)	25.01
305			Approximate Chi Square Value (0.05)	52.23
306	Adjusted Level of Significance	0.0312	Adjusted Chi Square Value	50.14
307				
308	**Assuming Gamma Distribution**			
309	95% Approximate Gamma UCL	53.65	95% Adjusted Gamma UCL	55.89
310				
311	**Lognormal GOF Test**			
312	Shapiro Wilk Test Statistic	0.926	**Shapiro Wilk Lognormal GOF Test**	
313	10% Shapiro Wilk Critical Value	0.895	Data appear Lognormal at 10% Significance Level	
314	Lilliefors Test Statistic	0.164	**Lilliefors Lognormal GOF Test**	
315	10% Lilliefors Critical Value	0.208	Data appear Lognormal at 10% Significance Level	
316	**Data appear Lognormal at 10% Significance Level**			
317				
318	**Lognormal Statistics**			
319	Minimum of Logged Data	2.054	Mean of logged Data	3.514
320	Maximum of Logged Data	4.385	SD of logged Data	0.658
321				
322	**Assuming Lognormal Distribution**			
323	95% H-UCL	63.37	90% Chebyshev (MVUE) UCL	63.58
324	95% Chebyshev (MVUE) UCL	73.82	97.5% Chebyshev (MVUE) UCL	88.03
325	99% Chebyshev (MVUE) UCL	115.9		
326				
327	**Nonparametric Distribution Free UCL Statistics**			
328	**Data appear to follow a Discernible Distribution**			
329				
330	**Nonparametric Distribution Free UCLs**			
331	95% CLT UCL	48.82	95% BCA Bootstrap UCL	48.87
332	95% Standard Bootstrap UCL	48.51	95% Bootstrap-t UCL	49.86
333	95% Hall's Bootstrap UCL	49.55	95% Percentile Bootstrap UCL	48.5
334	90% Chebyshev(Mean, Sd) UCL	56.33	95% Chebyshev(Mean, Sd) UCL	63.86
335	97.5% Chebyshev(Mean, Sd) UCL	74.32	99% Chebyshev(Mean, Sd) UCL	94.85
336				
337	**Suggested UCL to Use**			
338	95% Student's-t UCL	49.52		
339				
340	Note: Suggestions regarding the selection of a 95% UCL are provided to help the user to select the most appropriate 95% UCL.			
341	Recommendations are based upon data size, data distribution, and skewness using results from simulation studies.			
342	However, simulations results will not cover all Real World data sets; for additional insight the user may want to consult a statistician.			
343				
344	**Beryllium**			
345				
346	**General Statistics**			
347	Total Number of Observations	14	Number of Distinct Observations	10
348	Number of Detects	10	Number of Non-Detects	4
349	Number of Distinct Detects	9	Number of Distinct Non-Detects	1
350	Minimum Detect	0.11	Minimum Non-Detect	0.1

351	Maximum Detect	0.31	Maximum Non-Detect	0.1
352	Variance Detects	0.00421	Percent Non-Detects	28.57%
353	Mean Detects	0.215	SD Detects	0.0649
354	Median Detects	0.21	CV Detects	0.302
355	Skewness Detects	0.0489	Kurtosis Detects	-0.906
356	Mean of Logged Detects	-1.582	SD of Logged Detects	0.323
357				
358	**Normal GOF Test on Detects Only**			
359	Shapiro Wilk Test Statistic	0.961	**Shapiro Wilk GOF Test**	
360	1% Shapiro Wilk Critical Value	0.781	Detected Data appear Normal at 1% Significance Level	
361	Lilliefors Test Statistic	0.156	**Lilliefors GOF Test**	
362	1% Lilliefors Critical Value	0.304	Detected Data appear Normal at 1% Significance Level	
363	**Detected Data appear Normal at 1% Significance Level**			
364				
365	**Kaplan-Meier (KM) Statistics using Normal Critical Values and other Nonparametric UCLs**			
366	KM Mean	0.182	KM Standard Error of Mean	0.0207
367	90KM SD	0.0735	95% KM (BCA) UCL	0.214
368	95% KM (t) UCL	0.219	95% KM (Percentile Bootstrap) UCL	0.216
369	95% KM (z) UCL	0.216	95% KM Bootstrap t UCL	0.219
370	90% KM Chebyshev UCL	0.244	95% KM Chebyshev UCL	0.272
371	97.5% KM Chebyshev UCL	0.311	99% KM Chebyshev UCL	0.388
372				
373	**Gamma GOF Tests on Detected Observations Only**			
374	A-D Test Statistic	0.232	**Anderson-Darling GOF Test**	
375	5% A-D Critical Value	0.725	Detected data appear Gamma Distributed at 5% Significance Level	
376	K-S Test Statistic	0.146	**Kolmogorov-Smirnov GOF**	
377	5% K-S Critical Value	0.267	Detected data appear Gamma Distributed at 5% Significance Level	
378	**Detected data appear Gamma Distributed at 5% Significance Level**			
379				
380	**Gamma Statistics on Detected Data Only**			
381	k hat (MLE)	11.41	k star (bias corrected MLE)	8.052
382	Theta hat (MLE)	0.0188	Theta star (bias corrected MLE)	0.0267
383	nu hat (MLE)	228.2	nu star (bias corrected)	161
384	Mean (detects)	0.215		
385				
386	**Gamma ROS Statistics using Imputed Non-Detects**			
387	GROS may not be used when data set has > 50% NDs with many tied observations at multiple DLs			
388	GROS may not be used when kstar of detects is small such as <1.0, especially when the sample size is small (e.g., <15-20)			
389	For such situations, GROS method may yield incorrect values of UCLs and BTVs			
390	This is especially true when the sample size is small.			
391	For gamma distributed detected data, BTVs and UCLs may be computed using gamma distribution on KM estimates			
392	Minimum	0.0441	Mean	0.176
393	Maximum	0.31	Median	0.17
394	SD	0.0852	CV	0.485
395	k hat (MLE)	3.816	k star (bias corrected MLE)	3.046
396	Theta hat (MLE)	0.046	Theta star (bias corrected MLE)	0.0576
397	nu hat (MLE)	106.8	nu star (bias corrected)	85.28
398	Adjusted Level of Significance (β)	0.0312		
399	Approximate Chi Square Value (85.28, α)	65	Adjusted Chi Square Value (85.28, β)	62.65
400	95% Gamma Approximate UCL	0.23	95% Gamma Adjusted UCL	0.239

Estimates of Gamma Parameters using KM Estimates

Mean (KM)	0.182	SD (KM)	0.0735
Variance (KM)	0.0054	SE of Mean (KM)	0.0207
k hat (KM)	6.141	k star (KM)	4.873
nu hat (KM)	171.9	nu star (KM)	136.4
theta hat (KM)	0.0297	theta star (KM)	0.0374
80% gamma percentile (KM)	0.246	90% gamma percentile (KM)	0.293
95% gamma percentile (KM)	0.336	99% gamma percentile (KM)	0.426

Gamma Kaplan-Meier (KM) Statistics

Approximate Chi Square Value (136.43, α)	110.4	Adjusted Chi Square Value (136.43, β)	107.3
95% KM Approximate Gamma UCL	0.225	95% KM Adjusted Gamma UCL	0.231

Lognormal GOF Test on Detected Observations Only

Shapiro Wilk Test Statistic	0.951	**Shapiro Wilk GOF Test**
10% Shapiro Wilk Critical Value	0.869	Detected Data appear Lognormal at 10% Significance Level
Lilliefors Test Statistic	0.128	**Lilliefors GOF Test**
10% Lilliefors Critical Value	0.241	Detected Data appear Lognormal at 10% Significance Level

Detected Data appear Lognormal at 10% Significance Level

Lognormal ROS Statistics Using Imputed Non-Detects

Mean in Original Scale	0.181	Mean in Log Scale	-1.806
SD in Original Scale	0.0784	SD in Log Scale	0.464
95% t UCL (assumes normality of ROS data)	0.218	95% Percentile Bootstrap UCL	0.215
95% BCA Bootstrap UCL	0.215	95% Bootstrap t UCL	0.221
95% H-UCL (Log ROS)	0.237		

Statistics using KM estimates on Logged Data and Assuming Lognormal Distribution

KM Mean (logged)	-1.788	KM Geo Mean	0.167
KM SD (logged)	0.416	95% Critical H Value (KM-Log)	1.992
KM Standard Error of Mean (logged)	0.117	95% H-UCL (KM -Log)	0.23
KM SD (logged)	0.416	95% Critical H Value (KM-Log)	1.992
KM Standard Error of Mean (logged)	0.117		

DL/2 Statistics

DL/2 Normal		**DL/2 Log-Transformed**	
Mean in Original Scale	0.168	Mean in Log Scale	-1.986
SD in Original Scale	0.0943	SD in Log Scale	0.715
95% t UCL (Assumes normality)	0.212	95% H-Stat UCL	0.283

DL/2 is not a recommended method, provided for comparisons and historical reasons

Nonparametric Distribution Free UCL Statistics

Detected Data appear Normal Distributed at 1% Significance Level

Suggested UCL to Use

95% KM (t) UCL	0.219

Note: Suggestions regarding the selection of a 95% UCL are provided to help the user to select the most appropriate 95% UCL.

Recommendations are based upon data size, data distribution, and skewness using results from simulation studies.

However, simulations results will not cover all Real World data sets; for additional insight the user may want to consult a statistician.

Boron

General Statistics

Total Number of Observations	14	Number of Distinct Observations	4
Number of Detects	4	Number of Non-Detects	10
Number of Distinct Detects	4	Number of Distinct Non-Detects	1
Minimum Detect	2	Minimum Non-Detect	2
Maximum Detect	4.1	Maximum Non-Detect	2
Variance Detects	0.94	Percent Non-Detects	71.43%
Mean Detects	2.7	SD Detects	0.97
Median Detects	2.35	CV Detects	0.359
Skewness Detects	1.598	Kurtosis Detects	2.387
Mean of Logged Detects	0.95	SD of Logged Detects	0.328

Normal GOF Test on Detects Only

Shapiro Wilk Test Statistic	0.827	**Shapiro Wilk GOF Test**
1% Shapiro Wilk Critical Value	0.687	Detected Data appear Normal at 1% Significance Level
Lilliefors Test Statistic	0.291	**Lilliefors GOF Test**
1% Lilliefors Critical Value	0.413	Detected Data appear Normal at 1% Significance Level

Detected Data appear Normal at 1% Significance Level

Note GOF tests may be unreliable for small sample sizes

Kaplan-Meier (KM) Statistics using Normal Critical Values and other Nonparametric UCLs

KM Mean	2.2	KM Standard Error of Mean	0.169
90KM SD	0.549	95% KM (BCA) UCL	N/A
95% KM (t) UCL	2.5	95% KM (Percentile Bootstrap) UCL	N/A
95% KM (z) UCL	2.479	95% KM Bootstrap t UCL	N/A
90% KM Chebyshev UCL	2.708	95% KM Chebyshev UCL	2.939
97.5% KM Chebyshev UCL	3.258	99% KM Chebyshev UCL	3.886

Gamma GOF Tests on Detected Observations Only

A-D Test Statistic	0.444	**Anderson-Darling GOF Test**
5% A-D Critical Value	0.657	Detected data appear Gamma Distributed at 5% Significance Level
K-S Test Statistic	0.268	**Kolmogorov-Smirnov GOF**
5% K-S Critical Value	0.395	Detected data appear Gamma Distributed at 5% Significance Level

Detected data appear Gamma Distributed at 5% Significance Level

Note GOF tests may be unreliable for small sample sizes

Gamma Statistics on Detected Data Only

k hat (MLE)	11.83	k star (bias corrected MLE)	3.124
Theta hat (MLE)	0.228	Theta star (bias corrected MLE)	0.864
nu hat (MLE)	94.65	nu star (bias corrected)	25
Mean (detects)	2.7		

Gamma ROS Statistics using Imputed Non-Detects

GROS may not be used when data set has > 50% NDs with many tied observations at multiple DLs

GROS may not be used when kstar of detects is small such as <1.0, especially when the sample size is small (e.g., <15-20)

For such situations, GROS method may yield incorrect values of UCLs and BTVs

This is especially true when the sample size is small.			
For gamma distributed detected data, BTVs and UCLs may be computed using gamma distribution on KM estimates			
Minimum	0.01	Mean	0.893
Maximum	4.1	Median	0.0959
SD	1.3	CV	1.457
k hat (MLE)	0.325	k star (bias corrected MLE)	0.303
Theta hat (MLE)	2.742	Theta star (bias corrected MLE)	2.942
nu hat (MLE)	9.113	nu star (bias corrected)	8.494
Adjusted Level of Significance (β)	0.0312		
Approximate Chi Square Value (8.49, α)	3.024	Adjusted Chi Square Value (8.49, β)	2.607
95% Gamma Approximate UCL	2.507	95% Gamma Adjusted UCL	N/A
Estimates of Gamma Parameters using KM Estimates			
Mean (KM)	2.2	SD (KM)	0.549
Variance (KM)	0.301	SE of Mean (KM)	0.169
k hat (KM)	16.06	k star (KM)	12.66
nu hat (KM)	449.6	nu star (KM)	354.6
theta hat (KM)	0.137	theta star (KM)	0.174
80% gamma percentile (KM)	2.696	90% gamma percentile (KM)	3.02
95% gamma percentile (KM)	3.306	99% gamma percentile (KM)	3.887
Gamma Kaplan-Meier (KM) Statistics			
Approximate Chi Square Value (354.58, α)	311.9	Adjusted Chi Square Value (354.58, β)	306.6
95% KM Approximate Gamma UCL	2.501	95% KM Adjusted Gamma UCL	2.544
Lognormal GOF Test on Detected Observations Only			
Shapiro Wilk Test Statistic	0.87	**Shapiro Wilk GOF Test**	
10% Shapiro Wilk Critical Value	0.792	Detected Data appear Lognormal at 10% Significance Level	
Lilliefors Test Statistic	0.244	**Lilliefors GOF Test**	
10% Lilliefors Critical Value	0.346	Detected Data appear Lognormal at 10% Significance Level	
Detected Data appear Lognormal at 10% Significance Level			
Note GOF tests may be unreliable for small sample sizes			
Lognormal ROS Statistics Using Imputed Non-Detects			
Mean in Original Scale	1.298	Mean in Log Scale	-0.0324
SD in Original Scale	1.074	SD in Log Scale	0.801
95% t UCL (assumes normality of ROS data)	1.806	95% Percentile Bootstrap UCL	1.785
95% BCA Bootstrap UCL	1.858	95% Bootstrap t UCL	2.1
95% H-UCL (Log ROS)	2.314		
Statistics using KM estimates on Logged Data and Assuming Lognormal Distribution			
KM Mean (logged)	0.767	KM Geo Mean	2.153
KM SD (logged)	0.191	95% Critical H Value (KM-Log)	1.812
KM Standard Error of Mean (logged)	0.059	95% H-UCL (KM -Log)	2.413
KM SD (logged)	0.191	95% Critical H Value (KM-Log)	1.812
KM Standard Error of Mean (logged)	0.059		
DL/2 Statistics			
DL/2 Normal		**DL/2 Log-Transformed**	
Mean in Original Scale	1.486	Mean in Log Scale	0.272

SD in Original Scale	0.923	SD in Log Scale	0.473
95% t UCL (Assumes normality)	1.923	95% H-Stat UCL	1.909
DL/2 is not a recommended method, provided for comparisons and historical reasons			
Nonparametric Distribution Free UCL Statistics			
Detected Data appear Normal Distributed at 1% Significance Level			
Suggested UCL to Use			
95% KM (t) UCL	2.5		
Note: Suggestions regarding the selection of a 95% UCL are provided to help the user to select the most appropriate 95% UCL.			
Recommendations are based upon data size, data distribution, and skewness using results from simulation studies.			
However, simulations results will not cover all Real World data sets; for additional insight the user may want to consult a statistician.			
Cadmium			
General Statistics			
Total Number of Observations	14	Number of Distinct Observations	13
		Number of Missing Observations	0
Minimum	0.066	Mean	0.0994
Maximum	0.185	Median	0.0895
SD	0.0331	Std. Error of Mean	0.00884
Coefficient of Variation	0.333	Skewness	1.479
Normal GOF Test			
Shapiro Wilk Test Statistic	0.853	**Shapiro Wilk GOF Test**	
1% Shapiro Wilk Critical Value	0.825	Data appear Normal at 1% Significance Level	
Lilliefors Test Statistic	0.179	**Lilliefors GOF Test**	
1% Lilliefors Critical Value	0.263	Data appear Normal at 1% Significance Level	
Data appear Normal at 1% Significance Level			
Assuming Normal Distribution			
95% Normal UCL		**95% UCLs (Adjusted for Skewness)**	
95% Student's-t UCL	0.115	95% Adjusted-CLT UCL (Chen-1995)	0.118
		95% Modified-t UCL (Johnson-1978)	0.116
Gamma GOF Test			
A-D Test Statistic	0.516	**Anderson-Darling Gamma GOF Test**	
5% A-D Critical Value	0.734	Detected data appear Gamma Distributed at 5% Significance Level	
K-S Test Statistic	0.175	**Kolmogorov-Smirnov Gamma GOF Test**	
5% K-S Critical Value	0.229	Detected data appear Gamma Distributed at 5% Significance Level	
Detected data appear Gamma Distributed at 5% Significance Level			
Gamma Statistics			
k hat (MLE)	11.49	k star (bias corrected MLE)	9.079
Theta hat (MLE)	0.00864	Theta star (bias corrected MLE)	0.0109
nu hat (MLE)	321.8	nu star (bias corrected)	254.2
MLE Mean (bias corrected)	0.0994	MLE Sd (bias corrected)	0.033
		Approximate Chi Square Value (0.05)	218.3

	A	B	C	D	E	F	G	H	I	J	K	L
601					Adjusted Level of Significance	0.0312					Adjusted Chi Square Value	213.9
602												
603	Assuming Gamma Distribution											
604					95% Approximate Gamma UCL	0.116					95% Adjusted Gamma UCL	0.118
605												
606	Lognormal GOF Test											
607					Shapiro Wilk Test Statistic	0.923	Shapiro Wilk Lognormal GOF Test					
608					10% Shapiro Wilk Critical Value	0.895	Data appear Lognormal at 10% Significance Level					
609					Lilliefors Test Statistic	0.161	Lilliefors Lognormal GOF Test					
610					10% Lilliefors Critical Value	0.208	Data appear Lognormal at 10% Significance Level					
611	Data appear Lognormal at 10% Significance Level											
612												
613	Lognormal Statistics											
614					Minimum of Logged Data	-2.718					Mean of logged Data	-2.353
615					Maximum of Logged Data	-1.687					SD of logged Data	0.298
616												
617	Assuming Lognormal Distribution											
618					95% H-UCL	0.116					90% Chebyshev (MVUE) UCL	0.123
619					95% Chebyshev (MVUE) UCL	0.134					97.5% Chebyshev (MVUE) UCL	0.149
620					99% Chebyshev (MVUE) UCL	0.178						
621												
622	Nonparametric Distribution Free UCL Statistics											
623	Data appear to follow a Discernible Distribution											
624												
625	Nonparametric Distribution Free UCLs											
626					95% CLT UCL	0.114					95% BCA Bootstrap UCL	0.118
627					95% Standard Bootstrap UCL	0.114					95% Bootstrap-t UCL	0.124
628					95% Hall's Bootstrap UCL	0.135					95% Percentile Bootstrap UCL	0.115
629					90% Chebyshev(Mean, Sd) UCL	0.126					95% Chebyshev(Mean, Sd) UCL	0.138
630					97.5% Chebyshev(Mean, Sd) UCL	0.155					99% Chebyshev(Mean, Sd) UCL	0.187
631												
632	Suggested UCL to Use											
633					95% Student's-t UCL	0.115						
634												
635	Note: Suggestions regarding the selection of a 95% UCL are provided to help the user to select the most appropriate 95% UCL.											
636	Recommendations are based upon data size, data distribution, and skewness using results from simulation studies.											
637	However, simulations results will not cover all Real World data sets; for additional insight the user may want to consult a statistician.											
638												
639												
640	Chromium											
641												
642	General Statistics											
643					Total Number of Observations	14					Number of Distinct Observations	14
644											Number of Missing Observations	0
645					Minimum	13.7					Mean	34.06
646					Maximum	45.2					Median	34.35
647					SD	8.31					Std. Error of Mean	2.221
648					Coefficient of Variation	0.244					Skewness	-1.068
649												
650	Normal GOF Test											

	A	B	C	D	E	F	G	H	I	J	K	L
651					Shapiro Wilk Test Statistic	0.921	**Shapiro Wilk GOF Test**					
652					1% Shapiro Wilk Critical Value	0.825	Data appear Normal at 1% Significance Level					
653					Lilliefors Test Statistic	0.216	**Lilliefors GOF Test**					
654					1% Lilliefors Critical Value	0.263	Data appear Normal at 1% Significance Level					
655	**Data appear Normal at 1% Significance Level**											
656												
657	**Assuming Normal Distribution**											
658	**95% Normal UCL**						**95% UCLs (Adjusted for Skewness)**					
659					95% Student's-t UCL	37.99					95% Adjusted-CLT UCL (Chen-1995)	37.03
660											95% Modified-t UCL (Johnson-1978)	37.88
661												
662	**Gamma GOF Test**											
663					A-D Test Statistic	0.734	**Anderson-Darling Gamma GOF Test**					
664					5% A-D Critical Value	0.734	Detected data appear Gamma Distributed at 5% Significance Level					
665					K-S Test Statistic	0.258	**Kolmogorov-Smirnov Gamma GOF Test**					
666					5% K-S Critical Value	0.229	Data Not Gamma Distributed at 5% Significance Level					
667	**Detected data follow Appr. Gamma Distribution at 5% Significance Level**											
668												
669	**Gamma Statistics**											
670					k hat (MLE)	13.75					k star (bias corrected MLE)	10.85
671					Theta hat (MLE)	2.478					Theta star (bias corrected MLE)	3.14
672					nu hat (MLE)	384.9					nu star (bias corrected)	303.7
673					MLE Mean (bias corrected)	34.06					MLE Sd (bias corrected)	10.34
674											Approximate Chi Square Value (0.05)	264.4
675					Adjusted Level of Significance	0.0312					Adjusted Chi Square Value	259.5
676												
677	**Assuming Gamma Distribution**											
678					95% Approximate Gamma UCL	39.13					95% Adjusted Gamma UCL	39.86
679												
680	**Lognormal GOF Test**											
681					Shapiro Wilk Test Statistic	0.808	**Shapiro Wilk Lognormal GOF Test**					
682					10% Shapiro Wilk Critical Value	0.895	Data Not Lognormal at 10% Significance Level					
683					Lilliefors Test Statistic	0.277	**Lilliefors Lognormal GOF Test**					
684					10% Lilliefors Critical Value	0.208	Data Not Lognormal at 10% Significance Level					
685	**Data Not Lognormal at 10% Significance Level**											
686												
687	**Lognormal Statistics**											
688					Minimum of Logged Data	2.617					Mean of logged Data	3.491
689					Maximum of Logged Data	3.811					SD of logged Data	0.306
690												
691	**Assuming Lognormal Distribution**											
692					95% H-UCL	40.4					90% Chebyshev (MVUE) UCL	42.77
693					95% Chebyshev (MVUE) UCL	46.62					97.5% Chebyshev (MVUE) UCL	51.96
694					99% Chebyshev (MVUE) UCL	62.45						
695												
696	**Nonparametric Distribution Free UCL Statistics**											
697	**Data appear to follow a Discernible Distribution**											
698												
699	**Nonparametric Distribution Free UCLs**											
700					95% CLT UCL	37.71					95% BCA Bootstrap UCL	37.06

	A–E	F	G–K	L
701	95% Standard Bootstrap UCL	37.61	95% Bootstrap-t UCL	37.43
702	95% Hall's Bootstrap UCL	37.23	95% Percentile Bootstrap UCL	37.36
703	90% Chebyshev(Mean, Sd) UCL	40.72	95% Chebyshev(Mean, Sd) UCL	43.74
704	97.5% Chebyshev(Mean, Sd) UCL	47.93	99% Chebyshev(Mean, Sd) UCL	56.16
705				
706	**Suggested UCL to Use**			
707	95% Student's-t UCL	37.99		
708				
709	Note: Suggestions regarding the selection of a 95% UCL are provided to help the user to select the most appropriate 95% UCL.			
710	Recommendations are based upon data size, data distribution, and skewness using results from simulation studies.			
711	However, simulations results will not cover all Real World data sets; for additional insight the user may want to consult a statistician.			
712				
713	**Note: For highly negatively-skewed data, confidence limits (e.g., Chen, Johnson, Lognormal, and Gamma) may not be**			
714	**reliable. Chen's and Johnson's methods provide adjustments for positvely skewed data sets.**			
715				
716				
717	Cobalt			
718				
719	**General Statistics**			
720	Total Number of Observations	14	Number of Distinct Observations	14
721			Number of Missing Observations	0
722	Minimum	8.7	Mean	14.45
723	Maximum	21.7	Median	12.8
724	SD	4.56	Std. Error of Mean	1.219
725	Coefficient of Variation	0.316	Skewness	0.458
726				
727	**Normal GOF Test**			
728	Shapiro Wilk Test Statistic	0.903	**Shapiro Wilk GOF Test**	
729	1% Shapiro Wilk Critical Value	0.825	Data appear Normal at 1% Significance Level	
730	Lilliefors Test Statistic	0.189	**Lilliefors GOF Test**	
731	1% Lilliefors Critical Value	0.263	Data appear Normal at 1% Significance Level	
732	**Data appear Normal at 1% Significance Level**			
733				
734	**Assuming Normal Distribution**			
735	**95% Normal UCL**		**95% UCLs (Adjusted for Skewness)**	
736	95% Student's-t UCL	16.61	95% Adjusted-CLT UCL (Chen-1995)	16.61
737			95% Modified-t UCL (Johnson-1978)	16.63
738				
739	**Gamma GOF Test**			
740	A-D Test Statistic	0.458	**Anderson-Darling Gamma GOF Test**	
741	5% A-D Critical Value	0.734	Detected data appear Gamma Distributed at 5% Significance Level	
742	K-S Test Statistic	0.172	**Kolmogorov-Smirnov Gamma GOF Test**	
743	5% K-S Critical Value	0.229	Detected data appear Gamma Distributed at 5% Significance Level	
744	**Detected data appear Gamma Distributed at 5% Significance Level**			
745				
746	**Gamma Statistics**			
747	k hat (MLE)	11.05	k star (bias corrected MLE)	8.731
748	Theta hat (MLE)	1.307	Theta star (bias corrected MLE)	1.655
749	nu hat (MLE)	309.4	nu star (bias corrected)	244.5
750	MLE Mean (bias corrected)	14.45	MLE Sd (bias corrected)	4.889

	A	B	C	D	E	F	G	H	I	J	K	L
751											Approximate Chi Square Value (0.05)	209.3
752					Adjusted Level of Significance	0.0312					Adjusted Chi Square Value	204.9
753												
754	**Assuming Gamma Distribution**											
755					95% Approximate Gamma UCL	16.88					95% Adjusted Gamma UCL	17.23
756												
757	**Lognormal GOF Test**											
758					Shapiro Wilk Test Statistic	0.926	**Shapiro Wilk Lognormal GOF Test**					
759					10% Shapiro Wilk Critical Value	0.895	Data appear Lognormal at 10% Significance Level					
760					Lilliefors Test Statistic	0.152	**Lilliefors Lognormal GOF Test**					
761					10% Lilliefors Critical Value	0.208	Data appear Lognormal at 10% Significance Level					
762	**Data appear Lognormal at 10% Significance Level**											
763												
764	**Lognormal Statistics**											
765					Minimum of Logged Data	2.163					Mean of logged Data	2.625
766					Maximum of Logged Data	3.077					SD of logged Data	0.314
767												
768	**Assuming Lognormal Distribution**											
769					95% H-UCL	17.12					90% Chebyshev (MVUE) UCL	18.13
770					95% Chebyshev (MVUE) UCL	19.8					97.5% Chebyshev (MVUE) UCL	22.11
771					99% Chebyshev (MVUE) UCL	26.66						
772												
773	**Nonparametric Distribution Free UCL Statistics**											
774	**Data appear to follow a Discernible Distribution**											
775												
776	**Nonparametric Distribution Free UCLs**											
777					95% CLT UCL	16.45					95% BCA Bootstrap UCL	16.49
778					95% Standard Bootstrap UCL	16.43					95% Bootstrap-t UCL	17.07
779					95% Hall's Bootstrap UCL	16.48					95% Percentile Bootstrap UCL	16.45
780					90% Chebyshev(Mean, Sd) UCL	18.1					95% Chebyshev(Mean, Sd) UCL	19.76
781					97.5% Chebyshev(Mean, Sd) UCL	22.06					99% Chebyshev(Mean, Sd) UCL	26.57
782												
783	**Suggested UCL to Use**											
784					95% Student's-t UCL	16.61						
785												
786	Note: Suggestions regarding the selection of a 95% UCL are provided to help the user to select the most appropriate 95% UCL.											
787	Recommendations are based upon data size, data distribution, and skewness using results from simulation studies.											
788	However, simulations results will not cover all Real World data sets; for additional insight the user may want to consult a statistician.											
789												
790												
791	Copper											
792												
793	**General Statistics**											
794					Total Number of Observations	14					Number of Distinct Observations	13
795											Number of Missing Observations	0
796					Minimum	52.9					Mean	1189
797					Maximum	4020					Median	856
798					SD	1068					Std. Error of Mean	285.3
799					Coefficient of Variation	0.898					Skewness	1.479
800												

	A	B	C	D	E	F	G	H	I	J	K	L
801	Normal GOF Test											
802					Shapiro Wilk Test Statistic	0.866	Shapiro Wilk GOF Test					
803					1% Shapiro Wilk Critical Value	0.825	Data appear Normal at 1% Significance Level					
804					Lilliefors Test Statistic	0.176	Lilliefors GOF Test					
805					1% Lilliefors Critical Value	0.263	Data appear Normal at 1% Significance Level					
806	Data appear Normal at 1% Significance Level											
807												
808	Assuming Normal Distribution											
809	95% Normal UCL						95% UCLs (Adjusted for Skewness)					
810					95% Student's-t UCL	1695					95% Adjusted-CLT UCL (Chen-1995)	1779
811											95% Modified-t UCL (Johnson-1978)	1714
812												
813	Gamma GOF Test											
814					A-D Test Statistic	0.155	Anderson-Darling Gamma GOF Test					
815					5% A-D Critical Value	0.756	Detected data appear Gamma Distributed at 5% Significance Level					
816					K-S Test Statistic	0.111	Kolmogorov-Smirnov Gamma GOF Test					
817					5% K-S Critical Value	0.234	Detected data appear Gamma Distributed at 5% Significance Level					
818	Detected data appear Gamma Distributed at 5% Significance Level											
819												
820	Gamma Statistics											
821					k hat (MLE)	1.202					k star (bias corrected MLE)	0.992
822					Theta hat (MLE)	989.2					Theta star (bias corrected MLE)	1199
823					nu hat (MLE)	33.67					nu star (bias corrected)	27.79
824					MLE Mean (bias corrected)	1189					MLE Sd (bias corrected)	1194
825											Approximate Chi Square Value (0.05)	16.76
826					Adjusted Level of Significance	0.0312					Adjusted Chi Square Value	15.63
827												
828	Assuming Gamma Distribution											
829					95% Approximate Gamma UCL	1972					95% Adjusted Gamma UCL	2114
830												
831	Lognormal GOF Test											
832					Shapiro Wilk Test Statistic	0.945	Shapiro Wilk Lognormal GOF Test					
833					10% Shapiro Wilk Critical Value	0.895	Data appear Lognormal at 10% Significance Level					
834					Lilliefors Test Statistic	0.118	Lilliefors Lognormal GOF Test					
835					10% Lilliefors Critical Value	0.208	Data appear Lognormal at 10% Significance Level					
836	Data appear Lognormal at 10% Significance Level											
837												
838	Lognormal Statistics											
839					Minimum of Logged Data	3.968					Mean of logged Data	6.611
840					Maximum of Logged Data	8.299					SD of logged Data	1.162
841												
842	Assuming Lognormal Distribution											
843					95% H-UCL	3929					90% Chebyshev (MVUE) UCL	2756
844					95% Chebyshev (MVUE) UCL	3388					97.5% Chebyshev (MVUE) UCL	4266
845					99% Chebyshev (MVUE) UCL	5991						
846												
847	Nonparametric Distribution Free UCL Statistics											
848	Data appear to follow a Discernible Distribution											
849												
850	Nonparametric Distribution Free UCLs											

	A	B	C	D	E	F	G	H	I	J	K	L
851					95% CLT UCL	1659					95% BCA Bootstrap UCL	1795
852					95% Standard Bootstrap UCL	1642					95% Bootstrap-t UCL	1910
853					95% Hall's Bootstrap UCL	2059					95% Percentile Bootstrap UCL	1662
854					90% Chebyshev(Mean, Sd) UCL	2046					95% Chebyshev(Mean, Sd) UCL	2433
855					97.5% Chebyshev(Mean, Sd) UCL	2971					99% Chebyshev(Mean, Sd) UCL	4029
856												
857	**Suggested UCL to Use**											
858					95% Student's-t UCL	1695						
859												
860	Note: Suggestions regarding the selection of a 95% UCL are provided to help the user to select the most appropriate 95% UCL.											
861	Recommendations are based upon data size, data distribution, and skewness using results from simulation studies.											
862	However, simulations results will not cover all Real World data sets; for additional insight the user may want to consult a statistician.											
863												
864												
865	Iron											
866												
867	**General Statistics**											
868					Total Number of Observations	13					Number of Distinct Observations	12
869											Number of Missing Observations	1
870					Minimum	11500					Mean	29454
871					Maximum	46300					Median	28900
872					SD	9436					Std. Error of Mean	2617
873					Coefficient of Variation	0.32					Skewness	0.031
874												
875	**Normal GOF Test**											
876					Shapiro Wilk Test Statistic	0.948	**Shapiro Wilk GOF Test**					
877					1% Shapiro Wilk Critical Value	0.814	Data appear Normal at 1% Significance Level					
878					Lilliefors Test Statistic	0.179	**Lilliefors GOF Test**					
879					1% Lilliefors Critical Value	0.271	Data appear Normal at 1% Significance Level					
880	**Data appear Normal at 1% Significance Level**											
881												
882	**Assuming Normal Distribution**											
883	**95% Normal UCL**						**95% UCLs (Adjusted for Skewness)**					
884					95% Student's-t UCL	34118					95% Adjusted-CLT UCL (Chen-1995)	33782
885											95% Modified-t UCL (Johnson-1978)	34122
886												
887	**Gamma GOF Test**											
888					A-D Test Statistic	0.487	**Anderson-Darling Gamma GOF Test**					
889					5% A-D Critical Value	0.734	Detected data appear Gamma Distributed at 5% Significance Level					
890					K-S Test Statistic	0.172	**Kolmogorov-Smirnov Gamma GOF Test**					
891					5% K-S Critical Value	0.237	Detected data appear Gamma Distributed at 5% Significance Level					
892	**Detected data appear Gamma Distributed at 5% Significance Level**											
893												
894	**Gamma Statistics**											
895					k hat (MLE)	9.127					k star (bias corrected MLE)	7.072
896					Theta hat (MLE)	3227					Theta star (bias corrected MLE)	4165
897					nu hat (MLE)	237.3					nu star (bias corrected)	183.9
898					MLE Mean (bias corrected)	29454					MLE Sd (bias corrected)	11076
899											Approximate Chi Square Value (0.05)	153.5
900					Adjusted Level of Significance	0.0301					Adjusted Chi Square Value	149.6

Assuming Gamma Distribution

95% Approximate Gamma UCL	35280	95% Adjusted Gamma UCL	36212

Lognormal GOF Test

Shapiro Wilk Test Statistic	0.897	**Shapiro Wilk Lognormal GOF Test**
10% Shapiro Wilk Critical Value	0.889	Data appear Lognormal at 10% Significance Level
Lilliefors Test Statistic	0.201	**Lilliefors Lognormal GOF Test**
10% Lilliefors Critical Value	0.215	Data appear Lognormal at 10% Significance Level

Data appear Lognormal at 10% Significance Level

Lognormal Statistics

Minimum of Logged Data	9.35	Mean of logged Data	10.23
Maximum of Logged Data	10.74	SD of logged Data	0.368

Assuming Lognormal Distribution

95% H-UCL	36747	90% Chebyshev (MVUE) UCL	38851
95% Chebyshev (MVUE) UCL	43021	97.5% Chebyshev (MVUE) UCL	48810
99% Chebyshev (MVUE) UCL	60181		

Nonparametric Distribution Free UCL Statistics

Data appear to follow a Discernible Distribution

Nonparametric Distribution Free UCLs

95% CLT UCL	33758	95% BCA Bootstrap UCL	33638
95% Standard Bootstrap UCL	33580	95% Bootstrap-t UCL	34211
95% Hall's Bootstrap UCL	34927	95% Percentile Bootstrap UCL	33523
90% Chebyshev(Mean, Sd) UCL	37305	95% Chebyshev(Mean, Sd) UCL	40861
97.5% Chebyshev(Mean, Sd) UCL	45797	99% Chebyshev(Mean, Sd) UCL	55492

Suggested UCL to Use

95% Student's-t UCL	34118		

Note: Suggestions regarding the selection of a 95% UCL are provided to help the user to select the most appropriate 95% UCL.

Recommendations are based upon data size, data distribution, and skewness using results from simulation studies.

However, simulations results will not cover all Real World data sets; for additional insight the user may want to consult a statistician.

Lead

General Statistics

Total Number of Observations	14	Number of Distinct Observations	14
		Number of Missing Observations	0
Minimum	1.2	Mean	6.974
Maximum	17.9	Median	4.805
SD	5.242	Std. Error of Mean	1.401
Coefficient of Variation	0.752	Skewness	0.832

Normal GOF Test

Shapiro Wilk Test Statistic	0.903	**Shapiro Wilk GOF Test**

	A	B	C	D	E	F	G	H	I	J	K	L
951					1% Shapiro Wilk Critical Value	0.825	Data appear Normal at 1% Significance Level					
952					Lilliefors Test Statistic	0.224	**Lilliefors GOF Test**					
953					1% Lilliefors Critical Value	0.263	Data appear Normal at 1% Significance Level					
954	**Data appear Normal at 1% Significance Level**											
955												
956	**Assuming Normal Distribution**											
957	**95% Normal UCL**						**95% UCLs (Adjusted for Skewness)**					
958					95% Student's-t UCL	9.455					95% Adjusted-CLT UCL (Chen-1995)	9.612
959											95% Modified-t UCL (Johnson-1978)	9.507
960												
961	**Gamma GOF Test**											
962					A-D Test Statistic	0.27	**Anderson-Darling Gamma GOF Test**					
963					5% A-D Critical Value	0.748	Detected data appear Gamma Distributed at 5% Significance Level					
964					K-S Test Statistic	0.144	**Kolmogorov-Smirnov Gamma GOF Test**					
965					5% K-S Critical Value	0.232	Detected data appear Gamma Distributed at 5% Significance Level					
966	**Detected data appear Gamma Distributed at 5% Significance Level**											
967												
968	**Gamma Statistics**											
969					k hat (MLE)	1.789					k star (bias corrected MLE)	1.453
970					Theta hat (MLE)	3.899					Theta star (bias corrected MLE)	4.8
971					nu hat (MLE)	50.08					nu star (bias corrected)	40.68
972					MLE Mean (bias corrected)	6.974					MLE Sd (bias corrected)	5.786
973											Approximate Chi Square Value (0.05)	27.07
974					Adjusted Level of Significance	0.0312					Adjusted Chi Square Value	25.6
975												
976	**Assuming Gamma Distribution**											
977					95% Approximate Gamma UCL	10.48					95% Adjusted Gamma UCL	11.08
978												
979	**Lognormal GOF Test**											
980					Shapiro Wilk Test Statistic	0.95	**Shapiro Wilk Lognormal GOF Test**					
981					10% Shapiro Wilk Critical Value	0.895	Data appear Lognormal at 10% Significance Level					
982					Lilliefors Test Statistic	0.141	**Lilliefors Lognormal GOF Test**					
983					10% Lilliefors Critical Value	0.208	Data appear Lognormal at 10% Significance Level					
984	**Data appear Lognormal at 10% Significance Level**											
985												
986	**Lognormal Statistics**											
987					Minimum of Logged Data	0.182					Mean of logged Data	1.637
988					Maximum of Logged Data	2.885					SD of logged Data	0.861
989												
990	**Assuming Lognormal Distribution**											
991					95% H-UCL	13.75					90% Chebyshev (MVUE) UCL	12.51
992					95% Chebyshev (MVUE) UCL	14.92					97.5% Chebyshev (MVUE) UCL	18.26
993					99% Chebyshev (MVUE) UCL	24.82						
994												
995	**Nonparametric Distribution Free UCL Statistics**											
996	**Data appear to follow a Discernible Distribution**											
997												
998	**Nonparametric Distribution Free UCLs**											
999					95% CLT UCL	9.279					95% BCA Bootstrap UCL	9.554
1000					95% Standard Bootstrap UCL	9.221					95% Bootstrap-t UCL	10.02

	A	B	C	D	E	F	G	H	I	J	K	L
1001					95% Hall's Bootstrap UCL	9.732					95% Percentile Bootstrap UCL	9.3
1002					90% Chebyshev(Mean, Sd) UCL	11.18					95% Chebyshev(Mean, Sd) UCL	13.08
1003					97.5% Chebyshev(Mean, Sd) UCL	15.72					99% Chebyshev(Mean, Sd) UCL	20.91
1004												
1005						**Suggested UCL to Use**						
1006					95% Student's-t UCL	9.455						
1007												
1008	Note: Suggestions regarding the selection of a 95% UCL are provided to help the user to select the most appropriate 95% UCL.											
1009	Recommendations are based upon data size, data distribution, and skewness using results from simulation studies.											
1010	However, simulations results will not cover all Real World data sets; for additional insight the user may want to consult a statistician.											
1011												
1012												
1013	**Lithium**											
1014												
1015						**General Statistics**						
1016					Total Number of Observations	14					Number of Distinct Observations	14
1017											Number of Missing Observations	0
1018					Minimum	2.67					Mean	8.59
1019					Maximum	16.5					Median	9.615
1020					SD	3.992					Std. Error of Mean	1.067
1021					Coefficient of Variation	0.465					Skewness	0.19
1022												
1023						**Normal GOF Test**						
1024					Shapiro Wilk Test Statistic	0.955			**Shapiro Wilk GOF Test**			
1025					1% Shapiro Wilk Critical Value	0.825			Data appear Normal at 1% Significance Level			
1026					Lilliefors Test Statistic	0.166			**Lilliefors GOF Test**			
1027					1% Lilliefors Critical Value	0.263			Data appear Normal at 1% Significance Level			
1028						**Data appear Normal at 1% Significance Level**						
1029												
1030						**Assuming Normal Distribution**						
1031			**95% Normal UCL**						**95% UCLs (Adjusted for Skewness)**			
1032					95% Student's-t UCL	10.48					95% Adjusted-CLT UCL (Chen-1995)	10.4
1033											95% Modified-t UCL (Johnson-1978)	10.49
1034												
1035						**Gamma GOF Test**						
1036					A-D Test Statistic	0.422			**Anderson-Darling Gamma GOF Test**			
1037					5% A-D Critical Value	0.739			Detected data appear Gamma Distributed at 5% Significance Level			
1038					K-S Test Statistic	0.221			**Kolmogorov-Smirnov Gamma GOF Test**			
1039					5% K-S Critical Value	0.23			Detected data appear Gamma Distributed at 5% Significance Level			
1040						**Detected data appear Gamma Distributed at 5% Significance Level**						
1041												
1042						**Gamma Statistics**						
1043					k hat (MLE)	4.297					k star (bias corrected MLE)	3.424
1044					Theta hat (MLE)	1.999					Theta star (bias corrected MLE)	2.509
1045					nu hat (MLE)	120.3					nu star (bias corrected)	95.86
1046					MLE Mean (bias corrected)	8.59					MLE Sd (bias corrected)	4.643
1047											Approximate Chi Square Value (0.05)	74.28
1048					Adjusted Level of Significance	0.0312					Adjusted Chi Square Value	71.76
1049												
1050						**Assuming Gamma Distribution**						

95% Approximate Gamma UCL	11.09	95% Adjusted Gamma UCL	11.47
Lognormal GOF Test			
Shapiro Wilk Test Statistic	0.929	**Shapiro Wilk Lognormal GOF Test**	
10% Shapiro Wilk Critical Value	0.895	Data appear Lognormal at 10% Significance Level	
Lilliefors Test Statistic	0.234	**Lilliefors Lognormal GOF Test**	
10% Lilliefors Critical Value	0.208	Data Not Lognormal at 10% Significance Level	
Data appear Approximate Lognormal at 10% Significance Level			
Lognormal Statistics			
Minimum of Logged Data	0.982	Mean of logged Data	2.03
Maximum of Logged Data	2.803	SD of logged Data	0.54
Assuming Lognormal Distribution			
95% H-UCL	12.01	90% Chebyshev (MVUE) UCL	12.6
95% Chebyshev (MVUE) UCL	14.37	97.5% Chebyshev (MVUE) UCL	16.82
99% Chebyshev (MVUE) UCL	21.63		
Nonparametric Distribution Free UCL Statistics			
Data appear to follow a Discernible Distribution			
Nonparametric Distribution Free UCLs			
95% CLT UCL	10.34	95% BCA Bootstrap UCL	10.3
95% Standard Bootstrap UCL	10.29	95% Bootstrap-t UCL	10.57
95% Hall's Bootstrap UCL	10.5	95% Percentile Bootstrap UCL	10.26
90% Chebyshev(Mean, Sd) UCL	11.79	95% Chebyshev(Mean, Sd) UCL	13.24
97.5% Chebyshev(Mean, Sd) UCL	15.25	99% Chebyshev(Mean, Sd) UCL	19.21
Suggested UCL to Use			
95% Student's-t UCL	10.48		
Note: Suggestions regarding the selection of a 95% UCL are provided to help the user to select the most appropriate 95% UCL.			
Recommendations are based upon data size, data distribution, and skewness using results from simulation studies.			
However, simulations results will not cover all Real World data sets; for additional insight the user may want to consult a statistician.			
Manganese			
General Statistics			
Total Number of Observations	14	Number of Distinct Observations	13
		Number of Missing Observations	0
Minimum	136	Mean	312.6
Maximum	489	Median	295.5
SD	90.38	Std. Error of Mean	24.16
Coefficient of Variation	0.289	Skewness	0.0736
Normal GOF Test			
Shapiro Wilk Test Statistic	0.953	**Shapiro Wilk GOF Test**	
1% Shapiro Wilk Critical Value	0.825	Data appear Normal at 1% Significance Level	
Lilliefors Test Statistic	0.193	**Lilliefors GOF Test**	

	A–E	F	G–K	L
1101	1% Lilliefors Critical Value	0.263	Data appear Normal at 1% Significance Level	
1102	**Data appear Normal at 1% Significance Level**			
1103				
1104	**Assuming Normal Distribution**			
1105	**95% Normal UCL**		**95% UCLs (Adjusted for Skewness)**	
1106	95% Student's-t UCL	355.3	95% Adjusted-CLT UCL (Chen-1995)	352.8
1107			95% Modified-t UCL (Johnson-1978)	355.4
1108				
1109	**Gamma GOF Test**			
1110	A-D Test Statistic	0.472	**Anderson-Darling Gamma GOF Test**	
1111	5% A-D Critical Value	0.734	Detected data appear Gamma Distributed at 5% Significance Level	
1112	K-S Test Statistic	0.165	**Kolmogorov-Smirnov Gamma GOF Test**	
1113	5% K-S Critical Value	0.229	Detected data appear Gamma Distributed at 5% Significance Level	
1114	**Detected data appear Gamma Distributed at 5% Significance Level**			
1115				
1116	**Gamma Statistics**			
1117	k hat (MLE)	11.68	k star (bias corrected MLE)	9.227
1118	Theta hat (MLE)	26.75	Theta star (bias corrected MLE)	33.87
1119	nu hat (MLE)	327.1	nu star (bias corrected)	258.4
1120	MLE Mean (bias corrected)	312.6	MLE Sd (bias corrected)	102.9
1121			Approximate Chi Square Value (0.05)	222.1
1122	Adjusted Level of Significance	0.0312	Adjusted Chi Square Value	217.7
1123				
1124	**Assuming Gamma Distribution**			
1125	95% Approximate Gamma UCL	363.5	95% Adjusted Gamma UCL	371
1126				
1127	**Lognormal GOF Test**			
1128	Shapiro Wilk Test Statistic	0.912	**Shapiro Wilk Lognormal GOF Test**	
1129	10% Shapiro Wilk Critical Value	0.895	Data appear Lognormal at 10% Significance Level	
1130	Lilliefors Test Statistic	0.164	**Lilliefors Lognormal GOF Test**	
1131	10% Lilliefors Critical Value	0.208	Data appear Lognormal at 10% Significance Level	
1132	**Data appear Lognormal at 10% Significance Level**			
1133				
1134	**Lognormal Statistics**			
1135	Minimum of Logged Data	4.913	Mean of logged Data	5.701
1136	Maximum of Logged Data	6.192	SD of logged Data	0.318
1137				
1138	**Assuming Lognormal Distribution**			
1139	95% H-UCL	372.6	90% Chebyshev (MVUE) UCL	394.6
1140	95% Chebyshev (MVUE) UCL	431.3	97.5% Chebyshev (MVUE) UCL	482.2
1141	99% Chebyshev (MVUE) UCL	582.2		
1142				
1143	**Nonparametric Distribution Free UCL Statistics**			
1144	**Data appear to follow a Discernible Distribution**			
1145				
1146	**Nonparametric Distribution Free UCLs**			
1147	95% CLT UCL	352.3	95% BCA Bootstrap UCL	353.1
1148	95% Standard Bootstrap UCL	351.6	95% Bootstrap-t UCL	356.2
1149	95% Hall's Bootstrap UCL	355.1	95% Percentile Bootstrap UCL	350.9
1150	90% Chebyshev(Mean, Sd) UCL	385	95% Chebyshev(Mean, Sd) UCL	417.9

97.5% Chebyshev(Mean, Sd) UCL	463.4	99% Chebyshev(Mean, Sd) UCL	552.9

Suggested UCL to Use

95% Student's-t UCL	355.3		

Note: Suggestions regarding the selection of a 95% UCL are provided to help the user to select the most appropriate 95% UCL.

Recommendations are based upon data size, data distribution, and skewness using results from simulation studies.

However, simulations results will not cover all Real World data sets; for additional insight the user may want to consult a statistician.

Mercury

General Statistics

Total Number of Observations	14	Number of Distinct Observations	6
Number of Detects	5	Number of Non-Detects	9
Number of Distinct Detects	5	Number of Distinct Non-Detects	1
Minimum Detect	0.059	Minimum Non-Detect	0.04
Maximum Detect	0.132	Maximum Non-Detect	0.04
Variance Detects	9.4770E-4	Percent Non-Detects	64.29%
Mean Detects	0.0962	SD Detects	0.0308
Median Detects	0.09	CV Detects	0.32
Skewness Detects	0.0905	Kurtosis Detects	-2.139
Mean of Logged Detects	-2.385	SD of Logged Detects	0.333

Normal GOF Test on Detects Only

Shapiro Wilk Test Statistic	0.938	**Shapiro Wilk GOF Test**
1% Shapiro Wilk Critical Value	0.686	Detected Data appear Normal at 1% Significance Level
Lilliefors Test Statistic	0.208	**Lilliefors GOF Test**
1% Lilliefors Critical Value	0.396	Detected Data appear Normal at 1% Significance Level

Detected Data appear Normal at 1% Significance Level

Note GOF tests may be unreliable for small sample sizes

Kaplan-Meier (KM) Statistics using Normal Critical Values and other Nonparametric UCLs

KM Mean	0.0601	KM Standard Error of Mean	0.00943
90KM SD	0.0316	95% KM (BCA) UCL	0.0747
95% KM (t) UCL	0.0768	95% KM (Percentile Bootstrap) UCL	0.075
95% KM (z) UCL	0.0756	95% KM Bootstrap t UCL	0.0737
90% KM Chebyshev UCL	0.0884	95% KM Chebyshev UCL	0.101
97.5% KM Chebyshev UCL	0.119	99% KM Chebyshev UCL	0.154

Gamma GOF Tests on Detected Observations Only

A-D Test Statistic	0.269	**Anderson-Darling GOF Test**
5% A-D Critical Value	0.679	Detected data appear Gamma Distributed at 5% Significance Level
K-S Test Statistic	0.235	**Kolmogorov-Smirnov GOF**
5% K-S Critical Value	0.358	Detected data appear Gamma Distributed at 5% Significance Level

Detected data appear Gamma Distributed at 5% Significance Level

Note GOF tests may be unreliable for small sample sizes

Gamma Statistics on Detected Data Only

k hat (MLE)	11.74	k star (bias corrected MLE)	4.828
Theta hat (MLE)	0.0082	Theta star (bias corrected MLE)	0.0199

	A B C D E	F	G H I J K	L
1201	nu hat (MLE)	117.4	nu star (bias corrected)	48.28
1202	Mean (detects)	0.0962		
1203				
1204	**Gamma ROS Statistics using Imputed Non-Detects**			
1205	GROS may not be used when data set has > 50% NDs with many tied observations at multiple DLs			
1206	GROS may not be used when kstar of detects is small such as <1.0, especially when the sample size is small (e.g., <15-20)			
1207	For such situations, GROS method may yield incorrect values of UCLs and BTVs			
1208	This is especially true when the sample size is small.			
1209	For gamma distributed detected data, BTVs and UCLs may be computed using gamma distribution on KM estimates			
1210	Minimum	0.01	Mean	0.0458
1211	Maximum	0.132	Median	0.0273
1212	SD	0.0436	CV	0.953
1213	k hat (MLE)	1.214	k star (bias corrected MLE)	1.001
1214	Theta hat (MLE)	0.0377	Theta star (bias corrected MLE)	0.0457
1215	nu hat (MLE)	33.98	nu star (bias corrected)	28.03
1216	Adjusted Level of Significance (β)	0.0312		
1217	Approximate Chi Square Value (28.03, α)	16.95	Adjusted Chi Square Value (28.03, β)	15.82
1218	95% Gamma Approximate UCL	0.0757	95% Gamma Adjusted UCL	0.0811
1219				
1220	**Estimates of Gamma Parameters using KM Estimates**			
1221	Mean (KM)	0.0601	SD (KM)	0.0316
1222	Variance (KM)	9.9592E-4	SE of Mean (KM)	0.00943
1223	k hat (KM)	3.623	k star (KM)	2.895
1224	nu hat (KM)	101.5	nu star (KM)	81.05
1225	theta hat (KM)	0.0166	theta star (KM)	0.0208
1226	80% gamma percentile (KM)	0.086	90% gamma percentile (KM)	0.107
1227	95% gamma percentile (KM)	0.127	99% gamma percentile (KM)	0.171
1228				
1229	**Gamma Kaplan-Meier (KM) Statistics**			
1230	Approximate Chi Square Value (81.05, α)	61.3	Adjusted Chi Square Value (81.05, β)	59.03
1231	95% KM Approximate Gamma UCL	0.0794	95% KM Adjusted Gamma UCL	0.0825
1232				
1233	**Lognormal GOF Test on Detected Observations Only**			
1234	Shapiro Wilk Test Statistic	0.946	**Shapiro Wilk GOF Test**	
1235	10% Shapiro Wilk Critical Value	0.806	Detected Data appear Lognormal at 10% Significance Level	
1236	Lilliefors Test Statistic	0.207	**Lilliefors GOF Test**	
1237	10% Lilliefors Critical Value	0.319	Detected Data appear Lognormal at 10% Significance Level	
1238	**Detected Data appear Lognormal at 10% Significance Level**			
1239	**Note GOF tests may be unreliable for small sample sizes**			
1240				
1241	**Lognormal ROS Statistics Using Imputed Non-Detects**			
1242	Mean in Original Scale	0.0537	Mean in Log Scale	-3.155
1243	SD in Original Scale	0.0382	SD in Log Scale	0.716
1244	95% t UCL (assumes normality of ROS data)	0.0719	95% Percentile Bootstrap UCL	0.0702
1245	95% BCA Bootstrap UCL	0.0715	95% Bootstrap t UCL	0.079
1246	95% H-UCL (Log ROS)	0.0879		
1247				
1248	**Statistics using KM estimates on Logged Data and Assuming Lognormal Distribution**			
1249	KM Mean (logged)	-2.921	KM Geo Mean	0.0539
1250	KM SD (logged)	0.438	95% Critical H Value (KM-Log)	2.001

KM Standard Error of Mean (logged)	0.131	95% H-UCL (KM -Log)	0.0756
KM SD (logged)	0.438	95% Critical H Value (KM-Log)	2.001
KM Standard Error of Mean (logged)	0.131		

DL/2 Statistics

DL/2 Normal		DL/2 Log-Transformed	
Mean in Original Scale	0.0472	Mean in Log Scale	-3.366
SD in Original Scale	0.0416	SD in Log Scale	0.782
95% t UCL (Assumes normality)	0.0669	95% H-Stat UCL	0.0797

DL/2 is not a recommended method, provided for comparisons and historical reasons

Nonparametric Distribution Free UCL Statistics

Detected Data appear Normal Distributed at 1% Significance Level

Suggested UCL to Use

95% KM (t) UCL	0.0768		

Note: Suggestions regarding the selection of a 95% UCL are provided to help the user to select the most appropriate 95% UCL.

Recommendations are based upon data size, data distribution, and skewness using results from simulation studies.

However, simulations results will not cover all Real World data sets; for additional insight the user may want to consult a statistician.

Molybdenum

General Statistics

Total Number of Observations	14	Number of Distinct Observations	13
		Number of Missing Observations	0
Minimum	0.22	Mean	1.504
Maximum	9.87	Median	0.73
SD	2.491	Std. Error of Mean	0.666
Coefficient of Variation	1.656	Skewness	3.344

Normal GOF Test

Shapiro Wilk Test Statistic	0.514	**Shapiro Wilk GOF Test**
1% Shapiro Wilk Critical Value	0.825	Data Not Normal at 1% Significance Level
Lilliefors Test Statistic	0.332	**Lilliefors GOF Test**
1% Lilliefors Critical Value	0.263	Data Not Normal at 1% Significance Level

Data Not Normal at 1% Significance Level

Assuming Normal Distribution

95% Normal UCL		95% UCLs (Adjusted for Skewness)	
95% Student's-t UCL	2.682	95% Adjusted-CLT UCL (Chen-1995)	3.234
		95% Modified-t UCL (Johnson-1978)	2.782

Gamma GOF Test

A-D Test Statistic	0.972	**Anderson-Darling Gamma GOF Test**
5% A-D Critical Value	0.763	Data Not Gamma Distributed at 5% Significance Level
K-S Test Statistic	0.195	**Kolmogorov-Smirnov Gamma GOF Test**
5% K-S Critical Value	0.236	Detected data appear Gamma Distributed at 5% Significance Level

Detected data follow Appr. Gamma Distribution at 5% Significance Level

Gamma Statistics

k hat (MLE)	0.925	k star (bias corrected MLE)	0.775
Theta hat (MLE)	1.625	Theta star (bias corrected MLE)	1.941
nu hat (MLE)	25.91	nu star (bias corrected)	21.69
MLE Mean (bias corrected)	1.504	MLE Sd (bias corrected)	1.708
		Approximate Chi Square Value (0.05)	12.11
Adjusted Level of Significance	0.0312	Adjusted Chi Square Value	11.16

Assuming Gamma Distribution

95% Approximate Gamma UCL	2.694	95% Adjusted Gamma UCL	2.921

Lognormal GOF Test

Shapiro Wilk Test Statistic	0.924	**Shapiro Wilk Lognormal GOF Test**
10% Shapiro Wilk Critical Value	0.895	Data appear Lognormal at 10% Significance Level
Lilliefors Test Statistic	0.143	**Lilliefors Lognormal GOF Test**
10% Lilliefors Critical Value	0.208	Data appear Lognormal at 10% Significance Level

Data appear Lognormal at 10% Significance Level

Lognormal Statistics

Minimum of Logged Data	-1.514	Mean of logged Data	-0.222
Maximum of Logged Data	2.289	SD of logged Data	1.027

Assuming Lognormal Distribution

95% H-UCL	3.048	90% Chebyshev (MVUE) UCL	2.444
95% Chebyshev (MVUE) UCL	2.967	97.5% Chebyshev (MVUE) UCL	3.694
99% Chebyshev (MVUE) UCL	5.12		

Nonparametric Distribution Free UCL Statistics

Data appear to follow a Discernible Distribution

Nonparametric Distribution Free UCLs

95% CLT UCL	2.598	95% BCA Bootstrap UCL	3.556
95% Standard Bootstrap UCL	2.576	95% Bootstrap-t UCL	5.713
95% Hall's Bootstrap UCL	6.784	95% Percentile Bootstrap UCL	2.761
90% Chebyshev(Mean, Sd) UCL	3.5	95% Chebyshev(Mean, Sd) UCL	4.405
97.5% Chebyshev(Mean, Sd) UCL	5.66	99% Chebyshev(Mean, Sd) UCL	8.127

Suggested UCL to Use

95% Adjusted Gamma UCL	2.921		

The calculated UCLs are based on assumptions that the data were collected in a random and unbiased manner.

Please verify the data were collected from random locations.

If the data were collected using judgmental or other non-random methods,

then contact a statistician to correctly calculate UCLs.

When a data set follows an approximate distribution passing only one of the GOF tests,

it is suggested to use a UCL based upon a distribution passing both GOF tests in ProUCL

Note: Suggestions regarding the selection of a 95% UCL are provided to help the user to select the most appropriate 95% UCL.

Recommendations are based upon data size, data distribution, and skewness using results from simulation studies.

However, simulations results will not cover all Real World data sets; for additional insight the user may want to consult a statistician.

Nickel

General Statistics			
Total Number of Observations	14	Number of Distinct Observations	14
		Number of Missing Observations	0
Minimum	15.7	Mean	24.56
Maximum	32.4	Median	24.55
SD	5.251	Std. Error of Mean	1.403
Coefficient of Variation	0.214	Skewness	0.125
Normal GOF Test			
Shapiro Wilk Test Statistic	0.951	**Shapiro Wilk GOF Test**	
1% Shapiro Wilk Critical Value	0.825	Data appear Normal at 1% Significance Level	
Lilliefors Test Statistic	0.121	**Lilliefors GOF Test**	
1% Lilliefors Critical Value	0.263	Data appear Normal at 1% Significance Level	
Data appear Normal at 1% Significance Level			
Assuming Normal Distribution			
95% Normal UCL		**95% UCLs (Adjusted for Skewness)**	
95% Student's-t UCL	27.05	95% Adjusted-CLT UCL (Chen-1995)	26.92
		95% Modified-t UCL (Johnson-1978)	27.06
Gamma GOF Test			
A-D Test Statistic	0.255	**Anderson-Darling Gamma GOF Test**	
5% A-D Critical Value	0.734	Detected data appear Gamma Distributed at 5% Significance Level	
K-S Test Statistic	0.121	**Kolmogorov-Smirnov Gamma GOF Test**	
5% K-S Critical Value	0.228	Detected data appear Gamma Distributed at 5% Significance Level	
Detected data appear Gamma Distributed at 5% Significance Level			
Gamma Statistics			
k hat (MLE)	23.06	k star (bias corrected MLE)	18.17
Theta hat (MLE)	1.065	Theta star (bias corrected MLE)	1.352
nu hat (MLE)	645.8	nu star (bias corrected)	508.8
MLE Mean (bias corrected)	24.56	MLE Sd (bias corrected)	5.763
		Approximate Chi Square Value (0.05)	457.4
Adjusted Level of Significance	0.0312	Adjusted Chi Square Value	451
Assuming Gamma Distribution			
95% Approximate Gamma UCL	27.32	95% Adjusted Gamma UCL	27.71
Lognormal GOF Test			
Shapiro Wilk Test Statistic	0.957	**Shapiro Wilk Lognormal GOF Test**	
10% Shapiro Wilk Critical Value	0.895	Data appear Lognormal at 10% Significance Level	
Lilliefors Test Statistic	0.111	**Lilliefors Lognormal GOF Test**	
10% Lilliefors Critical Value	0.208	Data appear Lognormal at 10% Significance Level	
Data appear Lognormal at 10% Significance Level			

Lognormal Statistics			
Minimum of Logged Data	2.754	Mean of logged Data	3.179
Maximum of Logged Data	3.478	SD of logged Data	0.219
Assuming Lognormal Distribution			
95% H-UCL	27.51	90% Chebyshev (MVUE) UCL	28.91
95% Chebyshev (MVUE) UCL	30.88	97.5% Chebyshev (MVUE) UCL	33.6
99% Chebyshev (MVUE) UCL	38.96		
Nonparametric Distribution Free UCL Statistics			
Data appear to follow a Discernible Distribution			
Nonparametric Distribution Free UCLs			
95% CLT UCL	26.87	95% BCA Bootstrap UCL	26.88
95% Standard Bootstrap UCL	26.82	95% Bootstrap-t UCL	27.27
95% Hall's Bootstrap UCL	26.92	95% Percentile Bootstrap UCL	26.86
90% Chebyshev(Mean, Sd) UCL	28.77	95% Chebyshev(Mean, Sd) UCL	30.68
97.5% Chebyshev(Mean, Sd) UCL	33.33	99% Chebyshev(Mean, Sd) UCL	38.53
Suggested UCL to Use			
95% Student's-t UCL	27.05		
Note: Suggestions regarding the selection of a 95% UCL are provided to help the user to select the most appropriate 95% UCL.			
Recommendations are based upon data size, data distribution, and skewness using results from simulation studies.			
However, simulations results will not cover all Real World data sets; for additional insight the user may want to consult a statistician.			
Selenium			
General Statistics			
Total Number of Observations	14	Number of Distinct Observations	10
Number of Detects	9	Number of Non-Detects	5
Number of Distinct Detects	9	Number of Distinct Non-Detects	1
Minimum Detect	0.5	Minimum Non-Detect	0.2
Maximum Detect	5.73	Maximum Non-Detect	0.2
Variance Detects	3.849	Percent Non-Detects	35.71%
Mean Detects	2.593	SD Detects	1.962
Median Detects	1.9	CV Detects	0.757
Skewness Detects	0.592	Kurtosis Detects	-1.346
Mean of Logged Detects	0.638	SD of Logged Detects	0.898
Normal GOF Test on Detects Only			
Shapiro Wilk Test Statistic	0.885	**Shapiro Wilk GOF Test**	
1% Shapiro Wilk Critical Value	0.764	Detected Data appear Normal at 1% Significance Level	
Lilliefors Test Statistic	0.194	**Lilliefors GOF Test**	
1% Lilliefors Critical Value	0.316	Detected Data appear Normal at 1% Significance Level	
Detected Data appear Normal at 1% Significance Level			
Note GOF tests may be unreliable for small sample sizes			
Kaplan-Meier (KM) Statistics using Normal Critical Values and other Nonparametric UCLs			

	A–E	F	G–K	L
1451	KM Mean	1.739	KM Standard Error of Mean	0.531
1452	90KM SD	1.875	95% KM (BCA) UCL	2.601
1453	95% KM (t) UCL	2.68	95% KM (Percentile Bootstrap) UCL	2.614
1454	95% KM (z) UCL	2.613	95% KM Bootstrap t UCL	2.972
1455	90% KM Chebyshev UCL	3.333	95% KM Chebyshev UCL	4.055
1456	97.5% KM Chebyshev UCL	5.057	99% KM Chebyshev UCL	7.026
1457				
1458	**Gamma GOF Tests on Detected Observations Only**			
1459	A-D Test Statistic	0.312	**Anderson-Darling GOF Test**	
1460	5% A-D Critical Value	0.732	Detected data appear Gamma Distributed at 5% Significance Level	
1461	K-S Test Statistic	0.193	**Kolmogorov-Smirnov GOF**	
1462	5% K-S Critical Value	0.283	Detected data appear Gamma Distributed at 5% Significance Level	
1463	**Detected data appear Gamma Distributed at 5% Significance Level**			
1464	**Note GOF tests may be unreliable for small sample sizes**			
1465				
1466	**Gamma Statistics on Detected Data Only**			
1467	k hat (MLE)	1.738	k star (bias corrected MLE)	1.233
1468	Theta hat (MLE)	1.492	Theta star (bias corrected MLE)	2.104
1469	nu hat (MLE)	31.28	nu star (bias corrected)	22.19
1470	Mean (detects)	2.593		
1471				
1472	**Gamma ROS Statistics using Imputed Non-Detects**			
1473	GROS may not be used when data set has > 50% NDs with many tied observations at multiple DLs			
1474	GROS may not be used when kstar of detects is small such as <1.0, especially when the sample size is small (e.g., <15-20)			
1475	For such situations, GROS method may yield incorrect values of UCLs and BTVs			
1476	This is especially true when the sample size is small.			
1477	For gamma distributed detected data, BTVs and UCLs may be computed using gamma distribution on KM estimates			
1478	Minimum	0.01	Mean	1.671
1479	Maximum	5.73	Median	0.835
1480	SD	2.005	CV	1.2
1481	k hat (MLE)	0.38	k star (bias corrected MLE)	0.346
1482	Theta hat (MLE)	4.4	Theta star (bias corrected MLE)	4.829
1483	nu hat (MLE)	10.63	nu star (bias corrected)	9.687
1484	Adjusted Level of Significance (β)	0.0312		
1485	Approximate Chi Square Value (9.69, α)	3.747	Adjusted Chi Square Value (9.69, β)	3.272
1486	95% Gamma Approximate UCL	4.319	95% Gamma Adjusted UCL	4.946
1487				
1488	**Estimates of Gamma Parameters using KM Estimates**			
1489	Mean (KM)	1.739	SD (KM)	1.875
1490	Variance (KM)	3.515	SE of Mean (KM)	0.531
1491	k hat (KM)	0.86	k star (KM)	0.723
1492	nu hat (KM)	24.08	nu star (KM)	20.25
1493	theta hat (KM)	2.022	theta star (KM)	2.404
1494	80% gamma percentile (KM)	2.855	90% gamma percentile (KM)	4.331
1495	95% gamma percentile (KM)	5.848	99% gamma percentile (KM)	9.461
1496				
1497	**Gamma Kaplan-Meier (KM) Statistics**			
1498	Approximate Chi Square Value (20.25, α)	11.04	Adjusted Chi Square Value (20.25, β)	10.14
1499	95% KM Approximate Gamma UCL	3.19	95% KM Adjusted Gamma UCL	3.471
1500				

	A	B	C	D	E	F	G	H	I	J	K	L
1501	**Lognormal GOF Test on Detected Observations Only**											
1502	Shapiro Wilk Test Statistic					0.927	**Shapiro Wilk GOF Test**					
1503	10% Shapiro Wilk Critical Value					0.859	Detected Data appear Lognormal at 10% Significance Level					
1504	Lilliefors Test Statistic					0.172	**Lilliefors GOF Test**					
1505	10% Lilliefors Critical Value					0.252	Detected Data appear Lognormal at 10% Significance Level					
1506	**Detected Data appear Lognormal at 10% Significance Level**											
1507	**Note GOF tests may be unreliable for small sample sizes**											
1508												
1509	**Lognormal ROS Statistics Using Imputed Non-Detects**											
1510	Mean in Original Scale					1.744	Mean in Log Scale					-0.187
1511	SD in Original Scale					1.942	SD in Log Scale					1.39
1512	95% t UCL (assumes normality of ROS data)					2.663	95% Percentile Bootstrap UCL					2.571
1513	95% BCA Bootstrap UCL					2.64	95% Bootstrap t UCL					3.029
1514	95% H-UCL (Log ROS)					8.35						
1515												
1516	**Statistics using KM estimates on Logged Data and Assuming Lognormal Distribution**											
1517	KM Mean (logged)					-0.164	KM Geo Mean					0.848
1518	KM SD (logged)					1.273	95% Critical H Value (KM-Log)					3.268
1519	KM Standard Error of Mean (logged)					0.361	95% H-UCL (KM -Log)					6.047
1520	KM SD (logged)					1.273	95% Critical H Value (KM-Log)					3.268
1521	KM Standard Error of Mean (logged)					0.361						
1522												
1523	**DL/2 Statistics**											
1524	**DL/2 Normal**						**DL/2 Log-Transformed**					
1525	Mean in Original Scale					1.703	Mean in Log Scale					-0.412
1526	SD in Original Scale					1.976	SD in Log Scale					1.623
1527	95% t UCL (Assumes normality)					2.638	95% H-Stat UCL					14.49
1528	**DL/2 is not a recommended method, provided for comparisons and historical reasons**											
1529												
1530	**Nonparametric Distribution Free UCL Statistics**											
1531	**Detected Data appear Normal Distributed at 1% Significance Level**											
1532												
1533	**Suggested UCL to Use**											
1534	95% KM (t) UCL					2.68						
1535												
1536	Note: Suggestions regarding the selection of a 95% UCL are provided to help the user to select the most appropriate 95% UCL.											
1537	Recommendations are based upon data size, data distribution, and skewness using results from simulation studies.											
1538	However, simulations results will not cover all Real World data sets; for additional insight the user may want to consult a statistician.											
1539												
1540	**Silver**											
1541												
1542	**General Statistics**											
1543	Total Number of Observations					14	Number of Distinct Observations					11
1544	Number of Detects					11	Number of Non-Detects					3
1545	Number of Distinct Detects					10	Number of Distinct Non-Detects					1
1546	Minimum Detect					0.11	Minimum Non-Detect					0.1
1547	Maximum Detect					2.45	Maximum Non-Detect					0.1
1548	Variance Detects					0.743	Percent Non-Detects					21.43%
1549	Mean Detects					0.962	SD Detects					0.862
1550	Median Detects					0.8	CV Detects					0.896

Skewness Detects	0.746	Kurtosis Detects	-0.839
Mean of Logged Detects	-0.542	SD of Logged Detects	1.159
Normal GOF Test on Detects Only			
Shapiro Wilk Test Statistic	0.875	**Shapiro Wilk GOF Test**	
1% Shapiro Wilk Critical Value	0.792	Detected Data appear Normal at 1% Significance Level	
Lilliefors Test Statistic	0.17	**Lilliefors GOF Test**	
1% Lilliefors Critical Value	0.291	Detected Data appear Normal at 1% Significance Level	
Detected Data appear Normal at 1% Significance Level			
Kaplan-Meier (KM) Statistics using Normal Critical Values and other Nonparametric UCLs			
KM Mean	0.777	KM Standard Error of Mean	0.227
90KM SD	0.81	95% KM (BCA) UCL	1.146
95% KM (t) UCL	1.179	95% KM (Percentile Bootstrap) UCL	1.151
95% KM (z) UCL	1.151	95% KM Bootstrap t UCL	1.307
90% KM Chebyshev UCL	1.458	95% KM Chebyshev UCL	1.767
97.5% KM Chebyshev UCL	2.195	99% KM Chebyshev UCL	3.036
Gamma GOF Tests on Detected Observations Only			
A-D Test Statistic	0.332	**Anderson-Darling GOF Test**	
5% A-D Critical Value	0.749	Detected data appear Gamma Distributed at 5% Significance Level	
K-S Test Statistic	0.169	**Kolmogorov-Smirnov GOF**	
5% K-S Critical Value	0.262	Detected data appear Gamma Distributed at 5% Significance Level	
Detected data appear Gamma Distributed at 5% Significance Level			
Gamma Statistics on Detected Data Only			
k hat (MLE)	1.132	k star (bias corrected MLE)	0.884
Theta hat (MLE)	0.85	Theta star (bias corrected MLE)	1.089
nu hat (MLE)	24.89	nu star (bias corrected)	19.44
Mean (detects)	0.962		
Gamma ROS Statistics using Imputed Non-Detects			
GROS may not be used when data set has > 50% NDs with many tied observations at multiple DLs			
GROS may not be used when kstar of detects is small such as <1.0, especially when the sample size is small (e.g., <15-20)			
For such situations, GROS method may yield incorrect values of UCLs and BTVs			
This is especially true when the sample size is small.			
For gamma distributed detected data, BTVs and UCLs may be computed using gamma distribution on KM estimates			
Minimum	0.01	Mean	0.758
Maximum	2.45	Median	0.355
SD	0.858	CV	1.132
k hat (MLE)	0.551	k star (bias corrected MLE)	0.481
Theta hat (MLE)	1.375	Theta star (bias corrected MLE)	1.577
nu hat (MLE)	15.43	nu star (bias corrected)	13.46
Adjusted Level of Significance (β)	0.0312		
Approximate Chi Square Value (13.46, α)	6.202	Adjusted Chi Square Value (13.46, β)	5.561
95% Gamma Approximate UCL	1.644	95% Gamma Adjusted UCL	1.834
Estimates of Gamma Parameters using KM Estimates			
Mean (KM)	0.777	SD (KM)	0.81
Variance (KM)	0.656	SE of Mean (KM)	0.227

1601	k hat (KM)	0.921	k star (KM)	0.771
1602	nu hat (KM)	25.78	nu star (KM)	21.59
1603	theta hat (KM)	0.844	theta star (KM)	1.008
1604	80% gamma percentile (KM)	1.272	90% gamma percentile (KM)	1.906
1605	95% gamma percentile (KM)	2.555	99% gamma percentile (KM)	4.09
1606				
1607	**Gamma Kaplan-Meier (KM) Statistics**			
1608	Approximate Chi Square Value (21.59, α)	12.03	Adjusted Chi Square Value (21.59, β)	11.09
1609	95% KM Approximate Gamma UCL	1.395	95% KM Adjusted Gamma UCL	1.512
1610				
1611	**Lognormal GOF Test on Detected Observations Only**			
1612	Shapiro Wilk Test Statistic	0.913	**Shapiro Wilk GOF Test**	
1613	10% Shapiro Wilk Critical Value	0.876	Detected Data appear Lognormal at 10% Significance Level	
1614	Lilliefors Test Statistic	0.154	**Lilliefors GOF Test**	
1615	10% Lilliefors Critical Value	0.231	Detected Data appear Lognormal at 10% Significance Level	
1616	**Detected Data appear Lognormal at 10% Significance Level**			
1617				
1618	**Lognormal ROS Statistics Using Imputed Non-Detects**			
1619	Mean in Original Scale	0.764	Mean in Log Scale	-1.134
1620	SD in Original Scale	0.852	SD in Log Scale	1.568
1621	95% t UCL (assumes normality of ROS data)	1.168	95% Percentile Bootstrap UCL	1.139
1622	95% BCA Bootstrap UCL	1.157	95% Bootstrap t UCL	1.331
1623	95% H-UCL (Log ROS)	5.803		
1624				
1625	**Statistics using KM estimates on Logged Data and Assuming Lognormal Distribution**			
1626	KM Mean (logged)	-0.919	KM Geo Mean	0.399
1627	KM SD (logged)	1.217	95% Critical H Value (KM-Log)	3.167
1628	KM Standard Error of Mean (logged)	0.341	95% H-UCL (KM -Log)	2.438
1629	KM SD (logged)	1.217	95% Critical H Value (KM-Log)	3.167
1630	KM Standard Error of Mean (logged)	0.341		
1631				
1632	**DL/2 Statistics**			
1633	**DL/2 Normal**		**DL/2 Log-Transformed**	
1634	Mean in Original Scale	0.766	Mean in Log Scale	-1.068
1635	SD in Original Scale	0.85	SD in Log Scale	1.458
1636	95% t UCL (Assumes normality)	1.169	95% H-Stat UCL	4.286
1637	**DL/2 is not a recommended method, provided for comparisons and historical reasons**			
1638				
1639	**Nonparametric Distribution Free UCL Statistics**			
1640	**Detected Data appear Normal Distributed at 1% Significance Level**			
1641				
1642	**Suggested UCL to Use**			
1643	95% KM (t) UCL	1.179		
1644				
1645	Note: Suggestions regarding the selection of a 95% UCL are provided to help the user to select the most appropriate 95% UCL.			
1646	Recommendations are based upon data size, data distribution, and skewness using results from simulation studies.			
1647	However, simulations results will not cover all Real World data sets; for additional insight the user may want to consult a statistician.			
1648				
1649				
1650	**Strontium**			

1651				
1652	General Statistics			
1653	Total Number of Observations	14	Number of Distinct Observations	13
1654			Number of Missing Observations	0
1655	Minimum	10.4	Mean	17.86
1656	Maximum	34	Median	18.05
1657	SD	6.306	Std. Error of Mean	1.685
1658	Coefficient of Variation	0.353	Skewness	1.223
1659				
1660	Normal GOF Test			
1661	Shapiro Wilk Test Statistic	0.887	Shapiro Wilk GOF Test	
1662	1% Shapiro Wilk Critical Value	0.825	Data appear Normal at 1% Significance Level	
1663	Lilliefors Test Statistic	0.196	Lilliefors GOF Test	
1664	1% Lilliefors Critical Value	0.263	Data appear Normal at 1% Significance Level	
1665	Data appear Normal at 1% Significance Level			
1666				
1667	Assuming Normal Distribution			
1668	95% Normal UCL		95% UCLs (Adjusted for Skewness)	
1669	95% Student's-t UCL	20.85	95% Adjusted-CLT UCL (Chen-1995)	21.22
1670			95% Modified-t UCL (Johnson-1978)	20.94
1671				
1672	Gamma GOF Test			
1673	A-D Test Statistic	0.423	Anderson-Darling Gamma GOF Test	
1674	5% A-D Critical Value	0.735	Detected data appear Gamma Distributed at 5% Significance Level	
1675	K-S Test Statistic	0.168	Kolmogorov-Smirnov Gamma GOF Test	
1676	5% K-S Critical Value	0.229	Detected data appear Gamma Distributed at 5% Significance Level	
1677	Detected data appear Gamma Distributed at 5% Significance Level			
1678				
1679	Gamma Statistics			
1680	k hat (MLE)	9.598	k star (bias corrected MLE)	7.589
1681	Theta hat (MLE)	1.861	Theta star (bias corrected MLE)	2.354
1682	nu hat (MLE)	268.7	nu star (bias corrected)	212.5
1683	MLE Mean (bias corrected)	17.86	MLE Sd (bias corrected)	6.485
1684			Approximate Chi Square Value (0.05)	179.8
1685	Adjusted Level of Significance	0.0312	Adjusted Chi Square Value	175.8
1686				
1687	Assuming Gamma Distribution			
1688	95% Approximate Gamma UCL	21.12	95% Adjusted Gamma UCL	21.6
1689				
1690	Lognormal GOF Test			
1691	Shapiro Wilk Test Statistic	0.943	Shapiro Wilk Lognormal GOF Test	
1692	10% Shapiro Wilk Critical Value	0.895	Data appear Lognormal at 10% Significance Level	
1693	Lilliefors Test Statistic	0.188	Lilliefors Lognormal GOF Test	
1694	10% Lilliefors Critical Value	0.208	Data appear Lognormal at 10% Significance Level	
1695	Data appear Lognormal at 10% Significance Level			
1696				
1697	Lognormal Statistics			
1698	Minimum of Logged Data	2.342	Mean of logged Data	2.83
1699	Maximum of Logged Data	3.526	SD of logged Data	0.334
1700				

Assuming Lognormal Distribution

95% H-UCL	21.4	90% Chebyshev (MVUE) UCL	22.67
95% Chebyshev (MVUE) UCL	24.86	97.5% Chebyshev (MVUE) UCL	27.91
99% Chebyshev (MVUE) UCL	33.88		

Nonparametric Distribution Free UCL Statistics

Data appear to follow a Discernible Distribution

Nonparametric Distribution Free UCLs

95% CLT UCL	20.64	95% BCA Bootstrap UCL	21.14
95% Standard Bootstrap UCL	20.57	95% Bootstrap-t UCL	21.65
95% Hall's Bootstrap UCL	23.45	95% Percentile Bootstrap UCL	20.65
90% Chebyshev(Mean, Sd) UCL	22.92	95% Chebyshev(Mean, Sd) UCL	25.21
97.5% Chebyshev(Mean, Sd) UCL	28.39	99% Chebyshev(Mean, Sd) UCL	34.63

Suggested UCL to Use

95% Student's-t UCL	20.85		

Note: Suggestions regarding the selection of a 95% UCL are provided to help the user to select the most appropriate 95% UCL.

Recommendations are based upon data size, data distribution, and skewness using results from simulation studies.

However, simulations results will not cover all Real World data sets; for additional insight the user may want to consult a statistician.

Thallium

General Statistics

Total Number of Observations	14	Number of Distinct Observations	1
Number of Detects	0	Number of Non-Detects	14
Number of Distinct Detects	0	Number of Distinct Non-Detects	1

Warning: All observations are Non-Detects (NDs), therefore all statistics and estimates should also be NDs!

Specifically, sample mean, UCLs, UPLs, and other statistics are also NDs lying below the largest detection limit!

The Project Team may decide to use alternative site specific values to estimate environmental parameters (e.g., EPC, BTV).

The data set for variable Thallium was not processed!

Tin

General Statistics

Total Number of Observations	14	Number of Distinct Observations	13
		Number of Missing Observations	0
Minimum	0.31	Mean	0.784
Maximum	2.6	Median	0.51
SD	0.654	Std. Error of Mean	0.175
Coefficient of Variation	0.834	Skewness	2.047

Normal GOF Test

Shapiro Wilk Test Statistic	0.715	**Shapiro Wilk GOF Test**
1% Shapiro Wilk Critical Value	0.825	Data Not Normal at 1% Significance Level

	A–E	F	G–K	L
1751	Lilliefors Test Statistic	0.313	**Lilliefors GOF Test**	
1752	1% Lilliefors Critical Value	0.263	Data Not Normal at 1% Significance Level	
1753	**Data Not Normal at 1% Significance Level**			
1754				
1755	**Assuming Normal Distribution**			
1756	**95% Normal UCL**		**95% UCLs (Adjusted for Skewness)**	
1757	95% Student's-t UCL	1.094	95% Adjusted-CLT UCL (Chen-1995)	1.174
1758			95% Modified-t UCL (Johnson-1978)	1.11
1759				
1760	**Gamma GOF Test**			
1761	A-D Test Statistic	0.995	**Anderson-Darling Gamma GOF Test**	
1762	5% A-D Critical Value	0.745	Data Not Gamma Distributed at 5% Significance Level	
1763	K-S Test Statistic	0.234	**Kolmogorov-Smirnov Gamma GOF Test**	
1764	5% K-S Critical Value	0.231	Data Not Gamma Distributed at 5% Significance Level	
1765	**Data Not Gamma Distributed at 5% Significance Level**			
1766				
1767	**Gamma Statistics**			
1768	k hat (MLE)	2.367	k star (bias corrected MLE)	1.908
1769	Theta hat (MLE)	0.331	Theta star (bias corrected MLE)	0.411
1770	nu hat (MLE)	66.29	nu star (bias corrected)	53.41
1771	MLE Mean (bias corrected)	0.784	MLE Sd (bias corrected)	0.568
1772			Approximate Chi Square Value (0.05)	37.62
1773	Adjusted Level of Significance	0.0312	Adjusted Chi Square Value	35.87
1774				
1775	**Assuming Gamma Distribution**			
1776	95% Approximate Gamma UCL	1.113	95% Adjusted Gamma UCL	1.168
1777				
1778	**Lognormal GOF Test**			
1779	Shapiro Wilk Test Statistic	0.883	**Shapiro Wilk Lognormal GOF Test**	
1780	10% Shapiro Wilk Critical Value	0.895	Data Not Lognormal at 10% Significance Level	
1781	Lilliefors Test Statistic	0.183	**Lilliefors Lognormal GOF Test**	
1782	10% Lilliefors Critical Value	0.208	Data appear Lognormal at 10% Significance Level	
1783	**Data appear Approximate Lognormal at 10% Significance Level**			
1784				
1785	**Lognormal Statistics**			
1786	Minimum of Logged Data	-1.171	Mean of logged Data	-0.469
1787	Maximum of Logged Data	0.956	SD of logged Data	0.642
1788				
1789	**Assuming Lognormal Distribution**			
1790	95% H-UCL	1.151	90% Chebyshev (MVUE) UCL	1.161
1791	95% Chebyshev (MVUE) UCL	1.345	97.5% Chebyshev (MVUE) UCL	1.6
1792	99% Chebyshev (MVUE) UCL	2.101		
1793				
1794	**Nonparametric Distribution Free UCL Statistics**			
1795	**Data appear to follow a Discernible Distribution**			
1796				
1797	**Nonparametric Distribution Free UCLs**			
1798	95% CLT UCL	1.072	95% BCA Bootstrap UCL	1.157
1799	95% Standard Bootstrap UCL	1.061	95% Bootstrap-t UCL	1.419
1800	95% Hall's Bootstrap UCL	1.318	95% Percentile Bootstrap UCL	1.086

	A–E	F	G–K	L
1801	90% Chebyshev(Mean, Sd) UCL	1.309	95% Chebyshev(Mean, Sd) UCL	1.547
1802	97.5% Chebyshev(Mean, Sd) UCL	1.877	99% Chebyshev(Mean, Sd) UCL	2.525
1803				
1804	**Suggested UCL to Use**			
1805	95% H-UCL	1.151		
1806				
1807	Note: Suggestions regarding the selection of a 95% UCL are provided to help the user to select the most appropriate 95% UCL.			
1808	Recommendations are based upon data size, data distribution, and skewness using results from simulation studies.			
1809	However, simulations results will not cover all Real World data sets; for additional insight the user may want to consult a statistician.			
1810				
1811	**Tungsten**			
1812				
1813	**General Statistics**			
1814	Total Number of Observations	13	Number of Distinct Observations	11
1815			Number of Missing Observations	1
1816	Number of Detects	10	Number of Non-Detects	3
1817	Number of Distinct Detects	10	Number of Distinct Non-Detects	1
1818	Minimum Detect	0.22	Minimum Non-Detect	0.2
1819	Maximum Detect	1.68	Maximum Non-Detect	0.2
1820	Variance Detects	0.187	Percent Non-Detects	23.08%
1821	Mean Detects	0.608	SD Detects	0.433
1822	Median Detects	0.485	CV Detects	0.712
1823	Skewness Detects	1.881	Kurtosis Detects	4.212
1824	Mean of Logged Detects	-0.683	SD of Logged Detects	0.623
1825				
1826	**Normal GOF Test on Detects Only**			
1827	Shapiro Wilk Test Statistic	0.808	**Shapiro Wilk GOF Test**	
1828	1% Shapiro Wilk Critical Value	0.781	Detected Data appear Normal at 1% Significance Level	
1829	Lilliefors Test Statistic	0.204	**Lilliefors GOF Test**	
1830	1% Lilliefors Critical Value	0.304	Detected Data appear Normal at 1% Significance Level	
1831	**Detected Data appear Normal at 1% Significance Level**			
1832				
1833	**Kaplan-Meier (KM) Statistics using Normal Critical Values and other Nonparametric UCLs**			
1834	KM Mean	0.514	KM Standard Error of Mean	0.117
1835	90KM SD	0.399	95% KM (BCA) UCL	0.704
1836	95% KM (t) UCL	0.722	95% KM (Percentile Bootstrap) UCL	0.709
1837	95% KM (z) UCL	0.706	95% KM Bootstrap t UCL	0.871
1838	90% KM Chebyshev UCL	0.864	95% KM Chebyshev UCL	1.022
1839	97.5% KM Chebyshev UCL	1.242	99% KM Chebyshev UCL	1.675
1840				
1841	**Gamma GOF Tests on Detected Observations Only**			
1842	A-D Test Statistic	0.299	**Anderson-Darling GOF Test**	
1843	5% A-D Critical Value	0.733	Detected data appear Gamma Distributed at 5% Significance Level	
1844	K-S Test Statistic	0.127	**Kolmogorov-Smirnov GOF**	
1845	5% K-S Critical Value	0.269	Detected data appear Gamma Distributed at 5% Significance Level	
1846	**Detected data appear Gamma Distributed at 5% Significance Level**			
1847				
1848	**Gamma Statistics on Detected Data Only**			
1849	k hat (MLE)	2.849	k star (bias corrected MLE)	2.061
1850	Theta hat (MLE)	0.213	Theta star (bias corrected MLE)	0.295

	A	B	C	D	E	F	G	H	I	J	K	L
1851					nu hat (MLE)	56.98					nu star (bias corrected)	41.22
1852					Mean (detects)	0.608						
1853												
1854	**Gamma ROS Statistics using Imputed Non-Detects**											
1855	GROS may not be used when data set has > 50% NDs with many tied observations at multiple DLs											
1856	GROS may not be used when kstar of detects is small such as <1.0, especially when the sample size is small (e.g., <15-20)											
1857	For such situations, GROS method may yield incorrect values of UCLs and BTVs											
1858	This is especially true when the sample size is small.											
1859	For gamma distributed detected data, BTVs and UCLs may be computed using gamma distribution on KM estimates											
1860					Minimum	0.01					Mean	0.47
1861					Maximum	1.68					Median	0.38
1862					SD	0.457					CV	0.973
1863					k hat (MLE)	0.722					k star (bias corrected MLE)	0.607
1864					Theta hat (MLE)	0.651					Theta star (bias corrected MLE)	0.775
1865					nu hat (MLE)	18.78					nu star (bias corrected)	15.78
1866					Adjusted Level of Significance (β)	0.0301						
1867					Approximate Chi Square Value (15.78, α)	7.805					Adjusted Chi Square Value (15.78, β)	7.019
1868					95% Gamma Approximate UCL	0.95					95% Gamma Adjusted UCL	1.056
1869												
1870	**Estimates of Gamma Parameters using KM Estimates**											
1871					Mean (KM)	0.514					SD (KM)	0.399
1872					Variance (KM)	0.159					SE of Mean (KM)	0.117
1873					k hat (KM)	1.658					k star (KM)	1.327
1874					nu hat (KM)	43.12					nu star (KM)	34.5
1875					theta hat (KM)	0.31					theta star (KM)	0.387
1876					80% gamma percentile (KM)	0.805					90% gamma percentile (KM)	1.103
1877					95% gamma percentile (KM)	1.395					99% gamma percentile (KM)	2.059
1878												
1879	**Gamma Kaplan-Meier (KM) Statistics**											
1880					Approximate Chi Square Value (34.50, α)	22.06					Adjusted Chi Square Value (34.50, β)	20.66
1881					95% KM Approximate Gamma UCL	0.803					95% KM Adjusted Gamma UCL	0.858
1882												
1883	**Lognormal GOF Test on Detected Observations Only**											
1884					Shapiro Wilk Test Statistic	0.963	**Shapiro Wilk GOF Test**					
1885					10% Shapiro Wilk Critical Value	0.869	Detected Data appear Lognormal at 10% Significance Level					
1886					Lilliefors Test Statistic	0.113	**Lilliefors GOF Test**					
1887					10% Lilliefors Critical Value	0.241	Detected Data appear Lognormal at 10% Significance Level					
1888	**Detected Data appear Lognormal at 10% Significance Level**											
1889												
1890	**Lognormal ROS Statistics Using Imputed Non-Detects**											
1891					Mean in Original Scale	0.493					Mean in Log Scale	-1.038
1892					SD in Original Scale	0.434					SD in Log Scale	0.873
1893					95% t UCL (assumes normality of ROS data)	0.708					95% Percentile Bootstrap UCL	0.693
1894					95% BCA Bootstrap UCL	0.757					95% Bootstrap t UCL	0.818
1895					95% H-UCL (Log ROS)	1.007						
1896												
1897	**Statistics using KM estimates on Logged Data and Assuming Lognormal Distribution**											
1898					KM Mean (logged)	-0.897					KM Geo Mean	0.408
1899					KM SD (logged)	0.649					95% Critical H Value (KM-Log)	2.3
1900					KM Standard Error of Mean (logged)	0.19					95% H-UCL (KM -Log)	0.774

	A–E	F	G–K	L
1901	KM SD (logged)	0.649	95% Critical H Value (KM-Log)	2.3
1902	KM Standard Error of Mean (logged)	0.19		
1903				
1904	**DL/2 Statistics**			
1905	**DL/2 Normal**		**DL/2 Log-Transformed**	
1906	Mean in Original Scale	0.491	Mean in Log Scale	-1.057
1907	SD in Original Scale	0.436	SD in Log Scale	0.892
1908	95% t UCL (Assumes normality)	0.706	95% H-Stat UCL	1.027
1909	**DL/2 is not a recommended method, provided for comparisons and historical reasons**			
1910				
1911	**Nonparametric Distribution Free UCL Statistics**			
1912	**Detected Data appear Normal Distributed at 1% Significance Level**			
1913				
1914	**Suggested UCL to Use**			
1915	95% KM (t) UCL	0.722		
1916				
1917	Note: Suggestions regarding the selection of a 95% UCL are provided to help the user to select the most appropriate 95% UCL.			
1918	Recommendations are based upon data size, data distribution, and skewness using results from simulation studies.			
1919	However, simulations results will not cover all Real World data sets; for additional insight the user may want to consult a statistician.			
1920				
1921	**Uranium**			
1922				
1923	**General Statistics**			
1924	Total Number of Observations	14	Number of Distinct Observations	14
1925	Number of Detects	12	Number of Non-Detects	2
1926	Number of Distinct Detects	12	Number of Distinct Non-Detects	2
1927	Minimum Detect	0.052	Minimum Non-Detect	0.05
1928	Maximum Detect	0.348	Maximum Non-Detect	0.1
1929	Variance Detects	0.0102	Percent Non-Detects	14.29%
1930	Mean Detects	0.206	SD Detects	0.101
1931	Median Detects	0.197	CV Detects	0.49
1932	Skewness Detects	-0.0412	Kurtosis Detects	-1.464
1933	Mean of Logged Detects	-1.725	SD of Logged Detects	0.603
1934				
1935	**Normal GOF Test on Detects Only**			
1936	Shapiro Wilk Test Statistic	0.926	**Shapiro Wilk GOF Test**	
1937	1% Shapiro Wilk Critical Value	0.805	Detected Data appear Normal at 1% Significance Level	
1938	Lilliefors Test Statistic	0.213	**Lilliefors GOF Test**	
1939	1% Lilliefors Critical Value	0.281	Detected Data appear Normal at 1% Significance Level	
1940	**Detected Data appear Normal at 1% Significance Level**			
1941				
1942	**Kaplan-Meier (KM) Statistics using Normal Critical Values and other Nonparametric UCLs**			
1943	KM Mean	0.184	KM Standard Error of Mean	0.029
1944	90KM SD	0.104	95% KM (BCA) UCL	0.23
1945	95% KM (t) UCL	0.235	95% KM (Percentile Bootstrap) UCL	0.231
1946	95% KM (z) UCL	0.232	95% KM Bootstrap t UCL	0.236
1947	90% KM Chebyshev UCL	0.271	95% KM Chebyshev UCL	0.31
1948	97.5% KM Chebyshev UCL	0.365	99% KM Chebyshev UCL	0.472
1949				
1950	**Gamma GOF Tests on Detected Observations Only**			

	A–E	F	G–K	L
1951	A-D Test Statistic	0.462	**Anderson-Darling GOF Test**	
1952	5% A-D Critical Value	0.737	Detected data appear Gamma Distributed at 5% Significance Level	
1953	K-S Test Statistic	0.195	**Kolmogorov-Smirnov GOF**	
1954	5% K-S Critical Value	0.247	Detected data appear Gamma Distributed at 5% Significance Level	
1955	**Detected data appear Gamma Distributed at 5% Significance Level**			
1956				
1957	**Gamma Statistics on Detected Data Only**			
1958	k hat (MLE)	3.662	k star (bias corrected MLE)	2.802
1959	Theta hat (MLE)	0.0561	Theta star (bias corrected MLE)	0.0733
1960	nu hat (MLE)	87.88	nu star (bias corrected)	67.25
1961	Mean (detects)	0.206		
1962				
1963	**Gamma ROS Statistics using Imputed Non-Detects**			
1964	GROS may not be used when data set has > 50% NDs with many tied observations at multiple DLs			
1965	GROS may not be used when kstar of detects is small such as <1.0, especially when the sample size is small (e.g., <15-20)			
1966	For such situations, GROS method may yield incorrect values of UCLs and BTVs			
1967	This is especially true when the sample size is small.			
1968	For gamma distributed detected data, BTVs and UCLs may be computed using gamma distribution on KM estimates			
1969	Minimum	0.0282	Mean	0.183
1970	Maximum	0.348	Median	0.149
1971	SD	0.109	CV	0.597
1972	k hat (MLE)	2.347	k star (bias corrected MLE)	1.892
1973	Theta hat (MLE)	0.0779	Theta star (bias corrected MLE)	0.0967
1974	nu hat (MLE)	65.72	nu star (bias corrected)	52.97
1975	Adjusted Level of Significance (β)	0.0312		
1976	Approximate Chi Square Value (52.97, α)	37.25	Adjusted Chi Square Value (52.97, β)	35.51
1977	95% Gamma Approximate UCL	0.26	95% Gamma Adjusted UCL	0.273
1978				
1979	**Estimates of Gamma Parameters using KM Estimates**			
1980	Mean (KM)	0.184	SD (KM)	0.104
1981	Variance (KM)	0.0108	SE of Mean (KM)	0.029
1982	k hat (KM)	3.143	k star (KM)	2.517
1983	nu hat (KM)	88	nu star (KM)	70.48
1984	theta hat (KM)	0.0585	theta star (KM)	0.0731
1985	80% gamma percentile (KM)	0.268	90% gamma percentile (KM)	0.339
1986	95% gamma percentile (KM)	0.407	99% gamma percentile (KM)	0.554
1987				
1988	**Gamma Kaplan-Meier (KM) Statistics**			
1989	Approximate Chi Square Value (70.48, α)	52.15	Adjusted Chi Square Value (70.48, β)	50.06
1990	95% KM Approximate Gamma UCL	0.249	95% KM Adjusted Gamma UCL	0.259
1991				
1992	**Lognormal GOF Test on Detected Observations Only**			
1993	Shapiro Wilk Test Statistic	0.9	**Shapiro Wilk GOF Test**	
1994	10% Shapiro Wilk Critical Value	0.883	Detected Data appear Lognormal at 10% Significance Level	
1995	Lilliefors Test Statistic	0.199	**Lilliefors GOF Test**	
1996	10% Lilliefors Critical Value	0.223	Detected Data appear Lognormal at 10% Significance Level	
1997	**Detected Data appear Lognormal at 10% Significance Level**			
1998				
1999	**Lognormal ROS Statistics Using Imputed Non-Detects**			
2000	Mean in Original Scale	0.184	Mean in Log Scale	-1.897

	A–E	F	G–K	L
2001	SD in Original Scale	0.108	SD in Log Scale	0.712
2002	95% t UCL (assumes normality of ROS data)	0.235	95% Percentile Bootstrap UCL	0.231
2003	95% BCA Bootstrap UCL	0.229	95% Bootstrap t UCL	0.239
2004	95% H-UCL (Log ROS)	0.307		
2005				
2006	**Statistics using KM estimates on Logged Data and Assuming Lognormal Distribution**			
2007	KM Mean (logged)	-1.895	KM Geo Mean	0.15
2008	KM SD (logged)	0.68	95% Critical H Value (KM-Log)	2.316
2009	KM Standard Error of Mean (logged)	0.191	95% H-UCL (KM -Log)	0.293
2010	KM SD (logged)	0.68	95% Critical H Value (KM-Log)	2.316
2011	KM Standard Error of Mean (logged)	0.191		
2012				
2013	**DL/2 Statistics**			
2014	**DL/2 Normal**		**DL/2 Log-Transformed**	
2015	Mean in Original Scale	0.182	Mean in Log Scale	-1.956
2016	SD in Original Scale	0.111	SD in Log Scale	0.819
2017	95% t UCL (Assumes normality)	0.234	95% H-Stat UCL	0.349
2018	**DL/2 is not a recommended method, provided for comparisons and historical reasons**			
2019				
2020	**Nonparametric Distribution Free UCL Statistics**			
2021	**Detected Data appear Normal Distributed at 1% Significance Level**			
2022				
2023	**Suggested UCL to Use**			
2024	95% KM (t) UCL	0.235		
2025				
2026	Note: Suggestions regarding the selection of a 95% UCL are provided to help the user to select the most appropriate 95% UCL.			
2027	Recommendations are based upon data size, data distribution, and skewness using results from simulation studies.			
2028	However, simulations results will not cover all Real World data sets; for additional insight the user may want to consult a statistician.			
2029				
2030				
2031	**Vanadium**			
2032				
2033	**General Statistics**			
2034	Total Number of Observations	14	Number of Distinct Observations	14
2035			Number of Missing Observations	0
2036	Minimum	25.2	Mean	62.52
2037	Maximum	83.2	Median	65.35
2038	SD	15.08	Std. Error of Mean	4.032
2039	Coefficient of Variation	0.241	Skewness	-1.342
2040				
2041	**Normal GOF Test**			
2042	Shapiro Wilk Test Statistic	0.887	**Shapiro Wilk GOF Test**	
2043	1% Shapiro Wilk Critical Value	0.825	Data appear Normal at 1% Significance Level	
2044	Lilliefors Test Statistic	0.201	**Lilliefors GOF Test**	
2045	1% Lilliefors Critical Value	0.263	Data appear Normal at 1% Significance Level	
2046	**Data appear Normal at 1% Significance Level**			
2047				
2048	**Assuming Normal Distribution**			
2049	**95% Normal UCL**		**95% UCLs (Adjusted for Skewness)**	
2050	95% Student's-t UCL	69.66	95% Adjusted-CLT UCL (Chen-1995)	67.61

	A	B	C	D	E	F	G	H	I	J	K	L
2051											95% Modified-t UCL (Johnson-1978)	69.42
2052												
2053						**Gamma GOF Test**						
2054					A-D Test Statistic	1.034			**Anderson-Darling Gamma GOF Test**			
2055					5% A-D Critical Value	0.734			Data Not Gamma Distributed at 5% Significance Level			
2056					K-S Test Statistic	0.245			**Kolmogorov-Smirnov Gamma GOF Test**			
2057					5% K-S Critical Value	0.229			Data Not Gamma Distributed at 5% Significance Level			
2058						**Data Not Gamma Distributed at 5% Significance Level**						
2059												
2060						**Gamma Statistics**						
2061					k hat (MLE)	13.58					k star (bias corrected MLE)	10.72
2062					Theta hat (MLE)	4.603					Theta star (bias corrected MLE)	5.833
2063					nu hat (MLE)	380.3					nu star (bias corrected)	300.1
2064					MLE Mean (bias corrected)	62.52					MLE Sd (bias corrected)	19.1
2065											Approximate Chi Square Value (0.05)	261
2066					Adjusted Level of Significance	0.0312					Adjusted Chi Square Value	256.2
2067												
2068						**Assuming Gamma Distribution**						
2069					95% Approximate Gamma UCL	71.89					95% Adjusted Gamma UCL	73.25
2070												
2071						**Lognormal GOF Test**						
2072					Shapiro Wilk Test Statistic	0.773			**Shapiro Wilk Lognormal GOF Test**			
2073					10% Shapiro Wilk Critical Value	0.895			Data Not Lognormal at 10% Significance Level			
2074					Lilliefors Test Statistic	0.266			**Lilliefors Lognormal GOF Test**			
2075					10% Lilliefors Critical Value	0.208			Data Not Lognormal at 10% Significance Level			
2076						**Data Not Lognormal at 10% Significance Level**						
2077												
2078						**Lognormal Statistics**						
2079					Minimum of Logged Data	3.227					Mean of logged Data	4.098
2080					Maximum of Logged Data	4.421					SD of logged Data	0.309
2081												
2082						**Assuming Lognormal Distribution**						
2083					95% H-UCL	74.39					90% Chebyshev (MVUE) UCL	78.76
2084					95% Chebyshev (MVUE) UCL	85.91					97.5% Chebyshev (MVUE) UCL	95.85
2085					99% Chebyshev (MVUE) UCL	115.4						
2086												
2087						**Nonparametric Distribution Free UCL Statistics**						
2088						**Data appear to follow a Discernible Distribution**						
2089												
2090						**Nonparametric Distribution Free UCLs**						
2091					95% CLT UCL	69.15					95% BCA Bootstrap UCL	67.66
2092					95% Standard Bootstrap UCL	69.03					95% Bootstrap-t UCL	68.5
2093					95% Hall's Bootstrap UCL	68.09					95% Percentile Bootstrap UCL	68.61
2094					90% Chebyshev(Mean, Sd) UCL	74.62					95% Chebyshev(Mean, Sd) UCL	80.09
2095					97.5% Chebyshev(Mean, Sd) UCL	87.7					99% Chebyshev(Mean, Sd) UCL	102.6
2096												
2097						**Suggested UCL to Use**						
2098					95% Student's-t UCL	69.66						
2099												
2100						Note: Suggestions regarding the selection of a 95% UCL are provided to help the user to select the most appropriate 95% UCL.						

Recommendations are based upon data size, data distribution, and skewness using results from simulation studies.

However, simulations results will not cover all Real World data sets; for additional insight the user may want to consult a statistician.

Note: For highly negatively-skewed data, confidence limits (e.g., Chen, Johnson, Lognormal, and Gamma) may not be reliable. Chen's and Johnson's methods provide adjustments for positvely skewed data sets.

Zinc

	General Statistics		
Total Number of Observations	14	Number of Distinct Observations	14
		Number of Missing Observations	0
Minimum	32.9	Mean	42.3
Maximum	54.4	Median	42.5
SD	6.449	Std. Error of Mean	1.724
Coefficient of Variation	0.152	Skewness	0.159
	Normal GOF Test		
Shapiro Wilk Test Statistic	0.963	**Shapiro Wilk GOF Test**	
1% Shapiro Wilk Critical Value	0.825	Data appear Normal at 1% Significance Level	
Lilliefors Test Statistic	0.143	**Lilliefors GOF Test**	
1% Lilliefors Critical Value	0.263	Data appear Normal at 1% Significance Level	
	Data appear Normal at 1% Significance Level		
	Assuming Normal Distribution		
95% Normal UCL		**95% UCLs (Adjusted for Skewness)**	
95% Student's-t UCL	45.35	95% Adjusted-CLT UCL (Chen-1995)	45.21
		95% Modified-t UCL (Johnson-1978)	45.36
	Gamma GOF Test		
A-D Test Statistic	0.253	**Anderson-Darling Gamma GOF Test**	
5% A-D Critical Value	0.733	Detected data appear Gamma Distributed at 5% Significance Level	
K-S Test Statistic	0.162	**Kolmogorov-Smirnov Gamma GOF Test**	
5% K-S Critical Value	0.228	Detected data appear Gamma Distributed at 5% Significance Level	
	Detected data appear Gamma Distributed at 5% Significance Level		
	Gamma Statistics		
k hat (MLE)	46.03	k star (bias corrected MLE)	36.21
Theta hat (MLE)	0.919	Theta star (bias corrected MLE)	1.168
nu hat (MLE)	1289	nu star (bias corrected)	1014
MLE Mean (bias corrected)	42.3	MLE Sd (bias corrected)	7.029
		Approximate Chi Square Value (0.05)	941.1
Adjusted Level of Significance	0.0312	Adjusted Chi Square Value	931.7
	Assuming Gamma Distribution		
95% Approximate Gamma UCL	45.58	95% Adjusted Gamma UCL	46.03
	Lognormal GOF Test		
Shapiro Wilk Test Statistic	0.958	**Shapiro Wilk Lognormal GOF Test**	
10% Shapiro Wilk Critical Value	0.895	Data appear Lognormal at 10% Significance Level	

	A–E	F	G–K	L
2151	Lilliefors Test Statistic	0.171	**Lilliefors Lognormal GOF Test**	
2152	10% Lilliefors Critical Value	0.208	Data appear Lognormal at 10% Significance Level	
2153	**Data appear Lognormal at 10% Significance Level**			
2154				
2155	**Lognormal Statistics**			
2156	Minimum of Logged Data	3.493	Mean of logged Data	3.734
2157	Maximum of Logged Data	3.996	SD of logged Data	0.154
2158				
2159	**Assuming Lognormal Distribution**			
2160	95% H-UCL	45.69	90% Chebyshev (MVUE) UCL	47.54
2161	95% Chebyshev (MVUE) UCL	49.91	97.5% Chebyshev (MVUE) UCL	53.19
2162	99% Chebyshev (MVUE) UCL	59.66		
2163				
2164	**Nonparametric Distribution Free UCL Statistics**			
2165	**Data appear to follow a Discernible Distribution**			
2166				
2167	**Nonparametric Distribution Free UCLs**			
2168	95% CLT UCL	45.14	95% BCA Bootstrap UCL	44.92
2169	95% Standard Bootstrap UCL	45.03	95% Bootstrap-t UCL	45.58
2170	95% Hall's Bootstrap UCL	45.37	95% Percentile Bootstrap UCL	44.88
2171	90% Chebyshev(Mean, Sd) UCL	47.47	95% Chebyshev(Mean, Sd) UCL	49.81
2172	97.5% Chebyshev(Mean, Sd) UCL	53.06	99% Chebyshev(Mean, Sd) UCL	59.45
2173				
2174	**Suggested UCL to Use**			
2175	95% Student's-t UCL	45.35		
2176				
2177	Note: Suggestions regarding the selection of a 95% UCL are provided to help the user to select the most appropriate 95% UCL.			
2178	Recommendations are based upon data size, data distribution, and skewness using results from simulation studies.			
2179	However, simulations results will not cover all Real World data sets; for additional insight the user may want to consult a statistician.			
2180				

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